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Peter R. Mercer

A Compact Capstone Course in Classical Calculus

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For Cyril Sargent. Like him was I . . .

Preface

Most colleges and universities offer a general education mathematics course, informally referred to as *Mathematics for Poets*. Though it may have modest prerequisites, such a course often contains some very interesting material, like Euclid's theorem on the infinitude of primes, the Pythagorean Theorem, irrational numbers, infinities, Pick's Theorem, Euler's Polyhedral Formula, the Platonic Solids, etc.

The book you are holding could well have been titled

Mathematics for Poets Who Know Some Calculus,

as it also contains some interesting material (irrespective of this author's opinion!), as can be seen from a glance at the Table of Contents.

And there's more: the Binomial Theorem, the irrationality of e and π, the Harmonic Series, the Madhavan Series, Seidel's Product, other amazing series and products, the Kepler-Bouwkamp Constant, Euler's Constant, Apéry's Constant, Bernoulli's Inequality, Bernoulli Polynomials, Buffon's Needle, Jensen's Inequality, the Arithmetic Mean - Geometric Mean Inequality, Power Means . . .

Not only are these topics intrinsically agreeable, many are of use to working mathematicians.

A good number of the items covered are quite sophisticated–traditionally to be met only after substantial study. A novelty here is that the exposition is self-contained, and is accessible to a student who has had two one-semester courses in college calculus. (There is a pinch of multivariable calculus, which could readily be filled in "just in time" if necessary.) Many approaches taken are modifications of those appearing quite recently in the mathematics literature.

The book has 15 chapters, including a Prelude, three Interludes, and a Postlude. These last five are independent of subsequent chapters and contain no exercises, so could be postponed (not indefinitely!) at the reader's discretion.

Acknowledgments

I am profoundly grateful to my family – Pina, Sarah, Hannah, and John – who give me joy and strength, making all things possible. For this particular project I thank my excellent editors at Springer Mathematics, Chris Eder and Dorothy Mazlum. Several superb yet anonymous reviewers have improved the book considerably.

All this fine support notwithstanding, there surely remain typographical errors, oversights, even howlers, for which I take sole responsibility. As Shakespeare wrote in a certain prologue: *Piece out our imperfections with your thoughts*.

I shall be glad to be informed of any mistakes/omissions via email address below.

Buffalo, NY, USA
July 13, 2023

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Prelude: Viète's Product

1

The ancient Greeks, and Archimedes in particular (~ 250 BCE), obtained approximations for π geometrically, that is, by computing the perimeters of regular polygons *inscribed* in a circle and of regular polygons *outscribed* on a circle. The more sides the polygon has, the better the approximation. See Fig. 1.1.

Archimedes used 96-sided (yes) regular polygons to obtain the estimates

$$3.14085 \approx \frac{223}{71} < \pi < \frac{22}{7} \approx 3.14286\,.$$

Over 1800 years later, in 1593, Viète[1] obtained the first-ever *infinite product* for π, below. (Still, this was about 200 years before the French Revolution!)

Viète's Product is

$$\frac{2}{\pi} = \frac{\sqrt{2}}{2} \cdot \frac{\sqrt{2+\sqrt{2}}}{2} \cdot \frac{\sqrt{2+\sqrt{2+\sqrt{2}}}}{2} \cdot \frac{\sqrt{2+\sqrt{2+\sqrt{2+\sqrt{2}}}}}{2} \cdot \cdots . \tag{1.1}$$

Mathematicians generally regard Viète's discovery as the birth of modern mathematical analysis. This was the first explicit formula for the exact value of π. Though he did obtain it geometrically, Viète's Product gives approximations for π in a manner very different from Archimedes'.

If we take, for example, only the first six factors of the product on the right-hand side of (1.1) and solve for π, we get $\pi \approx 3.1412$—not bad for 1593.

Writing down more than just a few factors of (1.1) can be very confusing; you'll soon find yourself up to your elbows in root signs. But observe that each

[1] Françoise Viète (France, 1540–1603). His name is often Latinized, as Vieta. He may not have liked that.

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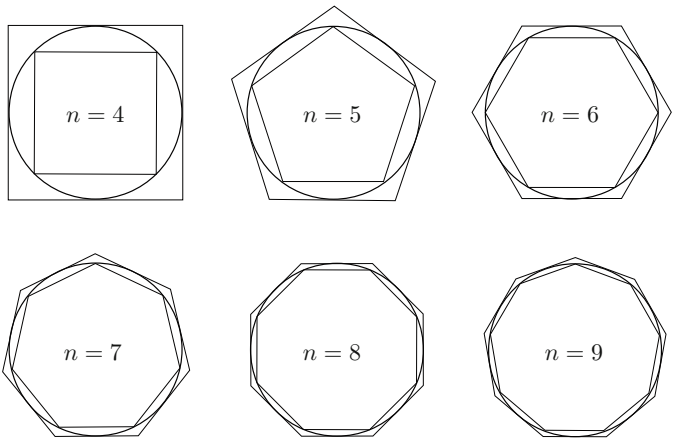


Fig. 1.1 The more sides the polygon has, the better it approximates a circle

numerator in (1.1), after the first one, is $\sqrt{2+\text{ the previous numerator}}$. And all the denominators are 2. So writing this more neatly, (1.1) reads

$$\frac{2}{\pi} \;=\; \frac{b_1}{2}\frac{b_2}{2}\frac{b_3}{2}\cdots \;=\; \prod_{n=1}^{\infty}\frac{b_n}{2}\,,$$

where

$$b_1 = \sqrt{2}\,, \quad \text{and} \quad b_{n+1} = \sqrt{2+b_n} \quad (n = 1, 2, 3, \ldots)\,.$$

A Word of Caution For a series

$$\sum_{n=1}^{N}\frac{a_n}{2} \;=\; \frac{1}{2}\sum_{n=1}^{N} a_n\,,$$

but for a product

$$\prod_{n=1}^{N}\frac{a_n}{2} \;=\; \frac{1}{2^N}\prod_{n=1}^{N} a_n \;\neq\; \frac{1}{2}\prod_{n=1}^{N} a_n\,.$$

In this prelude, we actually prove a bit more than (1.1), in line (1.5). But before doing so, we isolate a few facts which we shall use along the way.

1.1 Stuff We'll Need

- Taking $A = B = x/2$ in the angle-sum trig formula

$$\sin(A + B) \;=\; \sin(A)\cos(B) \;+\; \cos(A)\sin(B)$$

gives the half-angle formula

$$\sin(x) \;=\; 2\sin\Big(\frac{x}{2}\Big)\cos\Big(\frac{x}{2}\Big)\,. \tag{1.2}$$

- And taking $A = B = x/2$ in the (other) angle-sum trig formula

$$\cos(A + B) \;=\; \cos(A)\cos(B) \;-\; \sin(A)\cos(B)$$

gives

$$\cos(x) \;=\; \cos^2\Big(\frac{x}{2}\Big) - \sin^2\Big(\frac{x}{2}\Big) \;=\; \cos^2\Big(\frac{x}{2}\Big) - \Big[1 - \cos^2\Big(\frac{x}{2}\Big)\Big] \;=\; 2\cos^2\Big(\frac{x}{2}\Big) - 1\,.$$

Therefore

$$\cos\Big(\frac{x}{2}\Big) \;=\; \frac{\sqrt{2 + 2\cos(x)}}{2} \qquad \text{and} \qquad 2\cos\Big(\frac{x}{2}\Big) \;=\; \sqrt{2 + 2\cos(x)}\,. \tag{1.3}$$

[These trig identities, and many others, were known to the ancient Greeks but in quite a different form. The modern versions were discovered by the Persians (in modern Iran), around 960 CE.]

- We shall also use the important limit from freshman calculus:

$$\lim_{x \to 0} \frac{\sin(x)}{x} = 1\,. \tag{1.4}$$

(This says, in particular, that for $f(x) = \sin(x)$, we have $f'(0) = 1$.)

1.2 Proof of Viète's Product

Following [9], for example, we obtain a more general infinite product which was discovered by Euler[2] (pronounced OY-lur):

[2] The great Leonhard Euler (Switzerland, 1707–1783) was the most prolific of all mathematicians. And not just prolific at mathematics, he fathered 13 children.

$$\frac{\sin(\theta)}{\theta} \;=\; \cos\!\left(\tfrac{\theta}{2}\right)\cos\!\left(\tfrac{\theta}{4}\right)\cos\!\left(\tfrac{\theta}{8}\right)\cdots\cos\!\left(\tfrac{\theta}{2^n}\right)\cdots \;=\; \prod_{n=1}^{\infty}\cos\!\left(\tfrac{\theta}{2^n}\right). \tag{1.5}$$

Proof of (1.5). Using formula (1.2) repeatedly, we have

$$\begin{aligned}
\sin(\theta) &= 2\cos\!\left(\tfrac{\theta}{2}\right)\sin\!\left(\tfrac{\theta}{2}\right)\\
&= 2\cos\!\left(\tfrac{\theta}{2}\right)2\cos\!\left(\tfrac{\theta}{4}\right)\sin\!\left(\tfrac{\theta}{4}\right) \;=\; 2^2\cos\!\left(\tfrac{\theta}{2}\right)\cos\!\left(\tfrac{\theta}{2^2}\right)\sin\!\left(\tfrac{\theta}{4}\right)\\
&= 2^2\cos\!\left(\tfrac{\theta}{2}\right)\cos\!\left(\tfrac{\theta}{2^2}\right)2\cos\!\left(\tfrac{\theta}{8}\right)\sin\!\left(\tfrac{\theta}{8}\right) \;=\; 2^3\cos\!\left(\tfrac{\theta}{2}\right)\cos\!\left(\tfrac{\theta}{2^2}\right)\cos\!\left(\tfrac{\theta}{2^3}\right)\sin\!\left(\tfrac{\theta}{2^3}\right)\\
&\;\;\vdots\\
&\;\;\downarrow\\
&= 2^n\cos\!\left(\tfrac{\theta}{2}\right)\cos\!\left(\tfrac{\theta}{2^2}\right)\cos\!\left(\tfrac{\theta}{2^3}\right)\cdots\cos\!\left(\tfrac{\theta}{2^n}\right)\sin\!\left(\tfrac{\theta}{2^n}\right).
\end{aligned}$$

Now, so that we can use (1.4), we shrewdly write this as

$$\frac{\sin(\theta)}{\theta} \;=\; \cos\!\left(\tfrac{\theta}{2}\right)\cos\!\left(\tfrac{\theta}{2^2}\right)\cos\!\left(\tfrac{\theta}{2^3}\right)\cdots\cos\!\left(\tfrac{\theta}{2^n}\right)\frac{\sin\!\left(\frac{\theta}{2^n}\right)}{\frac{\theta}{2^n}}.$$

Then, letting $n\to\infty$ and using (1.4), we get $\frac{\sin(\theta/2^n)}{\theta/2^n}\to 1$ and we're done. □

Proof of Viète's Product (1.1). Setting $\theta=\pi/2$ in (1.5), we have

$$\frac{2}{\pi} \;=\; \cos\!\left(\tfrac{\pi}{4}\right)\cdot\cos\!\left(\tfrac{\pi}{8}\right)\cdot\cos\!\left(\tfrac{\pi}{16}\right)\cdot\cos\!\left(\tfrac{\pi}{32}\right)\cdots$$

Using formulas (1.3) repeatedly, we get

$$\begin{aligned}
\frac{2}{\pi} &= \cos\!\left(\tfrac{\pi}{4}\right)\cdot\cos\!\left(\tfrac{\pi}{8}\right)\cdot\cos\!\left(\tfrac{\pi}{16}\right)\cdot\cos\!\left(\tfrac{\pi}{32}\right)\cdots\\
&= \cos\!\left(\tfrac{\pi}{4}\right)\cdot\frac{\sqrt{2+2\cos\left(\frac{\pi}{4}\right)}}{2}\cdot\frac{\sqrt{2+2\cos\left(\frac{\pi}{8}\right)}}{2}\cdot\frac{\sqrt{2+2\cos\left(\frac{\pi}{16}\right)}}{2}\cdots\\
&= \frac{\sqrt{2}}{2}\cdot\frac{\sqrt{2+2\cos\left(\frac{\pi}{4}\right)}}{2}\cdot\frac{\sqrt{2+\sqrt{2+2\cos\left(\frac{\pi}{4}\right)}}}{2}\cdot\frac{\sqrt{2+\sqrt{2+2\cos\left(\frac{\pi}{8}\right)}}}{2}\cdots\\
&= \frac{b_1}{2}\cdot\frac{\sqrt{2+b_1}}{2}\cdot\frac{\sqrt{2+b_2}}{2}\cdot\frac{\sqrt{2+b_3}}{2}\cdots,
\end{aligned}$$

where $b_1 = \sqrt{2}$ and $b_{n+1} = \sqrt{2 + b_n}$ for $n = 1, 2, 3, \ldots$, as desired. □

▶ **Remarks** In a similar way, but taking $\theta = \pi/6$ in (1.5), one can show (try it!) that

$$\frac{3}{\pi} = \frac{\sqrt{2+\sqrt{3}}}{2} \cdot \frac{\sqrt{2+\sqrt{2+\sqrt{3}}}}{2} \cdot \frac{\sqrt{2+\sqrt{2+\sqrt{2+\sqrt{3}}}}}{2} \cdots .$$

For a proof of (1.1) which is close to Viète's original argument, see [3]. It uses no trigonometry, just the Pythagorean Theorem.

In [6], the sum of the series

$$\tan\left(\tfrac{\pi}{4}\right) + \frac{1}{2}\tan\left(\tfrac{\pi}{8}\right) + \frac{1}{4}\tan\left(\tfrac{\pi}{16}\right) + \frac{1}{8}\tan\left(\tfrac{\pi}{32}\right) + \cdots$$

is computed, just the way Euler did it. The starting point is (1.5).

1.3 Seidel's Product

The hyperbolic trig functions

$$\sinh(t) = \frac{e^t - e^{-t}}{2} \qquad \text{and} \qquad \cosh(t) = \frac{e^t + e^{-t}}{2}$$

have many properties which mirror those of the (circular) trig functions. In particular (verify!)

$$\lim_{x\to 0} \frac{\sinh(x)}{x} = 1 \quad \text{and} \quad \sinh(2x) = 2\sinh(x)\cosh(x).$$

So in exactly the same way that we obtained (1.5) above, we can get

$$\frac{\sinh(\theta)}{\theta} = \cosh\left(\tfrac{\theta}{2}\right)\cosh\left(\tfrac{\theta}{4}\right)\cosh\left(\tfrac{\theta}{8}\right)\cosh\left(\tfrac{\theta}{16}\right)\cdots . \tag{1.6}$$

Using this, we obtain a sort of a hyperbolic partner for Viète's Product:

$$\frac{1}{\ln(2)} = \frac{1+\sqrt{2}}{2} \cdot \frac{1+\sqrt{\sqrt{2}}}{2} \cdot \frac{1+\sqrt{\sqrt{\sqrt{2}}}}{2} \cdot \frac{1+\sqrt{\sqrt{\sqrt{\sqrt{2}}}}}{2} \cdots . \tag{1.7}$$

Proof of (1.7). In (1.6), set $\theta = \ln(x)$, so that $\frac{1}{2}\theta = \ln(x^{1/2})$, $\frac{1}{4}\theta = \ln(x^{1/4})$, $\frac{1}{8}\theta = \ln(x^{1/8})$, etc., $\sinh(\theta) = \frac{x-1/x}{2}$, and $\cosh(\theta) = \frac{x+1/x}{2}$:

$$\begin{aligned}\frac{x-1/x}{2\ln(x)} &= \frac{x^{1/2}+1/x^{1/2}}{2}\,\frac{x^{1/4}+1/x^{1/4}}{2}\,\frac{x^{1/8}+1/x^{1/8}}{2}\cdots\\ &= \frac{1+x}{2x^{1/2}}\,\frac{1+x^{1/2}}{2x^{1/4}}\,\frac{1+x^{1/4}}{2x^{1/8}}\cdots\\ &= \frac{1+x}{2}\,\frac{1}{x^{1/2+1/4+1/8+\cdots}}\prod_{n=1}^{\infty}\frac{1+x^{1/2^n}}{2}\,.\end{aligned}$$

The geometric series $1/2 + 1/4 + 1/8 + \cdots = 1$, and so

$$\frac{x-1/x}{2\ln(x)} = \frac{1+x}{2x}\prod_{n=1}^{\infty}\frac{1+x^{1/2^n}}{2},$$

or

$$\frac{x-1}{\ln(x)} = \prod_{n=1}^{\infty}\frac{1+x^{1/2^n}}{2}\,.$$

This is *Seidel's*[3] *Product* of 1871 (e.g., [8, 10]). Setting $x = 2$ here gives (1.7). □

1.4 Epilogue

Our proof of (1.1) leaned heavily on (1.5) with $\theta = \pi/2$:

$$\frac{2}{\pi} = \cos\left(\tfrac{\pi}{4}\right)\cos\left(\tfrac{\pi}{8}\right)\cos\left(\tfrac{\pi}{16}\right)\cos\left(\tfrac{\pi}{32}\right)\cdots = \prod_{n=2}^{\infty}\cos\left(\tfrac{\pi}{2^n}\right).$$

Incorporating in this product, *all* integer denominators 3 and larger gives the number

$$P = \prod_{n=3}^{\infty}\cos\left(\tfrac{\pi}{n}\right) \approx 0.114942\,.$$

[3] Philipp Ludwig von Seidel (Germany, 1821–1896) made substantial contributions to mathematical analysis and to optics—there's even an aberration named after him.

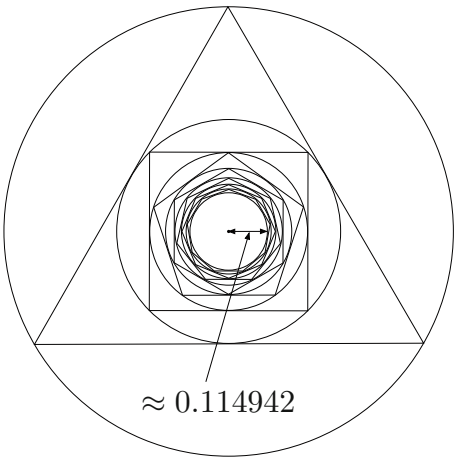


Fig. 1.2 The limiting radius here is $P = \prod_{n=3}^{\infty} \cos(\pi/n) \approx 0.114942$

This is known as the *Kepler-Bouwkamp constant* [1]. Bouwkamp himself wrote, in 1964, "So far as I know, a closed expression for P is not available and, in my opinion, is not likely to exist." None has yet been found. But what has Kepler to do with this? Well, consider the following construction:

> An equilateral triangle is inscribed in a circle of radius one, then a circle is inscribed in the triangle, then a square is inscribed in this circle, etc. Continue this procedure, increasing the number of sides of the regular polygon each time by one.

It turns out that successive circles' radii satisfy $r_n = r_{n-1}\cos(\pi/n)$ (verify this!), so the shrinking circles have a limiting radius of P [2, also [7]]. See Fig. 1.2.

So by Viète's Product (1.1), the construction above, but modified by using only 2^n-sided regular polygons each time, yields a limiting radius of $2/\pi \approx 0.636619$. (See [5] for more on this.) This modification is getting close to the idea behind Viète's original proof, though he did not use (1.5). Again, see [3].

About the connection with Kepler: Beginning with a circle of radius 1 and *outscribing* instead of inscribing, the limiting outer circle has radius of $1/P \approx 8.70004$. Okay, now have a look at Fig. 1.3, from Kepler's first major work in astronomy, the *Mysterium Cosmographicum*, of 1596. Pretty cool.

▶ **Remark** One way (of many!) to measure the roundness of certain plane figures is via *aspect ratio*: the ratio of the radius of the largest inscribed circle to the radius of the smallest outscribed circle. So a figure having aspect ratio near one might be deemed to be pretty darned round. For an n-sided regular polygon, this measure is precisely $\cos(\pi/n)\ldots$.

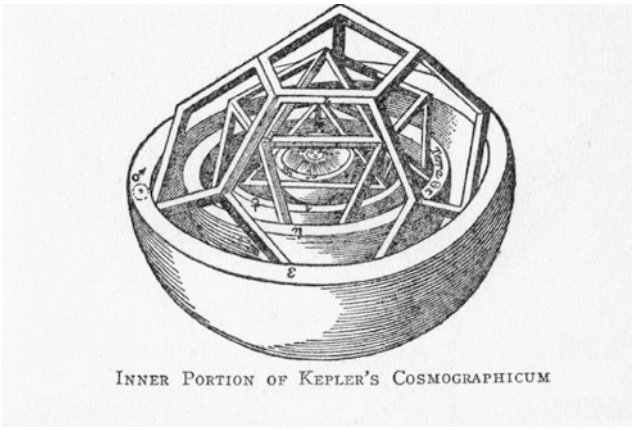


Fig. 1.3 Kepler's Cosmographicum, 1597 [4]

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Calculus Warm-Up

2

In this chapter we review a few particularly important ideas from calculus, in case you're a bit rusty (or forgot outright). We'll be using many of these, as we go forward. And we'll look at a thing or two that you have possibly not seen at all.

2.1 Taylor's Theorem

Taylor's Theorem *Let f and its derivatives f', f'', $f^{(3)}, \ldots, f^{(n)}$ be defined on some open interval I. Let $x_0 \in I$. Then for each $x \in I$, there is c between x and x_0 such that*

$$f(x) = f(x_0) + f'(x_0)(x - x_0) + \frac{f''(x_0)}{2}(x - x_0)^2 + \frac{f^{(3)}(x_0)}{3!}(x - x_0)^3$$

$$+ \cdots + \frac{f^{(n-1)}(x_0)}{(n-1)!}(x - x_0)^{n-1} + \frac{f^{(n)}(c)}{n!}(x - x_0)^n.$$

Note. The c above depends not only on x and x_0 but also on n. So we really should write $c = c_n$ here. A proof of Taylor's[1] Theorem can be found in just about any calculus textbook. (The paper [6] contains a particularly slick proof.)

The $n!$ is, of course, the *factorial function*: For $n \geq 1$,

$$n! = n(n-1)(n-2)\cdots(2)(1).$$

[1] Named for Brook Taylor (England, 1685–1731) but anticipated 40 years earlier by James Gregory (Scotland, 1638–1675).

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This is the number of different ways of arranging n objects. And we take $0! = 1$; indeed there is only one way to arrange zero objects—that is, to not arrange them at all. So the *number* of ways of arranging zero objects is $0! = 1$.

Taylor's Theorem with $n = 1$ is the Mean Value Theorem[2] and the Mean Value Theorem is really just Rolle's Theorem.[3]

The following useful formula can be proved without calculus (Exercise 2.1a), but let's prove it using Taylor's Theorem.

The Binomial Formula *For integers $m \geq 1$, we have*

$$(1+x)^m = \sum_{k=0}^{m} \binom{m}{k} x^k, \quad \text{where} \quad \binom{m}{k} = \frac{m!}{(m-k)!\,k!}.$$

Proof. Let $f(x) = (1+x)^m$. Then

$$f'(x) = m(1+x)^{m-1}, \quad f''(x) = m(m-1)(1+x)^{m-2}, \quad \text{etc.}$$

And for any $k \leq m$,

$$f^{(k)}(x) = m(m-1)\cdots(m-(k-1))(1+x)^{m-k}.$$

Therefore

$$f^{(k)}(0) = \begin{cases} m(m-1)\cdots(m-k+1) & \text{if } k \leq m \\ 0 & \text{if } k > m. \end{cases}$$

So Taylor's Theorem with $x_0 = 0$ and any $n > m$ gives

$$(1+x)^m = 1 + mx + \frac{m(m-1)}{2}x^2 + \frac{m(m-1)(m-2)}{3!}x^3 + \\ + \cdots + \frac{m(m-1)(n-2)\cdots 2}{(m-1)!}x^{m-1} + \frac{m!}{m!}x^m.$$

There's clearly a pattern here (remember that $0! = 1$):

$$(1+x)^m = \frac{m!}{(m-0)!0!} + \frac{m!}{(m-1)!1!}x + \frac{m!}{(m-2)!2!}x^2 + \frac{m!}{(m-3)!3!}x^3 +$$

[2] Due to Augustin-Louis Cauchy (France, 1789–1857). There are more mathematical concepts named for Cauchy than for any other mathematician.

[3] Named for Michel Rolle (France, 1652–1719) who obtained the theorem for polynomials.

$$+ \cdots + \frac{m!}{(m-(m-1))!(m-1)!}\, x^{m-1} + \frac{m!}{(m-m)!m!}\, x^m .$$

□

▶ **Remark** $\binom{m}{k} = \frac{m!}{(m-k)!\,k!}$ is the number of different ways of choosing k objects from a set of m objects ($k \le m$). Out loud (or in your own head), it is read "m choose k."

Fourier[4] used Taylor's Theorem to prove a classic fact, as follows:

Theorem 2.1
e is irrational.

Proof. Let $f(x) = \mathrm{e}^x$. This has, of course, $f^{(k)}(x) = \mathrm{e}^x$ for all k. So taking $x_0 = 0$ and $x = 1$ in Taylor's Theorem, there is c between 0 and 1 such that

$$\mathrm{e} = 1 + 1 + \frac{1}{2!} + \frac{1}{3!} + \cdots + \frac{1}{(n-1)!} + \frac{\mathrm{e}^c}{n!},$$

wherein $1 < \mathrm{e}^c < \mathrm{e}$, no matter what n is. Now, looking for a contradiction, we suppose that e is rational. That is, $\mathrm{e} = a/b$, where a and b are positive integers (since $\mathrm{e} > 0$). Then

$$\frac{a}{b} = 1 + 1 + \frac{1}{2!} + \cdots + \frac{1}{(n-1)!} + \frac{\mathrm{e}^c}{n!},$$

and so

$$(n-1)!\frac{a}{b} = (n-1)!\left(1 + 1 + \frac{1}{2!} + \cdots + \frac{1}{(n-1)!}\right) + (n-1)!\frac{e^c}{n!}$$

$$= (n-1)!\left(1 + 1 + \frac{1}{2!} + \cdots + \frac{1}{(n-1)!}\right) + \frac{\mathrm{e}^c}{n}.$$

If $n > b$, then the left-hand side is an integer. The first term on the right-hand side is an integer for any n. If $n > \mathrm{e}$, then the second term on the right-hand side is between 0 and 1. So choosing any $n > \max\{\mathrm{e}, b\}$ yields a contradiction, as desired. □

[4] Joseph Fourier (France, 1768–1830) made seminal contributions to applied mathematics and to heat conduction in particular. [Fourier believed that always being wrapped in blankets was good for one's health. He died after falling down the stairs—many say from tripping on his blankets.]

An important application of Taylor's Theorem is to obtain (when possible) a Taylor Series. For convenience take $x_0 = 0$, in which case a Taylor Series is called a *Maclaurin Series*.[5] Here's the story:

- *If a function f has derivatives of all orders $f'(0)$, $f''(0)$, $f^{(3)}(0), \ldots$ on I, and if for $x \in I$, $\frac{f^{(n)}(c_n)}{n!}x^n \to 0$ as $n \to \infty$, then f has a Maclaurin Series*

$$f(x) = f(0) + f'(0)x + \frac{f''(0)}{2}x^2 + \frac{f^{(3)}(0)}{3!}x^3 + \cdots = \sum_{n=0}^{\infty} \frac{f^{(n)}(0)}{n!}x^n.$$

A few familiar Maclaurin Series are listed below.

- $e^x = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \frac{x^4}{4!} + \cdots = \sum_{n=0}^{\infty} \frac{x^n}{n!}$ for $x \in \mathbb{R}$.
- $\sin(x) = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!} + \cdots = \sum_{n=0}^{\infty} \frac{(-1)^n}{(2n+1)!}x^{2n+1}$ for $x \in \mathbb{R}$.
- $\cos(x) = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \frac{x^6}{6!} + \cdots = \sum_{n=0}^{\infty} \frac{(-1)^n}{(2n)!}x^{2n}$ for $x \in \mathbb{R}$.
- $\frac{1}{1-x} = 1 + x + x^2 + x^3 + \cdots = \sum_{n=0}^{\infty} x^n$ for $x \in (-1, 1)$.
- $\ln(1+x) = x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} + \cdots = \sum_{n=0}^{\infty} \frac{(-1)^n}{n+1}x^{n+1}$ for $x \in (-1, 1]$.

2.2 Integration by Parts

Applying the Product Rule for derivatives to $f(x)G(x)$, and then integrating the result, we get

$$\int f(x)g(x)\,dx = f(x)G(x) - \int f'(x)G(x)\,dx, \quad \text{where } G'(x) = g(x).$$

To make use of this for actually evaluating an integral, one differentiates *part* of the integrand (the $f(x)$) and integrates[6] the other *part* (the $g(x)$). If the integral in the result looks simpler, then it is probably worth pursuing.

We write this in shorter form, for definite integrals, using "D" for *differentiate* and "I" for *integrate*, with the understanding that $G'(x) = g(x)$:

[5] Colin Maclaurin (Scotland, 1698–1746) was the world's youngest-ever university professor (age 19) until 2008, when scientist Alia Sabur (USA, 1989) became a professor at age 18.

[6] Rather than *integrates*, we should (perhaps) say *anti-differentiates*.

Integration by Parts Formula

$$\int_a^b \underbrace{f(x)}_{\mathrm{D}} \underbrace{g(x)}_{\mathrm{I}}\,dx \;=\; f(x)G(x)\Big|_a^b \;-\; \int_a^b f'(x)G(x)\,dx.$$

Example

$$\int_1^{\mathrm{e}} \frac{\ln(x)}{x^2}\,dx \;=\; \int_1^{\mathrm{e}} \underbrace{\ln(x)}_{\mathrm{D}}\,\underbrace{\frac{1}{x^2}}_{\mathrm{I}}\,dx \;=\; \ln(x)\frac{-1}{x}\Big|_1^{\mathrm{e}} \;-\; \int_1^{\mathrm{e}} \frac{1}{x}\frac{-1}{x}\,dx$$

$$= \frac{-1}{\mathrm{e}} + \int_1^{\mathrm{e}} \frac{1}{x^2}\,dx \;=\; \frac{-1}{\mathrm{e}} + \frac{-1}{x}\Big|_1^{\mathrm{e}} \;=\; \frac{-2}{\mathrm{e}} + 1 \;\approx\; 0.2642\,.$$

Example (We'll Need These Integrals in Chap. 11)

Let k be an integer. Then

$$\int_0^1 x\cos(k\pi x)\,dx \;=\; \int_0^1 \underbrace{x}_{\mathrm{D}}\,\underbrace{\cos(k\pi x)}_{\mathrm{I}}\,dx \;=\; x\frac{1}{k\pi}\sin(k\pi x)\Big|_0^1 \;-\; \int_0^1 1\frac{1}{k\pi}\sin(k\pi x)\,dx$$

$$= 0 \;-\; \frac{1}{k\pi}\int_0^1 \sin(k\pi x)\,dx \;=\; \frac{1}{k^2\pi^2}\cos(k\pi x)\Big|_0^1$$

$$= \frac{1}{k^2\pi^2}\left((-1)^k - 1\right) = \begin{cases} 0 & \text{if } k \text{ is even} \\ -\dfrac{2}{k^2\pi^2} & \text{if } k \text{ is odd}\,. \end{cases}$$

Example (We'll Need These Integrals in Chap. 12)

Let k be an integer. Then

$$\int_0^1 \left(x - \frac{1}{2}\right)\sin(k\pi x)dx = \int_0^1 \underbrace{\left(x - \frac{1}{2}\right)}_{\mathrm{D}}\,\underbrace{\sin(k\pi x)}_{\mathrm{I}}\,dx$$

$$= \left(x - \frac{1}{2}\right)\cdot\frac{-1}{k\pi}\cos(k\pi x)\Big|_0^1 \;+\; \int_0^1 \frac{1}{k\pi}\cos(k\pi x)dx$$

$$= \frac{1}{2k\pi}\left((-1)^{k+1} - 1\right) \;+\; \frac{1}{k^2\pi^2}\sin(k\pi x)\Big|_0^1$$

$$= \frac{1}{2k\pi}\left((-1)^{k+1} - 1\right) = \begin{cases} 0 & \text{if } k \text{ is odd} \\ -\frac{1}{k\pi} & \text{if } k \text{ is even}\,. \end{cases}$$

Reduction formulas often come from integration by parts. Here's one (which we shall meet again, in different guise, in Sect. 3.2).

Example
For integers $n \geq 0$, consider the (improper) definite integral

$$\Lambda_n = \int_0^1 \left(\ln\left(\frac{1}{x}\right)\right)^n dx = \int_0^1 (-\ln(x))^n dx. \text{ Clearly } \Lambda_0 = 1. \text{ Also}$$

$$\Lambda_n = \int_0^1 \underbrace{(-\ln(x))^n}_{\mathrm{D}} \underbrace{1}_{\mathrm{I}}\, dx \;=\; (-\ln(x))^n x \Big|_0^1 \;+\; \int_0^1 n(-\ln(x))^{n-1}\frac{1}{x}\, x\, dx\,.$$

Applying L'Hospital's Rule[7] to the first piece on the right-hand side, n times, we get (this is Exercise 2.3)

$$(-\ln(x))^n x \Big|_0^1 \;=\; 0 - \lim_{x\to 0^+} \frac{(-\ln(x))^n}{1/x} \;=\; \cdots \;=\; 0\,,$$

and so we obtain the reduction formula $\Lambda_n = \int_0^1 n(-\ln(x))^{n-1} dx \;=\; n\Lambda_{n-1}\,.$

Therefore

$$\Lambda_n \;=\; n\Lambda_{n-1} \;=\; n(n-1)\Lambda_{n-2} \;=\; \cdots \;=\; n(n-1)(n-2)\cdots(2)\cdot(1)\cdot\Lambda_0 \;=\; n!\,.$$

Thanks largely to [2], nowadays the following classic fact[8] is typically proved using integration by parts.

Theorem 2.2
π is irrational.

[7] Guillaume de L'Hospital (France, 1661–1704). The rule is in fact due to Johann Bernoulli (Switzerland, 1667–1748). L'Hospital was Bernoulli's student, who published (with permission!) his teacher's notes, in 1696.

[8] This was first proved, in 1761, by Johann Heinrich Lambert (Switzerland, 1728–1777). Lambert's father was a Taylor. No no—a *tailor.*

Proof (Also [3]). Let f be any *polynomial* of degree $2n$. Integrating by parts three times, we get $\int_0^\pi \underbrace{f(x)}_{\mathrm{D}} \underbrace{\sin(x)}_{\mathrm{I}}\,dx$

$$= -f(x)\cos(x)\Big|_0^\pi + \int_0^\pi \underbrace{f'(x)}_{\mathrm{D}} \underbrace{\cos(x)}_{\mathrm{I}}\,dx$$

$$= -f(x)\cos(x)\Big|_0^\pi + f'(x)\sin(x)\Big|_0^\pi - \int_0^\pi \underbrace{f''(x)}_{\mathrm{D}} \underbrace{\sin(x)}_{\mathrm{I}}\,dx$$

$$= -f(x)\cos(x)\Big|_0^\pi + f'(x)\sin(x)\Big|_0^\pi + f''(x)\cos(x)\Big|_0^\pi + \int_0^\pi f'''(x)\cos(x)dx\,.$$

We continue in this way $\big($in the next step, we would "do" $\int_0^\pi \underbrace{f'''(x)}_{\mathrm{D}} \underbrace{\cos(x)}_{\mathrm{I}}\,dx\big)$, and also use $\sin(0) = \sin(\pi) = 0$. Then since f is a polynomial of degree $2n$, the procedure terminates, and we get

$$\int_0^\pi f(x)\sin(x)dx \tag{2.1}$$

$$= -f(x)\cos(x)\Big|_0^\pi + f^{(2)}(x)\cos(x)\Big|_0^\pi - f^{(4)}(x)\cos(x)\Big|_0^\pi + \cdots \pm f^{(2n)}(x)\cos(x)\Big|_0^\pi\,.$$

Now, seeking a contradiction we assume that $\pi = a/b$ where a and b are positive integers (since $\pi > 0$). Consider then the polynomial of degree $2n$ given by

$$f_n(x) \;=\; \frac{b^n}{n!}\,x^n(\pi - x)^n\,.$$

Evidently $f_0(x) = 1$ and $f_1(x) = bx(\pi - x)$. By the Product Rule for derivatives, and then some manipulating, we get

$$f_n'(x) \;=\; \frac{b^n}{n!}\Big[nx^{n-1}(\pi - x)^n - x^n n(\pi - x)^{n-1}\Big]\,.$$

Then some simplifying gives

$$f_n'(x) \;=\; \frac{b^{n-1}}{(n-1)!}x^{n-1}(\pi - x)^{n-1}\big[b\pi - 2bx\big]$$

$$= f_{n-1}(x)\big[b\pi - 2bx\big].$$

Using this in conjunction with the Product Rule again, we get

$$f_n''(x) = f_{n-2}(x)\big[b\pi - 2bx\big]^2 - 2bf_{n-1}(x). \tag{2.2}$$

Since $f_0(x) = 1$ and $f_1(x) = bx(\pi - x)$, $f_0(\pi)$, $f_0(0)$, $f_1(\pi)$, and $f_1(0)$ are all integers. Then by the recursion (2.2), $f_n^{(2)}(\pi)$ and $f_n^{(2)}(0)$ are also integers, and so on: all $f_n^{(2k)}(\pi)$ and $f_n^{(2k)}(0)$ are integers. Therefore, looking at (2.1), $\int_0^\pi f_n(x)\sin(x)dx$ must be an integer. Okay, observe now that

$$0 < \int_0^\pi f_n(x)\sin(x)dx < \int_0^\pi f_n(x)\,dx < \pi\frac{b^n}{n!}\pi^n\pi^n = \pi\frac{(a\pi)^n}{n!}.$$

The right-hand side here $\to 0$ as $n \to \infty$. (This requires justification: Exercise 2.7.) Therefore, if we choose n very large, then $0 < \int_0^\pi f_n(x)\sin(x)\,dx < 1$, and so $\int_0^\pi f_n(x)\sin(x)\,dx$ *cannot* be an integer. This is a contradiction, as desired, and therefore π is irrational. □

2.3 The Fundamental Property of the Real Numbers

Suppose that we have a sequence $\{a_n\}$ which is *increasing*:

$$a_1 \le a_2 \le a_3 \le a_4 \le \cdots\cdots .$$

In short, $a_n \le a_{n+1}$ for $n = 1, 2, 3, \ldots$. Suppose also that there is some number M (perhaps very large) such that

$$a_n \le M \quad \text{for } n = 1, 2, 3, \ldots .$$

The number M is called an *upper bound* for $\{a_n\}$. See Fig. 2.1.

It is a fundamental property of the real numbers[9] that there must then be some number $A \le M$ to which the sequence $\{a_n\}$ *converges*. See Fig. 2.2.

The a_ns are getting closer and closer (in fact, as close as we please) to the *limit* A. No matter what number we choose to the left of A, call it X say, there will always

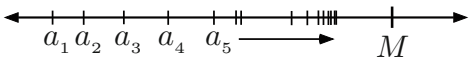


Fig. 2.1 M is an upper bound for $\{a_n\}$, that is, $a_n \le M$ for all n

[9] Either a property or an axiom, depending on your perspective. Either way, it's a fact.

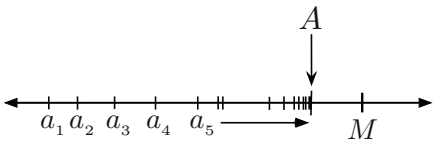


Fig. 2.2 The sequence $\{a_n\}$ converges to some number $A \leq M$

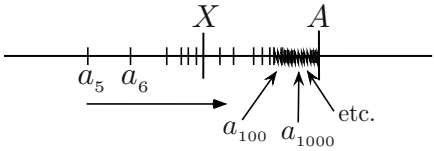


Fig. 2.3 $a_n \to A$: The sequence $\{a_n\}$ is *bunching up* toward A

be infinitely many of the a_ns between X and A:

$$X \leq a_n \leq A \quad \text{for infinitely many } n\text{'s}.$$

(And only finitely many a_ns will lie to the left of X.) See Fig. 2.3, a zoomed-in version of Fig. 2.2 with an X included.

This fundamental property gets used *all the time*—sometimes when we aren't even aware of it. It's worth repeating [1].

•• **The Fundamental Property of the Real Numbers:**
A sequence of real numbers, which is increasing and is bounded above, converges.
•• **Equivalently** (after multiplying everything by -1):
A sequence of real numbers, which is decreasing and is bounded below, converges.

Example
Let $a_1 = 1$. For $n = 1, 2, 3, 4, \ldots,$ consider the sequence

$$a_{n+1} = \sqrt{1 + \sqrt{1 + a_n}}\ .$$

One can show by induction (Exercise 2.8) that $\{a_n\}$ is an increasing sequence and that it is bounded above, by $M = \sqrt{3}$, say. Therefore, by the fundamental property, $\{a_n\}$ converges to some number $A \leq \sqrt{3}$. Letting $n \to \infty$ yields $A = \sqrt{1 + \sqrt{1 + A}}$. Simplification gives $A = \frac{1+\sqrt{5}}{2}$, the *golden mean* ≈ 1.618.

Here follows a very important application of the fundamental property.

Let $\{b_n\}$ be a sequence of positive numbers. An *alternating series* is a series that looks like

$$\sum_{k=1}^{\infty} (-1)^{k+1} b_k = b_1 - b_2 + b_3 - b_4 + b_5 - b_6 + \cdots\cdots .$$

As with any series, this may or may not converge. That is, the *sequence of partial sums*

$$\{S_n\} = \left\{\sum_{k=1}^{n}(-1)^{k+1}b_k\right\} = \{b_1\,,\ b_1-b_2\,,\ b_1-b_2+b_3\,,\ b_1-b_2+b_3-b_4\,,\dots\}$$

may or may not have a limit. The following test[10] gives sufficient conditions for an alternating series to converge.

The Alternating Series Test *Let*

$$b_1 \ge b_2 \ge b_3 \ge b_4 \ge \cdots \quad \text{and} \quad b_n \to 0 \quad \text{as} \quad n \to \infty\,.$$

Then the alternating series $\sum_{k=1}^{\infty}(-1)^{k+1}b_k$ *converges.*

Proof. We have

$$\begin{aligned} S_{2n} &= b_1 - b_2 + b_3 - b_4 + b_5 - b_6 + \cdots + b_{2n-1} - b_{2n} \\ &= (b_1 - b_2) + (b_3 - b_4) + (b_5 - b_6) + \cdots + (b_{2n-1} - b_{2n})\,. \end{aligned}$$

Therefore, since $b_k \ge b_{k+1}$, the sequence $\{S_{2n}\}$ is increasing: $S_{2n} \le S_{2(n+1)}$. But also,

$$S_{2n} = b_1 - (b_2 - b_3) - (b_4 - b_5) - \cdots - (b_{2n-2} - b_{2n-1}) - b_{2n}\,.$$

So, again since $b_k \ge b_{k+1}$, $\{S_{2n}\}$ is bounded above: $S_{2n} \le b_1$. Therefore by the fundamental property, $\{S_{2n}\}$ converges to some real number S (which is less than b_1). Now

$$S_{2n+1} = S_{2n} + b_{2n+1}\,,$$

and $b_{2n+1} \to 0$, so $\{S_{2n+1}\}$ must converge to S also. Therefore $S_n \to S$. □

The medieval philosopher d'Oresme[11] was the first to prove, in 1360, that the *Harmonic Series diverges*:[12]

[10] Sometimes called *Leibniz's test*, for philosopher/mathematician and co-discoverer of calculus Gottfried Wilhelm Leibniz (Germany, 1646–1716).

[11] Nicole d'Oresme (France, ∼1320–1382). He often gets Latinized too: Nicholas of Oresme.

[12] In music, the *nth harmonic* of a tone with wavelength 1 is a tone with wavelength $1/n$.

$$1 + \frac{1}{2} + \frac{1}{3} + \frac{1}{4} + \frac{1}{5} + \cdots = \sum_{n=1}^{\infty} \frac{1}{n} = \infty.$$

Here, in modern notation, is d'Oresme's proof: Let $S = \sum_{n=1}^{\infty} \frac{1}{n}$. Then

$$\begin{aligned} S &= 1 + \frac{1}{2} + \left(\frac{1}{3} + \frac{1}{4}\right) + \left(\frac{1}{5} + \frac{1}{6} + \frac{1}{7} + \frac{1}{8}\right) + \left(\frac{1}{9} + \cdots + \frac{1}{16}\right) + \left(\frac{1}{17} + \cdots \right. \\ &> 1 + \frac{1}{2} + \left(\frac{1}{4} + \frac{1}{4}\right) + \left(\frac{1}{8} + \frac{1}{8} + \frac{1}{8} + \frac{1}{8}\right) + \left(\frac{1}{16} + \cdots + \frac{1}{16}\right) + \cdots \\ &= 1 + \frac{1}{2} + \frac{1}{2} + \frac{1}{2} + \frac{1}{2} + \cdots = \infty. \end{aligned}$$

But by the Alternating Series Test, the *Alternating Harmonic Series converges*:

$$\sum_{k=1}^{\infty} (-1)^{k+1} \frac{1}{k} = 1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \cdots = S < \infty.$$

Clearly, $S < 1$. In fact,

$$1 - \frac{1}{2} < S < 1 - \frac{1}{2} + \frac{1}{3},$$

and

$$1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} < S < 1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \frac{1}{5},$$

and

$$1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \frac{1}{5} - \frac{1}{6} < S < 1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \frac{1}{5} - \frac{1}{6} + \frac{1}{7}, \quad \ldots \text{etc.}$$

This last estimate is approximately $0.62 < S < 0.76$. If you trust that the Maclaurin Series for $\ln(1 + x)$ that we quoted in Sect. 2.1 is correct (and you can), then you'll believe that $S = \ln(2) \approx 0.693$ (more on this later).

Exercises

2.1. Prove the Binomial Formula, two other ways:

(a) By induction.

(b) Differentiate; then tidy up, to show that $\frac{d}{dx}\sum_{k=0}^{m}\binom{m}{k}\frac{x^k}{(1+x)^m} = 0$. Conclude that $\sum_{k=0}^{m}\binom{m}{k}x^k = (1+x)^m$ [5].

2.2. **(a)** Use Taylor's Theorem to show that $1 - x^2/2 \le \cos(x) \le 1$ for all $x \in \mathbb{R}$.
(b) Use Taylor's Theorem to show that $x - x^3/6 \le \sin(x) \le x$ for $x \ge 0$.
(c) Show that the inequalities in (b) are reversed for $x \le 0$.
(d) Apply Taylor's Theorem to $f(x) = e^x$ to show that $n! > \left(\frac{n}{e}\right)^n$.

2.3. Use L'Hospital's Rule to verify that (indeed) $\lim_{x\to 0^+} x(\ln(x))^n = 0$.

2.4. Here are just a few integrals which we shall encounter as we go along. Refresh yourself—so that you can evaluate all of them!

$\int x^n dx$ $\int \sin(kx)dx$ $\int \cos(kx)dx$ $\int \tan(kx)dx$
$\int xe^{x^2}dx$ $\int x^2\cos(kx)dx$ $\int \cos(x)\cos(kx)dx$
$\int \frac{1}{1+x^2}dx$ $\int \frac{1}{1-x}dx$ $\int \ln(x)dx$ $\int \tan^{-1}(x)dx$

2.5. For integers $n \ge 2$, consider the definite integral $S_n = \int_0^{\pi/4} \sec^n(x)dx$.

(a) Show that $S_n = \frac{(\sqrt{2}\,)^{n-2}}{n-1} + \frac{n-2}{n-1}S_{n-2}$.
(b) Verify that $S_3 = \frac{\sqrt{2}}{2} + \frac{1}{2}\ln\left(\sqrt{2}+1\right)$.
(c) Verify that $S_6 = 28/15$.

2.6. Show that $\int_0^1 x^{n-1}\ln\left(\frac{1}{1-x}\right)dx = \frac{1}{n}\sum_{k=1}^{n}\frac{1}{k}$.

2.7. In our proof that π is irrational, we used the fact that for fixed $x > 0$, we have $x^n/n! \to 0$ as $n \to \infty$. Prove this.

2.8. Let $a_1 = 1$. Verify (as we claimed in Sect. 2.3) that

$$a_{n+1} = \sqrt{1+\sqrt{1+a_n}} \quad (n = 1, 2, 3, \ldots)$$

defines an increasing sequence which is bounded above.

2.9. Let $a_1 = 1$ and for $n = 1, 2, 3, \ldots$, set $a_{n+1} = \frac{1}{2}\left(a_n + \frac{2}{a_n}\right)$. Show that $\{a_n\}$ converges and find its limit. (This is the sequence got by applying Newton's Method[13] to $f(x) = x^2 - 2$, but it was known to the Babylonians ~ 1700 BCE.)

2.10 ([4]). Here's another proof that e is irrational, using the (proof of the) Alternating Series Test. Seeking a contradiction, suppose that e is rational. Then so must be $1/\mathrm{e}$, that is, $\frac{1}{\mathrm{e}} = \frac{k}{m}$ for positive integers k and m.

(a) Show that $x = -1$ in the Maclaurin Series for e^x gives

$$\frac{k}{m} - \sum_{n=0}^{m} \frac{(-1)^n}{n!} = \sum_{n=m+1}^{\infty} \frac{(-1)^n}{n!}.$$

(b) Multiply through by $(-1)^{m+1}m!$ to get something that looks like "$p = q$."
(c) Think about how this $p = q$ cannot be true—the desired contradiction.

References

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[13] Physicist/mathematician and co-discoverer of calculus Isaac Newton (England, 1642–1727) was born the same year Galileo died. Galileo was born the same year Michelangelo died ...

The Probability Integral and Gamma Function 3

The ubiquitous *Probability Integral*

$$\int_0^\infty e^{-x^2} dx = \frac{\sqrt{\pi}}{2} \tag{3.1}$$

was evaluated by Laplace[1] though Euler had probably done it earlier in a slightly different form [3, 10].

After verifying (3.1), and indicating a few other approaches in Exercises 3.3–3.5, we shall see a connection between (3.1), factorials $n!$, and a special function called the *Gamma Function*.

By a change of variables, and/or noticing that the integrand is an even function, (3.1) is equivalent to any of

$$\int_0^\infty e^{-x^2/2} dx = \sqrt{\frac{\pi}{2}}, \qquad \int_{-\infty}^\infty e^{-\pi x^2} dx = 1,$$
$$\int_{-\infty}^\infty e^{-x^2} dx = \sqrt{\pi}, \qquad \int_{-\infty}^\infty e^{-x^2/2} dx = \sqrt{2\pi}.$$

With $\phi(x) = \frac{1}{\sqrt{2\pi}} e^{-x^2/2}$, the last equality above gives $\int_{-\infty}^\infty \phi(x) dx = 1$. And since $e^{-x^2/2} > 0$ for all x, ϕ serves as a probability distribution function. This is the

[1] Pierre-Simon, marquis de Laplace (France, 1749–1827) first evaluated (3.1) in 1774, the same year that Louis XVI ascended to the throne. He (Louis, that is) was executed 19 years later, during the French Revolution.

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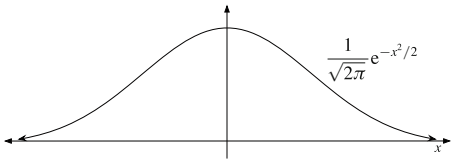


Fig. 3.1 The normal distribution $\leftrightarrow$ the classic bell curve

standardized Gaussian[2] *distribution function*.

It's called *standardized* because it has mean $\mu = 0$ and variance $\sigma^2 = 1$. See Exercise 3.1; the $1/2$ in ϕ's exponent is what makes $\sigma = 1$. A random variable which has a Gaussian distribution function (which many do!) is said to be *normally* distributed. The graph $y = \phi(x)$ has the shape of the classic bell curve (Fig. 3.1).

3.1 Evaluating the Probability Integral

Consider, as in [4], the volume obtained by revolving the curve $z = e^{-y^2}$ for $y \geq 0$, i.e., $y = \sqrt{-\ln(z)}$ for $0 < z \leq 1$, about the z-axis. Using the method of circular disks, this volume is (see Fig. 3.2)

$$\begin{aligned} V &= \pi \int_{0^+}^{1} y^2\, dz = \pi \int_{0^+}^{1} (-\ln(z))\, dz = -\pi \int_{0^+}^{1} (\ln(z))\, dz \\ &= -\pi \Big[z\ln(z) - z \Big]_{0^+}^{1} = -\pi(-1) \quad \text{(using L'Hospital's Rule)} \\ &= \pi\,. \end{aligned}$$

On the other hand, have a look at Fig. 3.3. This suggests that the volume is also

$$V = \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} e^{-(x^2+y^2)}\, dxdy = \int_{-\infty}^{\infty} e^{-x^2} dx \int_{-\infty}^{\infty} e^{-y^2}\, dy = \left(2\int_{0}^{\infty} e^{-x^2} dx \right)^2 .$$

Therefore we must have $\pi = \left(2\int_0^\infty e^{-x^2} dx \right)^2$, so that $\int_0^\infty e^{-x^2} dx = \frac{\sqrt{\pi}}{2}$. ✓

[2] Carl Friedrich Gauss (Germany, 1777–1855) gave credit for (3.1) to Laplace. Gauss was arguably the world's greatest mathematician. He had all the others looking over their shoulders, even long after he died.

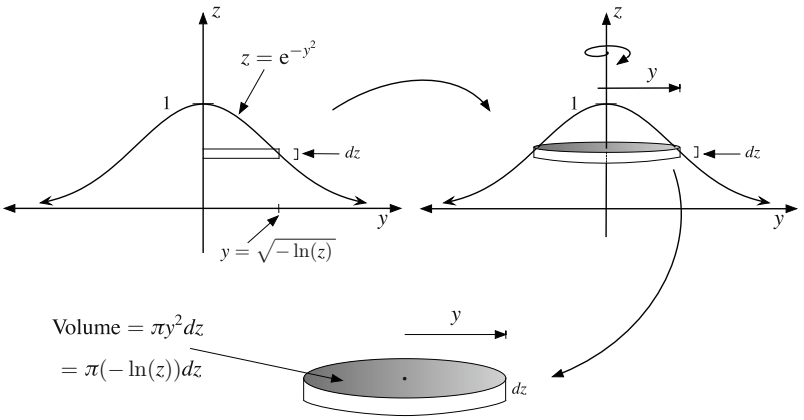
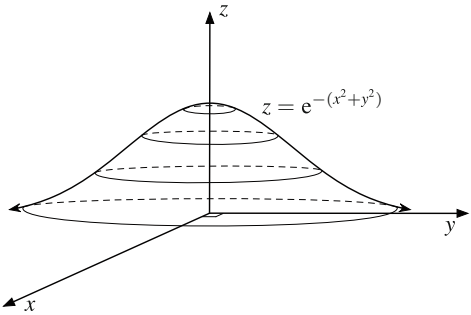


Fig. 3.2 Horizontal disks; each has volume $\pi y^2 dz$

Fig. 3.3 Revolving $z = e^{-y^2}$ about the z-axis



▶ **Remark** The volume can also be obtained by revolving the same curve $z = e^{-y^2}$ about the z-axis but using instead the method of cylindrical shells [8]; see Exercise 3.3. In [2] this volume is obtained by the method of cross sections!

3.2 Interpolating the Factorial

Consider, as Euler did, the integrals

$$\Delta(n) = \int_0^\infty t^n e^{-t} dt \quad (\text{for } n = 0, 1, 2, 3, \ldots).$$

In Theorem 3.1, we show that these integrals indeed exist. Believing this for now, let's go ahead and integrate $\Delta(n)$ by parts, for $n \geq 1$, and use L'Hospital's Rule:

$$\int_0^\infty \underbrace{t^n}_{\mathrm{D}}\,\underbrace{\mathrm{e}^{-t}}_{\mathrm{I}}\,dt \;=\; -t^n\mathrm{e}^{-t}\Big|_0^\infty \;+\; n\int_0^\infty t^{n-1}\mathrm{e}^{-t}dt \;=\; n\int_0^\infty t^{n-1}\mathrm{e}^{-t}dt\,.$$

This yields the reduction formula

$$\Delta(n) \;=\; n\cdot\Delta(n-1).$$

Continuing on down, we come to

$$\Delta(n) \;=\; n\Delta(n-1) \;=\; n(n-1)(n-2)\cdots(3)(2)(1)\Delta(0)\,.$$

Then, since

$$\Delta(0) \;=\; \int_0^\infty \mathrm{e}^{-t}dt \;=\; -\mathrm{e}^{-t}\Big|_0^\infty \;=\; 1,$$

we get

$$\Delta(n) \;=\; n(n-1)(n-2)\cdots(2)(1)(1) \;=\; n! \qquad (n=0,1,2,3,\ldots).$$

This is of course the *factorial function* which we met in Chap. 2: it's the number of ways of arranging n objects. (And same as Λ_n from Sect. 2.2, with $t=-\ln(x)$.)

Now here's something. In the definition of $\Delta(n)$ above, n did not need to be *just* a nonnegative integer. Indeed, we can extend the domain to include *all* $x>-1$:

Theorem 3.1
Let $x>-1$. Then the integral

$$\Delta(x) \;=\; \int_0^\infty t^x e^{-t}dt \quad \textit{exists}\,.$$

Proof. The only trouble that the integrand may present is near $t=0$ and "near" $t=\infty$, i.e., t very large. To address these separately, we write

$$\Delta(x) \;=\; \int_0^1 t^x\mathrm{e}^{-t}dt \;+\; \int_1^\infty t^x\mathrm{e}^{-t}dt \;:=\; I_1 \;+\; I_2\,.$$

I_1: For $t \in [0, 1]$, $\mathrm{e}^{-t} \leq 1$, and so

$$I_1 = \int_0^1 t^x \mathrm{e}^{-t} dt \leq \int_0^1 t^x dt = \frac{1}{x+1} t^{x+1} \Big|_{t=0}^1 < \infty, \quad \text{since } x > -1.$$

I_2: Let $\alpha > 1$. For *any* real number x, we have (L'Hospital strikes again)

$$t^{x+\alpha} \mathrm{e}^{-t} \to 0 \quad \text{as} \quad t \to \infty.$$

So for t large enough ($t \geq \beta$, say), we shall have $t^x \mathrm{e}^{-t} \leq \frac{1}{t^\alpha}$. Then

$$\begin{aligned} I_2 &= \int_1^\beta t^x \mathrm{e}^{-t} dt + \int_\beta^\infty t^x \mathrm{e}^{-t} dt \\ &\leq \int_1^\beta t^x \mathrm{e}^{-t} dt + \int_\beta^\infty \frac{1}{t^\alpha} dt \\ &= \int_1^\beta t^x \mathrm{e}^{-t} dt + \frac{1}{-\alpha+1} \frac{1}{t^{\alpha-1}} \Big|_{t=\beta}^\infty < \infty, \quad \text{since } \alpha > 1. \end{aligned}$$

Together, if $x > -1$, then $\Delta(x) = I_1 + I_2 < \infty$, and the theorem is proved. □

Theorem 3.1 clears our conscience, so that we may talk freely about $\Delta(n)$. But much more significant is that the theorem demonstrates $\Delta(x)$ *as an interpolant of the factorial function*. So we may reasonably write

$$x! = \Delta(x) \quad (\text{for } \textit{any } x > -1).$$

If we are happy with this particular way of interpolating $n!$ (a little more on this in Sect. 3.3), then the reduction formula above continues to hold for $x > 0$:

$$\Delta(x) = x \cdot \Delta(x-1).$$

Taking $x = 1/2$, for example,

$$\Delta(1/2) = \frac{1}{2}\Delta(-1/2) = \frac{1}{2} \int_0^\infty t^{-1/2} \mathrm{e}^{-t} dt \quad \text{and letting } t = u^2:$$

$$\Delta(1/2) \;=\; \int_0^\infty e^{-u^2}\,du \;=\; \frac{\sqrt{\pi}}{2}, \qquad \text{using (3.1).}$$

Therefore, as strange as it may seem, we may write

$$\frac{1}{2}! \;=\; \frac{\sqrt{\pi}}{2}.$$

3.3 Why This One ?

There are infinitely many functions which interpolate the factorial function. Just about any squiggly line that passes through $(0, 0!)$, $(1, 1!)$, $(2, 2!)$, $(3, 3!)$, etc. might have as strong a claim as Δ has. But it takes some real and/or complex analysis, which we do not go into here, to see why Δ is the right one to use.

Still, here's something cool, which sort of *suggests* why Δ is the right one. For $n = 1, 2, 3, \cdots$, consider the integrals defined by

$$L_n \;=\; \int_0^1 \left(1 - x^2\right)^n dx\,.$$

Integrating by parts, beginning with $L_n = \int_0^1 \underbrace{\left(1 - x^2\right)^n}_{\mathrm{D}} \underbrace{1}_{\mathrm{I}}\, dx$ gives

$$L_n = (1 - x^2)^n\, x\Big|_0^1 - \int_0^1 n\left(1 - x^2\right)^{n-1}(-2x)x\,dx$$

$$= 0 + 2n \int_0^1 \left(1 - x^2\right)^{n-1} x^2\,dx.$$

Continuing in this way (the 0s just keep on coming), we get

$$L_n \;=\; 2n \int_0^1 \underbrace{\left(1 - x^2\right)^{n-1}}_{\mathrm{D}} \underbrace{x^2}_{\mathrm{I}}\, dx$$

$$
\begin{aligned}
&= \frac{2\cdot n\cdot 2\cdot (n-1)}{3}\int_0^1 \underbrace{\left(1-x^2\right)^{n-2}}_{\mathrm{D}}\underbrace{x^4}_{\mathrm{I}}\,dx \\
&= \frac{2\cdot n\cdot 2\cdot (n-1)\cdot 2\cdot (n-2)}{3\cdot 5}\int_0^1 \underbrace{\left(1-x^2\right)^{n-3}}_{\mathrm{D}}\underbrace{x^6}_{\mathrm{I}}\,dx \\
&\ \ \vdots \\
&\ \ \downarrow \\
&= \frac{2\cdot n\cdot 2\cdot (n-1)\cdot 2\cdot (n-2)\cdot 2\cdot (n-3)\cdots 2\cdot (n-(n-1))}{3\cdot 5\cdot 7\cdots (2n-1)}\int_0^1 x^{2n}\left(1-x^2\right)^{n-n}dx \\
&= \frac{2n(2n-2)(2n-4)\cdots (6)(4)(2)}{3\cdot 5\cdot 7\cdots (2n+1)}\,. \qquad (3.2)
\end{aligned}
$$

We shall appeal to (3.2) in the next chapter. But right now, observe that

$$
\begin{aligned}
\frac{2n(2n-2)(2n-4)\cdots (6)(4)(2)}{3\cdot 5\cdot 7\cdots (2n+1)} &= \frac{2^n\cdot n!}{3\cdot 5\cdot 7\cdots (2n+1)} \\
&= \frac{2^n\cdot n!}{(2n+1)!}\,2n(2n-2)(2n-4)\cdots (4)(2) \\
&= \frac{2^{2n}(n!)^2}{(2n+1)!}\,,
\end{aligned}
$$

and so we also have

$$
L_n = \frac{2^{2n}(n!)^2}{(2n+1)!}\,. \qquad (3.3)
$$

Okay, this next part is illegal, since n is a natural number. But still (shhh...), *if we did set* $n = 1/2$ in the definition of L_n, we'd get

$$
\int_0^1 \sqrt{1-x^2}\,dx\,,
$$

which is one quarter of the area inside the unit circle, which is $\pi/4$. Then with $n = 1/2$, (3.3) gives

$$\frac{\pi}{4} \;=\; L_{1/2} \;=\; \frac{2^1((1/2)!)^2}{(1+1)!} \;=\; \left(\frac{1}{2}!\right)^2 .$$

That is,

$$\frac{\sqrt{\pi}}{2} \;=\; \frac{1}{2}! \qquad \text{as before—not bad !}$$

(This last ! is an exclamation point.)

3.4 The Gamma Function

Mathematicians typically use a shifted version of Δ to interpolate the factorial. This is the *Gamma Function*, as first defined by Legendre:[3] $\Gamma(x) = \Delta(x-1)$. That is,

$$\Gamma(x) \;=\; \int_0^\infty t^{x-1}\mathrm{e}^{-t}\,dt \qquad (\text{for } x > 0)\,.$$

The shift in the integrand from x to $x-1$ is a notational convenience, which is completely natural in many contexts. Still, "it continues to plague many people" [1].

Here, the $\Delta(x) = x \cdot \Delta(x-1)$ above translates to

$$\Gamma(x+1) \;=\; x \cdot \Gamma(x) \qquad (x > 0),$$

and the $\Delta(x) = x!$ translates to

$$\Gamma(x+1) \;=\; x! \qquad (x \geq 0).$$

In particular,

$$\Gamma(n) \;=\; (n-1)! \qquad (n = 1, 2, 3, \ldots).$$

Also, $\Gamma\left(\frac{3}{2}\right) = \Delta\left(\frac{1}{2}\right) = \frac{\sqrt{\pi}}{2}$ and $\Gamma\left(\frac{3}{2}\right) = \frac{1}{2}\Gamma(\frac{1}{2})$ together give

$$\Gamma\left(\tfrac{1}{2}\right) \;=\; \sqrt{\pi}\,.$$

Using Γ, one can extend (e.g., [12]) certain formulas for the natural numbers, like the binomial coefficient $\binom{n}{k} = \frac{n!}{(n-k)!\,k!}$, to real numbers, using instead

[3] Adrien-Marie Legendre (France, 1752–1833) adopted this definition, and the symbol Γ for it, around 1811. At about the same time, Napoleon was gearing up for a certain war.

$$\binom{x}{y} = \frac{\Delta(x)}{\Delta(x-y)\Delta(y)} = \frac{\Gamma(x+1)}{\Gamma(x-y+1)\Gamma(y+1)} \quad (x, y > -1;\ x - y > -1).$$

3.5 Epilogue

It is a consequence of several classic theorems due to Liouville[4] that

$$\int \mathrm{e}^{-x^2} dx$$

cannot be expressed as a sum, product, and/or composition of finitely many polynomial, rational, trigonometric, and/or exponential functions, and/or their inverses.

Still [7, 13], the great mathematician/physicist Lord Kelvin (Belfast, in what is now Northern Ireland, 1824–1907) once asserted to his students that

> A mathematician is one to whom $\int_{-\infty}^{\infty} \mathrm{e}^{-x^2} dx$ is as obvious as $2 \times 2 = 4$ is to you.

On the other hand [6], the great pure mathematician J.E. Littlewood (England, 1885–1977) wrote

> Many things are not accessible to intuition at all, the value of $\int_0^{\infty} \mathrm{e}^{-x^2} dx$, for instance.

You decide ...

Exercises

3.1.

(a) Verify that the mean $\mu = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} x \mathrm{e}^{-x^2/2} dx = 0$.

(b) Verify that the variance

$$\sigma^2 = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} (x-\mu)^2 \mathrm{e}^{-x^2/2} dx = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} x^2 \mathrm{e}^{-x^2/2} dx - \mu^2 = 1.$$

[4] Joseph Liouville (France, 1809–1882) was the first to prove that transcendental numbers exist, by actually finding one. Thirty years later, Georg Cantor (Germany, 1845–1918) showed that transcendental numbers are almost everywhere!

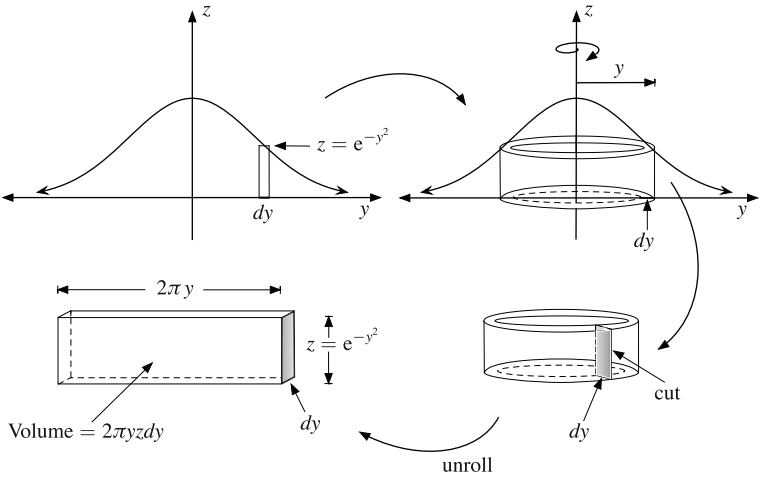


Fig. 3.4 (For Exercise 3.3) Cylindrical shells; each has volume $2\pi yz dy$

3.2. Let $f(x) = \frac{1}{b-a}$ for all $x \in [a, b]$ and $f(x) = 0$ otherwise.

(a) Agree that this is a probability distribution function (the *uniform* distribution).

(b) Find the mean $\mu = \int_{-\infty}^{\infty} x f(x) dx$.

(c) Find the variance $\sigma^2 = \int_{-\infty}^{\infty} (x - \mu)^2 f(x) dx$.

3.3 ([8]). Use the method of cylindrical disks to verify (again!) that the volume obtained by revolving the curve $z = e^{-y^2}$, for $y \geq 0$, about the z-axis, is π cubic units. See Fig. 3.4.

3.4. Evaluate

$$\int_0^{\infty} \int_0^{\infty} e^{-(x^2+y^2)} dx\, dy$$

using polar coordinates (as you may have done if you've taken Calculus 3). This method was discovered by Poisson[5] [5]. Conclude, again, that $\int_0^{\infty} e^{-x^2} dx = \frac{\sqrt{\pi}}{2}$.

[5] Siméon Denis Poisson (France, 1781–1840). If you know much mathematics, statistics, or physics, then you have heard of Poisson. Apparently his family wanted him to be a doctor.

3.5 (E.g., [7]). Fill in the details of Laplace's proof[6] of (3.1), as follows:

(a) In

$$\left(\int_0^\infty \mathrm{e}^{-x^2}dx\right)^2 = \int_0^\infty\int_0^\infty \mathrm{e}^{-(x^2+y^2)}dxdy\,,$$

set $y = xu$, to get $\displaystyle\int_0^\infty\int_0^\infty \mathrm{e}^{-(x^2+y^2)}dxdy = \int_0^\infty\int_0^\infty \mathrm{e}^{-x^2(1+u^2)}x\,dxdu\,.$

(b) Now set $t = x^2$, to get $\displaystyle\int_0^\infty\int_0^\infty \mathrm{e}^{-x^2(1+u^2)}x\,dx\,du = \cdots\cdots = \frac{\pi}{4}\,.$

(c) Conclude, again, that $\displaystyle\int_0^\infty \mathrm{e}^{-x^2}dx = \frac{\sqrt{\pi}}{2}.$

3.6. Evaluate $\Gamma(5/2)$, $\Gamma(7/2)$. Show that $\Gamma(x) = \dfrac{\Gamma(x+n)}{x(x+1)\cdots(x+n-1)}$ $(x > 0)$.

3.7. In [11], the $L_n = \frac{2}{3}\cdot\frac{4}{5}\cdot\cdots\cdot\frac{2n}{2n+1}$ that we obtained in line (3.2) is obtained quite differently, more or less as follows: Fill in the details.

(a) Observe that $-(1-x)$ is an antiderivative of 1. Use this to integrate by parts

$$2^N = 1 + N\int_0^1 (1+x)^{N-1}dx = 1 + N\int_0^1 \underbrace{(1+x)^{N-1}}_{\mathrm{D}}\underbrace{1}_{\mathrm{I}}\,dx\,,$$

to get

$$2^N = 1 + N + N(N-1)\int_0^1 (1+x)^{N-2}(1-x)dx\,.$$

(b) Then with $2^N = 1 + N + N(N-1)\displaystyle\int_0^1 \underbrace{(1+x)^{N-2}}_{\mathrm{D}}\underbrace{(1-x)}_{\mathrm{I}}\,dx$, integrate by parts $n-1$ more times, to get

[6] This was not the first proof Laplace found—that one was more complicated [5].

$$2^N = 1 + N + \frac{N(N-1)}{2!} + \frac{N(N-1)(N-2)}{3!} + \cdots$$
$$+ \frac{N(N-1)(N-2)\cdots(N-(n-1))}{n!}$$
$$+ \frac{N(N-1)(N-2)\cdots(N-n)}{n!}\int_0^1 (1+x)^{N-(n+1)}(1-x)^n dx.$$

(c) Take $N = 2n+1$ to get

$$2^{2n+1} = \sum_{k=0}^{n} \binom{2n+1}{k} + \frac{(2n+1)!}{(n!)^2}\int_0^1 (1-x^2)^n dx\,.$$

(d) Use the Binomial Formula (Sect. 3.1) to expand $2^{2n+1} = (1+1)^{2n+1}$.

3.8. We saw in line (3.2) that $L_n = \frac{2}{3}\cdot\frac{4}{5}\cdot\cdots\cdot\frac{2n}{2n+1}$. Evaluate (for natural numbers n and m)

$$\int_0^1 (1-x)^n (1+x)^m dx\,.$$

[Of course your answer should match L_n, when $m = n$.]

3.9 ([9]). For natural numbers n, consider the integrals $\int_0^1 (1-x)^n x^n\,dx\,.$

(a) Integrate by parts repeatedly, beginning with $\int_0^1 \underbrace{(1-x)^n}_{\mathrm{D}}\,\underbrace{x^n}_{\mathrm{I}}\,dx$, to show that

$$\int_0^1 (1-x)^n x^n\,dx = \frac{n!}{(n+1)(n+2)\cdots(2n)(2n+1)} = \frac{(n!)^2}{(2n+1)!}\,.$$

(b) Set $n = 1/2$ on both sides to (sort of) conclude, again, that

$$\frac{\pi}{8} = \frac{(\frac{1}{2}!)^2}{2!} \quad\Rightarrow\quad \frac{1}{2}! = \frac{\sqrt{\pi}}{2}\,.$$

3.10.

(a) Let m and n be natural numbers. Evaluate $\int_0^1 (1-x)^n x^m \, dx$.

[Of course your answer should match Exercise 3.9 (a), when $n = m$.]

(b) Verify that for $n \geq m$, $\binom{n}{m} \int_0^1 (1-x)^{n-m} x^m dx \quad \frac{1}{n+1}$.

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Wallis's Product

4

In Sect. 3.3, line (3.2), we evaluated the integrals (for $n = 1, 2, 3, \cdots$)

$$L_n = \int_0^1 (1-t^2)^n dt = \frac{2}{3}\cdot\frac{4}{5}\cdot\frac{6}{7}\cdot\cdots\cdot\frac{2n}{2n+1} = \prod_{k=1}^{n}\frac{2k}{2k+1}.$$

The substitution $t = \sin(x)$, so that $dt = \cos(x)dx$, gives

$$L_n = \int_0^{\pi/2} (1-\sin^2(x))^n \cos(x)\,dx = \int_0^{\pi/2} \cos^{2n+1}(x)\,dx\,.$$

So conspicuous by their absence perhaps are the integrals

$$\int_0^{\pi/2} \cos^{2n}(x)\,dx\,.$$

We shall see that these integrals as well (i.e., with even powers too) will go a very long way toward obtaining the wonderful

Wallis's Product[1]

$$\frac{\pi}{2} = \frac{2\cdot 2}{1\cdot 3}\cdot\frac{4\cdot 4}{3\cdot 5}\cdot\frac{6\cdot 6}{5\cdot 7}\cdot\frac{8\cdot 8}{7\cdot 9}\cdot\cdots\cdot = \prod_{n=1}^{\infty}\frac{2n}{2n-1}\frac{2n}{2n+1}. \tag{4.1}$$

[1] John Wallis (England, 1616–1703) obtained his product in 1655, 11 years before the Great Fire of London. Over in North America, the number of original Thirteen Colonies was ten.

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Wallis's own proof of (4.1) is ingenious but necessarily complicated—he had hardly any calculus at his disposal. Fair enough: It hadn't really been discovered yet [3]. We shall prove (4.1) using lots of calculus (c.f. [2]).

4.1 Prep Work

For $N = 0, 1, 2, \ldots$, set

$$I_N \;=\; \int_0^{\pi/2} \cos^N(x)\,dx\,.$$

In particular, $I_{2n} = \int_0^{\pi/2} \cos^{2n}(x)dx$ and $I_{2n+1} = \int_0^{\pi/2} \cos^{2n+1}(x)dx$ $(= L_n)$.

Integrating by parts, via

$$I_N \;=\; \int_0^{\pi/2} \underbrace{\cos^{N-1}(x)}_{\mathrm{D}}\,\underbrace{\cos(x)}_{\mathrm{I}}\,dx\,,$$

we get

$$\begin{aligned} I_N \;&=\; \cos^{N-1}(x)\sin(x)\Big|_0^{\pi/2} \;-\; \int_0^{\pi/2} (N-1)\cos^{N-2}(x)(-\sin(x))\sin(x)\,dx \\ &=\; 0 \;+\; (N-1)\int_0^{\pi/2} \cos^{N-2}(x)\sin^2(x)\,dx \\ &=\; (N-1)\int_0^{\pi} \cos^{N-2}(x)\left[1-\cos^2(x)\right]dx \\ &=\; (N-1)\int_0^{\pi} \cos^{N-2}(x)\,dx \;-\; (N-1)\int_0^{\pi} \cos^{N}(x)\,dx. \end{aligned}$$

This reads

$$I_N \;=\; (N-1)I_{N-2} \;-\; (N-1)I_N\,,$$

and solving for I_N we obtain the reduction formula

$$I_N \;=\; \frac{N-1}{N}\, I_{N-2}\,.$$

The subscripts go down in steps of two. Using the formula repeatedly, with $N = 2n$ (even), we get

$$\begin{aligned} I_{2n} \;&=\; \frac{2n-1}{2n}\cdot I_{2n-2} \\ &=\; \frac{2n-1}{2n}\cdot\frac{2n-3}{2n-2}\cdot I_{2n-4} \\ &\;\;\vdots \\ &\;\;\downarrow \\ &=\; \frac{2n-1}{2n}\cdot\frac{2n-3}{2n-2}\cdot\frac{2n-5}{2n-4}\cdot\cdots\cdot\frac{5}{6}\cdot\frac{3}{4}\cdot\frac{1}{2}\cdot I_0\,. \end{aligned}$$

Writing this product with its factors in the opposite order, and using the obvious fact that

$$I_0 \;=\; \int_0^{\pi/2} 1\,dt \;=\; \pi/2\,,$$

we get

$$I_{2n} \;=\; \frac{\pi}{2}\cdot\frac{1}{2}\cdot\frac{3}{4}\cdot\frac{5}{6}\cdots\frac{2n-3}{2n-2}\cdot\frac{2n-1}{2n} \;=\; \frac{\pi}{2}\prod_{k=1}^{n}\frac{2k-1}{2k}\,.$$

As mentioned, we have already obtained $I_{2n+1} = L_n$. Alternatively however, we could proceed just as above, again using $I_N = \frac{N-1}{N}\, I_{N-2}$, but beginning with $n = 2n+1$, and then going down in steps of two—ultimately to

$$I_1 \;=\; \int_0^{\pi/2} \cos(x)dx \;=\; \sin(x)\Big|_0^{\pi/2} \;=\; 1\,.$$

In summary then, we have obtained the following:

Lemma 4.1

$I_0 = \pi/2,\ I_1 = 1,$ *and for* $n = 1, 2, 3, \ldots,$

(continued)

Lemma 4.1 (continued)

$$I_{2n} = \frac{\pi}{2}\cdot\frac{1}{2}\cdot\frac{3}{4}\cdot\frac{5}{6}\cdots\frac{2n-1}{2n} = \frac{\pi}{2}\prod_{k=1}^{n}\frac{2k-1}{2k}$$

and

$$I_{2n+1} = 1\cdot\frac{2}{3}\cdot\frac{4}{5}\cdots\frac{2n-2}{2n-1}\cdot\frac{2n}{2n+1} = \prod_{k=1}^{n}\frac{2k}{2k+1}.$$

4.2 Proof of Wallis's Product

Lemma 4.1 provides most of the heavy lifting, in proving Wallis's Product (4.1). Looking at the definition of I_N, since $0 \le \cos(t) \le 1$ on $[0, \pi/2]$, the I_Ns decrease as N increases. That is,

$$I_{2n+2} \le I_{2n+1} \le I_{2n}.$$

By Lemma 4.1, this reads

$$\frac{\pi}{2}\prod_{k=1}^{n+1}\frac{2k-1}{2k} \le \prod_{k=1}^{n}\frac{2k}{2k+1} \le \frac{\pi}{2}\prod_{k=1}^{n}\frac{2k-1}{2k}.$$

Dividing through by $\prod_{k=1}^{n}\frac{2k-1}{2k}$, this becomes

$$\frac{\pi}{2}\frac{2n+1}{2n+2} \le \prod_{k=1}^{n}\frac{2k}{2k-1}\frac{2k}{2k+1} \le \frac{\pi}{2}.$$

And finally, letting $n \to \infty$, we obtain (4.1). ✓

Note. This is not the only path to Wallis's Product; we consider others in Exercises 4.2 and 4.3. For a neat geometric interpretation of Wallis's Product, see [4].

4.3 An Example

Suppose that a fair coin is tossed (heads or tails) $2n$ times. We estimate, for n very large, the probability that we get exactly n tails.

The number of different ways of obtaining n tails in $2n$ tosses is $\binom{2n}{n}$, the so-called *central binomial coefficient*. The total number of possible outcomes in $2n$ tosses is 2^{2n}. Therefore the probability that we get n tails in $2n$ tosses is

$$\binom{2n}{n}\frac{1}{2^{2n}} = \frac{(2n)!}{(n!)^2}\frac{1}{2^{2n}}.$$

In line (3.3) of Sect. 3.3, we obtained

$$L_n = \frac{2^{2n}(n!)^2}{(2n+1)!} = \frac{(n!)^2\,2^{2n}}{(2n)!}\frac{1}{(2n+1)}.$$

But, as we've seen, $L_n = I_{2n+1}$, and so

$$\binom{2n}{n}\frac{1}{2^{2n}} = \frac{1}{(2n+1)I_{2n+1}}.$$

Now by Wallis's Product (4.1),

$$\lim_{n\to\infty}\left((2n+1)\,I_{2n+1}^2\right) = \frac{\pi}{2}.$$

Therefore, for n large, the probability of getting n tails in $2n$ tosses is

$$\frac{1}{(2n+1)I_{2n+1}} = \frac{1}{\sqrt{2n+1}}\frac{1}{\sqrt{2n+1}\,I_{2n+1}} \approx \frac{1}{\sqrt{2n+1}}\frac{1}{\sqrt{\frac{\pi}{2}}} \approx \frac{1}{\sqrt{\pi n}}.$$

▶ **Remark** The computations above contain the fact (verify!) that

$$\frac{2^{2n}}{\sqrt{2n+1}}\cdot\frac{(n!)^2}{(2n)!} \to \sqrt{\frac{\pi}{2}} \quad \text{as } n\to\infty. \tag{4.2}$$

We shall appeal to (4.2) in Chap. 13.

Exercises

4.1. Prove by induction (for $n = 2,\ 3,\ 4,\ \ldots$)

$$\frac{1}{\sqrt{4n+1}} < \frac{1}{2}\cdot\frac{3}{4}\cdot\frac{5}{6}\cdot\cdots\cdot\frac{2n-3}{2n-2}\cdot\frac{2n-1}{2n} < \frac{1}{\sqrt{3n+1}}.$$

4.2 ([1]). Here's a slightly different way to obtain Wallis's Product (4.1). Again

$$I_N = \int_0^{\pi/2} \cos^N(x)\,dx\,, \quad I_{2n} = \frac{\pi}{2}\prod_{k=1}^{n}\frac{2k-1}{2k}\,, \quad I_{2n+1} = \prod_{k=1}^{n}\frac{2k}{2k+1}\,.$$

(a) Use $0 \le \cos(t) \le 1$ on $[0, \pi/2]$, to show that

$$1 \;\le\; \frac{I_{2n}}{I_{2n+1}} \;\le\; \frac{I_{2n-1}}{I_{2n+1}} \;=\; 1+\frac{1}{2n}\,.$$

(b) Substitute for I_{2n} and I_{2n+1}; then clean up a bit to get

$$1 \le \frac{(2n+1)(2n-1)}{2n\cdot 2n}\cdot\frac{(2n-1)(2n-3)}{(2n-2)(2n-2)}\cdots\frac{5\cdot 3}{4\cdot 4}\cdot\frac{3\cdot 1}{2\cdot 2}\cdot\frac{\pi}{2} \le 1+\frac{1}{2n}\,.$$

(Sometimes this, or similar, is called *Wallis's Inequality*.)

(c) Let $n \to \infty$ to obtain (4.1).

4.3. Prove Wallis's Product (4.1) as we did in the text, or as in Exercise 4.2, but beginning instead with the integrals

$$I_N \;=\; \int_0^{\pi/2} \sin^N(x)\,dx\,.$$

4.4.

(a) Show that $I_{N-1}I_N = \frac{\pi}{2N}$.

(b) Show that $\frac{\pi}{2(N+1)} < I_N^2 < \frac{\pi}{2N}$.

(c) Improve the right-hand side of (b) by showing that $I_N^2 \le I_{N-1}I_{N+1}$.

4.5 ([2]). Show that for $N \in \mathbb{N}$, $\displaystyle\prod_{n=1}^{N}\frac{(2n)^2}{(2n)^2-1} < \frac{\pi}{2} < \prod_{n=1}^{N}\frac{(2n)^2}{(2n)^2-1}\,e^{\frac{1}{4n+2}}$.

Also, show that the right side is closer to $\pi/2$ than is the left side.

4.6 ([5]). Show that

$$\pi \;=\; \frac{\displaystyle\prod_{n=1}^{\infty}\left(1+\frac{1}{4n^2-1}\right)}{\displaystyle\sum_{n=1}^{\infty}\frac{1}{4n^2-1}}\,.$$

4.7. In Sect. 4.3 we saw that for n large, $\binom{2n}{n}\frac{1}{2^{2n}} \approx \frac{1}{\sqrt{\pi n}}$. Show, a little more precisely, that

$$\frac{1}{\sqrt{\pi(n+1/2)}} \leq \binom{2n}{n}\frac{1}{2^{2n}} \leq \frac{1}{\sqrt{\pi n}}.$$

4.8.

(a) To pass the time, Vladimir and Estragon each toss a coin n times. What is the probability, for n large, that they get the same number of tails?
(b) Show that

$$\binom{2n}{n} = \sum_{k=0}^{n}\binom{n}{k}^2.$$

[A hint, perhaps: Remember that $\binom{2n}{n}$ is called the *central* binomial coefficient.]

4.9. Suppose that we start at $x = 0$ on the number line; then a fair coin is tossed. If it lands heads, x becomes $x + 1$, and if it lands tails, x becomes $x - 1$. Then we toss the coin repeatedly, continuing with this process. After n tosses (odd), the probability of $x = 0$ is zero. If n is even, what is it? What is the expected value of $|x|$, as $n \to \infty$?

4.10. Show that

$$\sum_{n=0}^{\infty}\binom{2n}{n}x^n = \frac{1}{\sqrt{1-4x}} \qquad (\text{for } |x| < 1/4).$$

[$\frac{1}{\sqrt{1-4x}}$ is called the *generating function* for the central binomial coefficient $\binom{2n}{n}$.]

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Interlude: How Big Is a Ball ?

5

Recall from Sect. 3.4 that the **Gamma Function** is defined by

$$\Gamma(x) = \int_0^\infty t^{x-1} e^{-t}\, dt \qquad (\text{for } x > 0)\,.$$

We have seen that $\Gamma(1) = 1$ and, using integration by parts, that

$$\Gamma(x+1) = x \cdot \Gamma(x) \qquad (\text{for } x > 0),$$

or equivalently

$$\Gamma(x) = (x-1)\Gamma(x-1) \qquad (\text{for } x > 1).$$

Either way, for $n = 0, 1, 2, 3, \ldots$ we have

$$\Gamma(n+1) = n!\,.$$

We have also seen via the Probability Integral (Sect. 3.4) that $\Gamma\left(\frac{1}{2}\right) = \sqrt{\pi}$.

In this interlude we shall use the Gamma Function, and some related things, to investigate the size of a ball in $\mathbb{R}^n$, for varying values of n.

5.1 Stuff We'll Need

Recall from Sect. 4.1 that for

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$$I_N = \int_0^{\pi/2} \cos^N(x)\,dx \qquad (N = 0,\ 1,\ 2,\ 3,\ \ldots),$$

we had $I_0 = \frac{\pi}{2}$ and $I_1 = 1$, and Lemma 4.1 gave

$$I_{2n} = \frac{\pi}{2}\prod_{k=1}^{n}\frac{2k-1}{2k} \qquad \text{and} \qquad I_{2n+1} = \prod_{k=1}^{n}\frac{2k}{2k+1}\,. \tag{5.1}$$

We shall use a couple of additional facts about the I_Ns, as contained in the following:

Lemma 5.1
For $N = 1, 2, 3, \ldots$, we have **(a)** $I_{N-1}\cdot I_N = \frac{\pi}{2N}$ *and consequently* **(b)** $\frac{\pi/2}{N+1} \le (I_N)^2 \le \frac{\pi/2}{N}$.

Proof. Multiplying the formulas (5.1) together, there is a lot of cancellation:

$$I_{2n}\cdot I_{2n+1} = \frac{\pi}{2}\frac{1}{2n+1} \qquad \text{and} \qquad I_{2n-1}\cdot I_{2n} = \frac{\pi}{2}\frac{1}{2n}\,.$$

Either way, we get (a). Now for (b), since $0 \le \cos(t) \le 1$ on $[0, \pi/2]$, the I_Ns decrease as N increases. Therefore

$$I_{N+1}I_N \le (I_N)^2 \le I_{N-1}I_N\,.$$

Using (a) here, (b) follows. □

5.2 Okay, so How Big Is a Ball?

A very good question—but we need to agree on what we're talking about.[1] The 1-volume (i.e., length in $\mathbb{R}^1$) of an interval of radius (i.e., half-length) equal to r is $V_1(r) = 2r$. The 2-volume (i.e., area in $\mathbb{R}^2$) inside a circle of radius r is $V_2(r) = \pi r^2$.

[1] In 2015, several weeks before NASA's New Horizons spacecraft was to provide unprecedented close-up observations of Pluto, principal investigator Alan Stern said, "Pluto is about the size of the United States." Its largest moon, Charon, "is about the size of Texas" [7, 8].

We aim to find the rest of the constants a_n such that $V_n(r) = a_n r^n$, for $n = 3, 4, 5, \ldots$. You may already know, for example, that $a_3 = \frac{4}{3}\pi$, but this will come out in what follows. (Almost nobody knows, at least offhand, what a_4 is.) As we shall see, these numbers a_n have a couple of properties that may surprise.

The ball centered at the origin of

$$\mathbb{R}^n \;=\; \big\{(x_1,\, x_2,\, \cdots,\, x_n) : \, x_j \in \mathbb{R}\big\}$$

with radius r is given by $\sum\limits_{j=1}^{n} x_j^2 \le r^2$. So its n-volume is, by definition,

$$\begin{aligned} V_n(r) \;&=\; \underbrace{\int\int\cdots\int}_{\sum_{j=1}^n x_j^2 \le r^2} 1 \cdot dx_1 dx_2 \cdots dx_{n-1} dx_n . \\ &=\; \int_{-r}^{r} \Big(\underbrace{\int\int\cdots\int}_{\sum_{j=1}^{n-1} x_j^2 \le r^2 - x_n^2} 1 \cdot dx_1 dx_2 \cdots dx_{n-1} \Big) dx_n . \end{aligned}$$

The quantity inside the brackets $\big(\cdots\big)$ above is the volume of the ball in $\mathbb{R}^{n-1}$ with radius $\sqrt{r^2 - x_n^2}$. Therefore we have

$$V_n(r) \;=\; \int_{-r}^{r} V_{n-1}\left(\sqrt{r^2 - x_n^2}\right) dx_n \;=\; \int_{-r}^{r} a_{n-1}\left(\sqrt{r^2 - x_n^2}\right)^{n-1} dx_n \,.$$

This last integrand is an even function, and so

$$V_n(r) \;=\; 2\int_{0}^{r} a_{n-1}\left(\sqrt{r^2 - x_n^2}\right)^{n-1} dx_n \,.$$

Now the change of variable $x_n = r\sin(t)$, so that $dx_n = r\cos(t)dt$, gives

$$V_n(r) \;=\; 2a_{n-1} r^n \int_{0}^{\pi/2} \cos^n(t) dt \;=\; 2a_{n-1}\, r^n I_n \,.$$

To obtain the constants a_n, we take $r = 1$ and write simply $V_n = V_n(1) = a_n$. Then

$$V_n \;=\; 2 \cdot V_{n-1} \cdot I_n \,. \tag{5.2}$$

Starting with $V_1 = 2$, we can use (5.2) to calculate any V_n, i.e., the volume of the unit ball in any $\mathbb{R}^n$. For example,

$$V_2 \;=\; 2 \cdot V_1 \cdot I_2 \;=\; 2 \cdot 2 \cdot \frac{\pi}{4} \;=\; \pi\,,$$

$$V_3 \;=\; 2 \cdot V_2 \cdot I_3 \;=\; 2 \cdot \pi \cdot \frac{2}{3} \;=\; \frac{4\pi}{3}\,.$$

Etcetera:

$$V_4 = 2 \cdot \frac{4\pi}{3} \cdot \frac{3\pi}{16} = \frac{\pi^2}{2}\,, \qquad V_5 = 2 \cdot \frac{\pi^2}{2} \cdot \frac{8}{15} = \frac{8\pi^2}{15}\,, \qquad V_6 = 2 \cdot \frac{8}{15}\pi^2 \cdot \frac{5\pi}{32} = \frac{\pi^3}{6}\,, \ldots.$$

Now here's something [3]: the V_ns increase up until $\frac{V_n}{V_{n-1}} = 2I_n < 1$, which occurs when $n = 6$: $V_6 < V_5$. We saw in the proof of Lemma 5.1(b) that the I_ns are decreasing, and so from $n = 5$ onward, so are the V_ns. That is, the largest unit ball in any $\mathbb{R}^n$ is the one in $\mathbb{R}^5$. (It has 5-volume $\frac{8}{15}\pi^2$.) Moreover, $I_n \to 0$ as $n \to \infty$ (by Lemma 5.1(b)), so for n large, the V_ns get just as small as you please.

▶ **Remark** The surface area σ_n of the unit ball in $\mathbb{R}^n$ satisfies

$$\sigma_n \;=\; n V_n\,.$$

An analysis similar to the one above (e.g., [3, 6]) shows that the unit ball having largest surface area lies in $\mathbb{R}^7$. And again, for n large, the surface areas get as small as you like.

5.3 But How Big Is a Ball, in a Really Big Place?

Another good question. Here we obtain (c.f. [1,5]) a closed-form expression for V_n. That is, a formula for V_n that does not contain any V_k, for $k < n$.

Using (5.2), namely,

$$V_n \;=\; 2 \cdot V_{n-1} \cdot I_n\,,$$

two times, we have

$$V_n \;=\; 2 \cdot V_{n-1} \cdot I_n \;=\; 4 \cdot V_{n-2} \cdot I_{n-1} \cdot I_n\,.$$

By Lemma 5.1(a), $I_{n-1} \cdot I_n \;=\; \frac{\pi}{2n}$, and so

$$V_n \;=\; \frac{2\pi}{n} \cdot V_{n-2}\,. \tag{5.3}$$

Now, since the Gamma Function satisfies $\Gamma(x) = (x-1)\Gamma(x-1)$, we have

$$\frac{\pi^{\frac{n}{2}}}{\frac{n}{2}\,\Gamma(\frac{n}{2})} = \frac{\pi^{\frac{n}{2}}}{\frac{n}{2}(\frac{n}{2}-1)\,\Gamma(\frac{n}{2}-1)} = \frac{2\pi}{n}\cdot\frac{\pi^{\frac{n-2}{2}}}{(\frac{n-2}{2})\,\Gamma(\frac{n-2}{2})}.$$

Therefore Eq. (5.3) has solution

$$V_n = \frac{\pi^{\frac{n}{2}}}{\frac{n}{2}\cdot\Gamma(\frac{n}{2})}.$$

With n being a nonnegative integer, it is reasonable to want to write V_n in terms of factorials. For any nonnegative integer m, $\Gamma(m+1) = m\Gamma(m) = m!$, and so for n *even*, we get $V_n = \pi^{\frac{n}{2}}/\frac{n}{2}!$

For n odd we have to do more work. In the denominator of V_n, we use $\Gamma(m) = (m-1)\Gamma(m-1)$ repeatedly, $\Gamma(1/2) = \sqrt{\pi}$, and copious algebraic gymnastics:

$$\begin{aligned}
\frac{n}{2}\cdot\Gamma(\tfrac{n}{2}) &= \frac{n}{2}\cdot\Gamma(\tfrac{n-1}{2}+\tfrac{1}{2}) \\
&= \frac{n}{2}\cdot\left(\tfrac{n-1}{2}-\tfrac{1}{2}\right)\cdot\Gamma(\tfrac{n-1}{2}-\tfrac{1}{2}) \\
&= \frac{n}{2}\cdot\left(\tfrac{n-1}{2}-\tfrac{1}{2}\right)\cdot\left(\tfrac{n-1}{2}-\tfrac{3}{2}\right)\cdot\Gamma(\tfrac{n-1}{2}-\tfrac{3}{2}) \\
&\;\;\vdots \\
&\;\;\downarrow \\
&= \frac{n}{2}\cdot\left(\tfrac{n-1}{2}-\tfrac{1}{2}\right)\cdot\left(\tfrac{n-1}{2}-\tfrac{3}{2}\right)\cdot\cdots\cdot\left(\tfrac{3}{2}\right)\cdot\left(\tfrac{1}{2}\right)\cdot\Gamma(\tfrac{1}{2}) \\
&= \frac{1}{2^{(n+1)/2}}\cdot n\cdot(n-2)\cdot(n-4)\cdot\cdots\cdot(3)\cdot(1)\cdot\sqrt{\pi} \\
&= \frac{1}{2^{(n+1)/2}}\cdot\frac{n!}{(n-1)\cdot(n-3)\cdot(n-5)\cdot\cdots\cdot(4)\cdot(2)}\cdot\sqrt{\pi} \\
&= \frac{1}{2^{(n+1)/2}}\cdot\frac{1}{2^{(n-1)/2}}\cdot\frac{n!}{\frac{n-1}{2}!}\cdot\sqrt{\pi} = \frac{1}{2^n}\cdot\frac{n!}{\frac{n-1}{2}!}\cdot\sqrt{\pi}.
\end{aligned}$$

Turning this upside down, and then bringing in the $\pi^{\frac{n}{2}}$, we get the formula for V_n, for n odd. In summary, we have

$$V_n = \begin{cases} \dfrac{(\sqrt{\pi})^n}{\frac{n}{2}!} & \text{if } n \text{ is even} \\ \dfrac{2^n\,(\sqrt{\pi})^{n-1}\;\frac{n-1}{2}!}{n!} & \text{if } n \text{ is odd.} \end{cases}$$

✓

▶ **Remark** The expressions above for V_n are indeed closed-form expressions. But the truth is to actually compute $n!$, one needs $(n-1)!$. So to calculate V_n, one really does require the V_ns for smaller n. An extremely useful approximation of $n!$, for n large, is available—it's called *Stirling's Formula*. We shall meet Stirling's Formula in Chap. 13 and again in Chap. 15.

▶ **Remark (How big is a really big ball, in a really big place?)** All this depended on taking $r = 1$. For general $r > 0$, the formula is

$$V_n(r) = \frac{\pi^{\frac{n}{2}}}{\frac{n}{2}\,\Gamma(\frac{n}{2})} r^n .$$

For example, the largest ball of radius $r = 2$ in any $\mathbb{R}^n$ is the one in $\mathbb{R}^{24}$. (Its 24-volume is $\frac{\pi^{12}}{12!}2^{24} \approx 32373$.) After that the n-volumes eventually get as small as you please [4]. See the excellent article [2] for some interesting musings.

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Convexity—Tangents

6

In his 1906 *Acta Mathematica* paper "Sur les fonctions convexes et les inégalités entre les valeurs moyennes," J. Jensen[1] wrote:

> It seems to me that the notion of *convex function* is about as fundamental as these: *positive function*, and *increasing function*. If I am not mistaken in this, the notion will have to find its place in the elementary exhibitions of the theory of real functions.

Well, it turned out that Mr. Jensen was not mistaken.

Let f be a function defined on some interval I. If $f''(x) \geq 0$ for all $x \in I$, we shall say that f is **convex** on I. Now since $f'' = (f')'$, f being convex means that f' is increasing on I. In freshman calculus, the graph of such a function is often called *concave upward*. The idea is that the graph of a convex function is "bending upwards" as we move along the interval I, from left to right.

For $f'' > 0$, we say f is *strictly convex*. For $f'' \leq 0$, we say f is *concave* and for $f'' < 0$, we say f is *strictly concave*. Evidently, f is (strictly) concave $\Leftrightarrow -f$ is (strictly) convex. So as long as you know how to multiply by -1, you really only need to know about one of these.

One can verify that $f'' \geq 0$ (or > 0 or ≤ 0 or < 0, as the case may be) for the following functions:

- x^r is convex on $[0, \infty)$ if $r \geq 1$.
- x^3 is strictly concave on $(-\infty, 0)$ and strictly convex on $(0, \infty)$.
- e^x is strictly convex on $(-\infty, +\infty)$.
- $\ln(x)$ is strictly concave on $(0, \infty)$.
- $\sin(x)$ is strictly concave on $(0, \pi)$.
- $\cos(x)$ is strictly concave on $(0, \pi/2)$ and strictly convex on $(\pi/2, \pi)$.

[1] Johan Ludwig William Valdemar Jensen (Denmark, 1859–1925) was a telephone engineer who did mathematics in his spare time. Apparently he was very good at both.

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- $\tan(x)$ is strictly concave on $(-\pi/2, 0)$ and strictly convex on $(0, \pi/2)$.

In this chapter we explore a few of the many benefits of convexity and see how they are related to some of the things that we have already done. With the appearance of f'' in the definition of *convex*, it should come as no surprise that Taylor's Theorem, with $n = 2$, is a useful device.

6.1 Convexity and Tangent Lines

The following lemma says that the graph of a convex function always lies above (or on) its tangent lines. See Fig. 6.1.

Convex Tangents Lemma *Let f be such that $f'' \geq 0$ on an open interval I and let $x_0 \in I$. Then for each $x \in I$,*

$$f(x) \;\geq\; f(x_0) + f'(x_0)(x - x_0).$$

If $f'' > 0$, then equality occurs only at $x = x_0$. (For $f'' \leq 0$ the inequality is reversed.)

Proof. Let $x_0 \in I$. By Taylor's Theorem (with $n = 2$), for $x \in I$ there is c between x and x_0 such that

$$f(x) = f(x_0) + f'(x_0)(x - x_0) + \frac{f''(c)}{2}(x - x_0)^2.$$

Now $y = f(x_0) + f'(x_0)(x - x_0)$ is an equation for the tangent line to $y = f(x)$, at $x = x_0$. So we need only observe that $f''(c) \geq 0$ and we're done. □

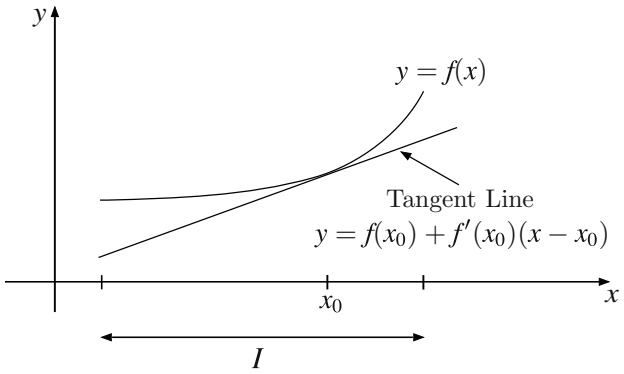


Fig. 6.1 The graph of a convex function lies above (or on) its tangent lines

Below we gather a few consequences of the Convex Tangents Lemma. The first of these, in particular, is an incredibly useful little fact.

Lemma 6.1
$e^x \geq 1 + x$ *for all* $x \in \mathbb{R}$, *and equality occurs only when* $x = 0$.

Proof. Apply the Convex Tangents Lemma to $f(x) = \mathrm{e}^x$, at $x = 0$. □

Lemma 6.2
$\left(1 + \frac{1}{t}\right)^t \leq e \leq \left(1 + \frac{1}{t}\right)^{t+1}$ *for all* $t > 0$.

Proof. For the left-hand inequality, replace x with $1/t$ in Lemma 6.1. For the right-hand inequality, replace x with $-1/(t+1)$ in Lemma 6.1. □

Lemma 6.3
$\lim_{t\to\infty} \left(1 + \frac{1}{t}\right)^t = e$ *and* $\lim_{u\to\infty} \left(1 + \frac{x}{u}\right)^u = e^x$.

Proof. Dividing Lemma 6.2 by $\left(1 + \frac{1}{t}\right)^t$ gives the first part: $\lim_{t\to\infty} \left(1 + \frac{1}{t}\right)^t = \mathrm{e}$. For $b > 1$, b^x is continuous for all x, and so $\left(\lim_{t\to\infty} \left(1 + \frac{1}{t}\right)^t\right)^x = \mathrm{e}^x = \lim_{t\to\infty} \left(1 + \frac{1}{t}\right)^{tx}$. Now set $u = tx$. □

Lemma 6.4
$\ln(x) \leq x - 1$ *for* $x > 0$, *and equality occurs only for* $x = 1$.

Proof. Apply the Convex Tangents Lemma to $f(x) = -\ln(x)$, at $x = 1$. □

We point out that Lemma 6.1 implies Lemma 6.4 via the change of variable $x \mapsto x - 1$. Lemma 6.4 implies Lemma 6.1 (for $x > -1$) via $x \mapsto x + 1$. (Lemma 6.1 is trivial for $x \leq -1$.)

Example
We saw in Sect. 2.3 that the Harmonic Series diverges:

$$1 + \frac{1}{2} + \frac{1}{3} + \frac{1}{4} + \frac{1}{5} + \cdots = \sum_{n=1}^{\infty} \frac{1}{n} = \infty . \tag{6.1}$$

Here's another proof, using the bazooka that is calculus (e.g., [1, 7]). We have

$$\ln(N) = \sum_{n=1}^{N-1} \big[\ln(n+1) - \ln(n) \big] = \sum_{n=1}^{N-1} \ln\Big(\frac{n+1}{n}\Big) = \sum_{n=1}^{N-1} \ln\Big(1 + \frac{1}{n}\Big).$$

So by Lemma 6.4,

$$\ln(N) \le \sum_{n=1}^{N-1} \frac{1}{n} < \sum_{n=1}^{N} \frac{1}{n} .$$

Since $\ln(N) \to \infty$ as $N \to \infty$, we must also have $\sum_{n=1}^{N} \frac{1}{n} \to \infty$. ✓

[Almost equivalent with this is applying the *Limit Comparison Test* to $\sum_{n=1}^{\infty} \ln\Big(1 + \frac{1}{n}\Big)$ and $\sum_{n=1}^{\infty} \frac{1}{n}$. L'Hospital strikes again!]

Bernoulli's Inequality[2] *For $\alpha \ge 1$ and $x \ge -1$, we have*

$$(1+x)^{\alpha} \ge 1 + \alpha x .$$

Proof. For such α, apply the Convex Tangents Lemma to $f(x) = (1+x)^{\alpha}$ at $x = 0$. (If $\alpha > 1$, equality occurs only for $x = 0$.) □

Example
Let $x \in \mathbb{R}$ and choose n large enough that $x/n > -1$. Then in Bernoulli's Inequality, we may replace x with x/n and α with n to get

$$\Big(1 + \frac{x}{n}\Big)^{n} \ge 1 + n\frac{x}{n} = 1 + x .$$

Having $n \to \infty$ gives, once again, $e^{x} \ge 1 + x$. ✓

6.2 Wallis's Product and Probability Integral, Again

In Chap. 4 we met Wallis's Product (4.1), in short,

[2] Named for Jacob Bernoulli (Switzerland, 1655–1705), Johann's brother. But it was anticipated almost 20 years earlier, by Newton's teacher Isaac Barrow (England, 1630–1677) [8].

$$(2n+1)\prod_{k=1}^{n}\left(\frac{2k}{2k+1}\right)^2 \;\to\; \frac{\pi}{2} \quad \text{as} \quad n\to\infty\,. \tag{6.2}$$

This is equivalent with (verify!)

$$\sqrt{n}\prod_{k=1}^{n}\frac{2k}{2k+1} \;\to\; \frac{\sqrt{\pi}}{2} \quad \text{as} \quad n\to\infty\,.$$

We also computed, in Lemma 4.1,

$$I_{2n+1} \;=\; \int_0^{\pi/2}\cos^{2n+1}(t)\,dt \;=\; \prod_{k=1}^{n}\frac{2k}{2k+1}\,.$$

Therefore, equivalent with (6.2) is

$$\sqrt{n}\;I_{2n+1} \;\to\; \frac{\sqrt{\pi}}{2} \quad \text{as} \quad n\to\infty\,. \tag{6.3}$$

In Chap. 3 we met the Probability Integral (3.1)

$$\int_0^{\infty}\mathrm{e}^{-x^2}dx \;=\; \frac{\sqrt{\pi}}{2}\,, \tag{6.4}$$

which we write more conveniently as

$$\int_0^{\sqrt{n}}\mathrm{e}^{-x^2}dx \;\to\; \frac{\sqrt{\pi}}{2} \quad \text{as} \quad n\to\infty\,. \tag{6.5}$$

So it appears that something might be afoot. It is. Leaning heavily on convexity, we show presently that (6.5) $\Leftrightarrow$ (6.3), and therefore (6.4) $\Leftrightarrow$ (6.2). That is,

Wallis's Product and the Probability Integral are equivalent.

The plan of attack is inspired largely by [6] (which was anticipated in [9, 11]; see also [2]). Lemma 6.3 says that for n large, $\left(1+x/n\right)^n \approx \mathrm{e}^x$, so the idea is to approximate the e^{-x^2} in the integrand of (6.5), with $\left(1-x^2/n\right)^n$. This is made possible by the following.

Lemma 6.5
For $0 \le x \le \sqrt{n}$, we have

$$0 \;\le\; \mathrm{e}^{-x^2} - \left(1 - \frac{x^2}{n}\right)^n \;\le\; \frac{4}{\mathrm{e}^2}\frac{1}{n}\,.$$

Proof. Lemma 6.1 gives $\left(1+\frac{x^2}{n}\right)^n \le \mathrm{e}^{x^2}$, then replacing x^2 with $-x^2$, we get the left-hand inequality (for all x). For the right-hand inequality, we use $\left(1+\frac{x^2}{n}\right)^n \le \mathrm{e}^{x^2}$ again:

$$\begin{aligned}
\mathrm{e}^{-x^2} - \left(1 - \frac{x^2}{n}\right)^n &= \mathrm{e}^{-x^2}\left[\,1 - \mathrm{e}^{x^2}\left(1 - \frac{x^2}{n}\right)^n\,\right] \\
&\le \mathrm{e}^{-x^2}\left[\,1 - \left(1 + \frac{x^2}{n}\right)^n\left(1 - \frac{x^2}{n}\right)^n\,\right] \\
&= \mathrm{e}^{-x^2}\left[\,1 - \left(1 - \frac{x^4}{n^2}\right)^n\,\right].
\end{aligned}$$

Restricting to $x \le \sqrt{n}$ we can apply Bernoulli's Inequality:

$$\mathrm{e}^{-x^2}\left[\,1 - \left(1 - \frac{x^4}{n^2}\right)^n\,\right] \;\le\; \mathrm{e}^{-x^2}\left[\,1 - \left(1 - n\frac{x^4}{n^2}\right)\,\right] \;=\; \mathrm{e}^{-x^2}\,\frac{x^4}{n}\,.$$

Finally (Exercise 6.10), the maximum value of $\mathrm{e}^{-u}u^2$ for all $u \ge 0$ is $4/\mathrm{e}^2$ (which occurs at $u = 2$). So the maximum value of $\mathrm{e}^{-x^2}x^4$ for all $x \ge 0$ is also $4/\mathrm{e}^2$ (which occurs at $x = \sqrt{2}$), and the proof is complete. □

Proof of (6.5) ⇔ (6.3). In Lemma 6.5, we integrate from 0 to $\sqrt{n}$ to get

$$0 \;\le\; \int_0^{\sqrt{n}} \mathrm{e}^{-x^2}\,dx \;-\; \int_0^{\sqrt{n}} \left(1 - \frac{x^2}{n}\right)^n dx \;\le\; \int_0^{\sqrt{n}} \frac{4}{\mathrm{e}^2}\frac{1}{n}\,dx \;=\; \frac{4}{\mathrm{e}^2}\frac{1}{\sqrt{n}}\,.$$

In the second integral from the left, we substitute $x = \sqrt{n}\sin(t)$. Then

$$\int_0^{\sqrt{n}} \left(1 - \frac{x^2}{n}\right)^n dx \;=\; \sqrt{n}\int_0^{\pi/2} \cos^{2n+1}(t)\,dt \;=\; \sqrt{n}\;I_{2n+1}\,.$$

Therefore

$$0 \;\le\; \int_0^{\sqrt{n}} \mathrm{e}^{-x^2}\,dx \;-\; \sqrt{n}\,I_{2n+1} \;\le\; \frac{4}{\mathrm{e}^2}\,\frac{1}{\sqrt{n}}\,,$$

and so

$$\lim_{n\to\infty} \int_0^{\sqrt{n}} \mathrm{e}^{-x^2}\,dx \;=\; \lim_{n\to\infty} \left(\sqrt{n}\,I_{2n+1}\right),$$

which gives (6.5) $\Leftrightarrow$ (6.3). □

Exercises

6.1.

(a) Let f and g be convex on I. Show that $f+g$ and $\max\{f, g\}$ are convex on I.
(b) Let f be convex on I and let g be increasing on $(-\infty, +\infty)$. Show that $g \circ f$ is convex on I.
(c) Show that $f(x) = x\ln(x)$ is strictly convex on $(0, +\infty)$.
(d) Show that $f(x) = \ln(\frac{\mathrm{e}^{2x}-1}{\mathrm{e}^x-1})$ is strictly convex on $(-\infty, +\infty)$. Can we replace the number 2 here with *any* number and arrive at the same conclusion?

6.2.

(a) Show that if f is defined on an open interval containing x and f'' is continuous at x, then

$$f''(x) \;=\; \lim_{h\to 0} \frac{f(x+h) - 2f(x) + f(x-h)}{h^2}\,.$$

(b) Show that this still holds if we only assume that $f''(x)$ exists.

6.3 (This One Is Tricky [4, 5]). Lemma 6.2 is $\left(1+\frac{1}{t}\right)^t \le \mathrm{e} \le \left(1+\frac{1}{t}\right)^{t+1}$ for $t > 0$. Find the maximum value for α and the minimum value for β such that for all $t > 0$,

$$\left(1+\frac{1}{t}\right)^{t+\alpha} \le \mathrm{e} \le \left(1+\frac{1}{t}\right)^{t+\beta}.$$

[Answer: $\alpha = 1/\ln(2) - 1 \approx 0.4427$, $\beta = 1/2$.]

6.4. Apply the Convex Tangents Lemma to $f(t) = \ln(t)$ at $t = x/\mathrm{e}$ to show that

$$x^{\mathrm{e}} < \mathrm{e}^{x}, \quad \text{and in particular,} \quad \pi^{\mathrm{e}} < \mathrm{e}^{\pi}.$$

6.5. Bernoulli's Inequality in its earliest form (the pre-calculus version) is as follows. *Let* $n \in \mathbb{N}$. *Then*

$$x^n \geq 1 + n(x-1) \quad \textit{for } x \geq 0\,. \tag{6.6}$$

(a) Prove (6.6) by induction.
(b) Prove (6.6) by verifying, and then analyzing, the identity

$$x^n - n(x-1) - 1 = (x-1)\big[x^{n-1} + x^{n-2} + \cdots + x - (n-1)\big].$$

6.6.

(a) Show that Bernoulli's Inequality is reversed, for $0 \leq \alpha \leq 1$.
(b) Use this to obtain *Young's Inequality*: *For* $x, y > 0$,

$$x^{1-\alpha} y^{\alpha} \leq (1-\alpha)x + \alpha y\,.$$

6.7. Show, as follows, that for $0 < x < 1$, the sequence $\{x^n\}$ converges to 0. Write $x = \frac{1}{1+q}$, for some $q > 0$. Apply Bernoulli's Inequality to $(1+q)^n$; then take reciprocals. How about $|x| < 1$?

6.8. Show that (6.2) implies $\sqrt{n} \prod_{k=1}^{n-1} \frac{2k-1}{2k} \to \frac{1}{\sqrt{\pi}}$ as $n \to \infty$.

Conclude that

$$\sqrt{n}\, I_{2n-2} = \sqrt{n} \int_0^{\pi/2} \cos^{2n-2}(x)\, dx \to \frac{1}{\sqrt{\pi}} \quad \text{as} \quad n \to \infty\,.$$

6.9 ([3,10,12]). Fill in the details of the following slick proof that (6.2) $\Rightarrow$ (6.5) (i.e., that Wallis's Product can be used to evaluate the Probability Integral).

(a) Use Lemma 6.1: $1 + x \leq \mathrm{e}^{x}$ to obtain

$$1 - x^2 \leq \mathrm{e}^{-x^2} \leq \frac{1}{1+x^2} \quad \text{for all } x \in \mathbb{R}\,.$$

Taking nth powers here, and then integrating from 0 to 1, gives

$$\int_0^1 (1-x^2)^n\, dx \leq \int_0^1 \mathrm{e}^{-nx^2} dx \leq \int_0^1 \frac{1}{(1+x^2)^n} dx\,.$$

(b) Set $x = \sin(t)$ in the first integral, $x = u/\sqrt{n}$ in the second, and $x = \tan(t)$ in the third, to get

$$I_{2n+1} \;\le\; \frac{1}{\sqrt{n}} \int_0^{\sqrt{n}} \mathrm{e}^{-u^2}\, du \;\le\; I_{2n-2}\,.$$

(c) Now use (6.2) on the two outside pieces; then manipulate to obtain (6.5).

6.10. Verify, as claimed in Sect. 6.2, that the maximum value of $\mathrm{e}^{-u}u^2$ for all $u \ge 0$ is $4/\mathrm{e}^2$ (which occurs at $u = 2$). Conclude that the maximum value of $\mathrm{e}^{-x^2}x^4$ for all $x \ge 0$ is also $4/\mathrm{e}^2$ (which occurs at $x = \sqrt{2}$).

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Some Important Series

7

We have seen that the Harmonic Series diverges:

$$\sum_{n=1}^{\infty} \frac{1}{n} \;=\; \infty\,.$$

Here's another proof which uses calculus: Since $y = 1/x$ decreases on $[1, \infty)$, we have (see Fig. 7.1)

$$1 + \frac{1}{2} + \frac{1}{3} + \frac{1}{4} + \frac{1}{5} + \cdots \;>\; \int_1^{\infty} \frac{1}{x} dx \;=\; \ln(x)\Big|_1^{\infty} \;=\; \infty\,. \qquad \checkmark$$

In this chapter we look at several important series which are closely related to the Harmonic Series.

7.1 *p*-Series

Once it is established that the Harmonic Series diverges, it is easy to see that a p-Series $\sum_{n=1}^{\infty} \frac{1}{n^p}$ diverges also for $p < 1$, since for such p, $\frac{1}{n^p} \geq \frac{1}{n}$.

On the other hand, an argument closely akin to the one just above gives the following:

The original version of the chapter has been revised. A correction to this chapter can be found at https://doi.org/10.1007/978-3-031-43914-8_16.

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P. R. Mercer, *A Compact Capstone Course in Classical Calculus*, Compact Textbooks in Mathematics, https://doi.org/10.1007/978-3-031-43914-8_7

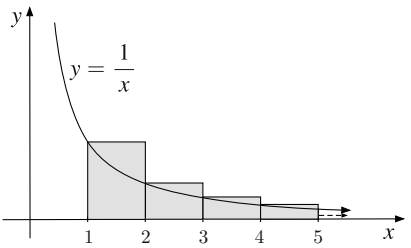


Fig. 7.1 The total area of all of the shaded rectangles is $> \int_1^\infty \frac{1}{x} dx$

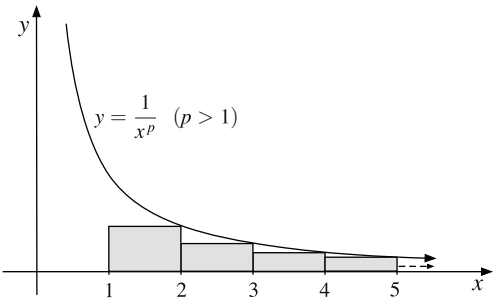


Fig. 7.2 The total area of all of the shaded rectangles is $< \int_1^\infty \frac{1}{x^p} dx$

Lemma *A p-Series converges for* $p > 1$. *That is,*

$$\sum_{n=1}^{\infty} \frac{1}{n^p} \; < \; \infty \quad \textit{for} \quad p > 1 \, .$$

Proof. See Fig. 7.2. The function $y = 1/x^p$ decreases on $[1, \infty)$, and so we have

$$1 \; + \; \frac{1}{2^p} \; + \; \frac{1}{3^p} \; + \; \frac{1}{4^p} \; + \; \frac{1}{5^p} \; + \cdots \; < \; 1 \; + \int_1^\infty \frac{1}{x^p} dx \, .$$

Then since $p > 1$,

$$1 \; + \int_1^\infty \frac{1}{x^p} \, dx \; = \; 1 + \frac{1}{-p+1} \, \frac{1}{x^{p-1}} \Big|_1^\infty \; = \; \frac{p}{p-1} \; < \; \infty \, .$$

□

▶ **Remark** The two proofs above (for $\sum \frac{1}{n}$ and $\sum \frac{1}{n^p}$) taken together are essentially what is known as the Integral Test, which you probably met in calculus 2. We outline a couple more proofs in Exercises 7.4 and 7.5.

7.2 Alternating Harmonic Series and Madhavan Series, Etc.

Okay, the Harmonic Series diverges—but its *alternating* version converges, as we saw by applying the Alternating Series Test in Sect. 2.3. P. Mengoli[1] was the first to find its sum, as follows:

The Alternating Harmonic Series

$$1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \frac{1}{5} - \frac{1}{6} + \cdots = \sum_{n=1}^{\infty} \frac{(-1)^{n-1}}{n} = \ln(2). \tag{7.1}$$

Gregory, and also Leibniz, summed a closely related series (7.2), which traditionally bears their names. But they were anticipated about 250 years earlier by Madhava,[2] so we name it for him. Following [13] (see also [2, 7, 10]), we show in tandem with (7.1) the following:

The Madhavan Series

$$1 - \frac{1}{3} + \frac{1}{5} - \frac{1}{7} + \frac{1}{9} - \frac{1}{11} + \cdots = \sum_{n=1}^{\infty} \frac{(-1)^{n-1}}{2n-1} = \frac{\pi}{4}. \tag{7.2}$$

Proof of (7.1) and (7.2). For $N = 0, 1, 2, \ldots$, set

$$T_N = \int_0^{\pi/4} \tan^N(x)\, dx\,.$$

So then $T_0 = \frac{\pi}{4}$ and $T_1 = \int_0^{\pi/4} \tan(x)\, dx = \ln(\sec(x))\Big|_0^{\pi/4} = \frac{\ln(2)}{2}$. For $N \geq 2$,

$$T_N = \int_0^{\pi/4} \tan^{N-2}(x)\tan^2(x)\, dx = \int_0^{\pi/4} \tan^{N-2}(x)\left[\sec^2(x) - 1\right] dx$$

[1] Pietro Mengoli (Italy, 1626–1686) summed the Alternating Harmonic Series in 1650, when Isaac Newton was 8 years old.

[2] Madhava of Sangamagrama (India, c. 1350–c. 1425) anticipated several other well-known series.

$$= \int_0^{\pi/4} \tan^{N-2}(x)\sec^2(x)\,dx \;-\; \int_0^{\pi/4} \tan^{N-2}(x)\,dx\,.$$

The substitution $u = \tan(x)$ in the first integral $\left[\text{wherein } du = \sec^2(x)dx\right]$ gives

$$T_N \;=\; \frac{1}{N-1}u^{N-1}\Big|_0^1 \;-\; \int_0^{\pi/4} \tan^{N-2}(x)\,dx\,,$$

which yields the reduction formula

$$T_N \;=\; \frac{1}{N-1} \;-\; T_{N-2}\,.$$

The subscripts go down in steps of two. Using this repeatedly, with $N = 2n$ (even) and $N = 2n+1$ (odd), all the way down to T_0 and to T_1, respectively, we get

$$T_{2n} = \frac{1}{2n-1} - \frac{1}{2n-3} + \frac{1}{2n-5} - \cdots \pm \frac{1}{3} \mp 1 \pm \frac{\pi}{4}\,,$$

and

$$T_{2n+1} = \frac{1}{2n} - \frac{1}{2n-2} + \frac{1}{2n-4} - \cdots \pm \frac{1}{2} \mp \frac{\ln(2)}{2}\,.$$

So if we can show that $T_N \to 0$ as $N \to \infty$, then we'll be all set. (Knowing what the graph of $y = \tan(x)$ looks like on $[0, \pi/4]$, it shouldn't be surprising that this is so.)

The substitution $u = \tan(x)$ in T_N gives

$$T_N \;=\; \int_0^1 \frac{u^N}{1+u^2}\,du\,.$$

Now $0 \le (1-u)^2$ and so $2u \le 1+u^2$ for all $u \in \mathbb{R}$. Therefore, for $u \in [0,1]$, we have

$$2u \;\le\; 1+u^2 \;\le\; 2\,,$$

which yields for T_N

$$\int_0^1 \frac{u^N}{2}\,du \;\le\; T_N \;\le\; \int_0^1 \frac{u^N}{2u}\,du\,,$$

i.e.,

$$\frac{1}{2}\int_0^1 u^N du \;\le\; T_N \;\le\; \frac{1}{2}\int_0^1 u^{N-1} du\,.$$

Evaluating the integrals, this reads

$$\frac{1}{2(N+1)} \;\le\; T_N \;\le\; \frac{1}{2N}\,,$$

an so indeed, $T_N \to 0$ as $N \to \infty$. We're all set. □

We close this section with another series for ln(2), but having positive terms, which converges faster than does (7.1). This is due to—who else?—Euler. First, we integrate by parts repeatedly:

$$\ln(2) = \int_0^1 \underbrace{\frac{1}{1+x}}_{\mathrm{D}} \underbrace{1}_{\mathrm{I}} dx = \frac{1}{1+x}\,x\,\Big|_0^1 + \int_0^1 \frac{1}{(1+x)^2}\,x\,dx$$

$$= \frac{1}{2} + \int_0^1 \underbrace{\frac{1}{(1+x)^2}}_{\mathrm{D}} \underbrace{x}_{\mathrm{I}} dx$$

$$= \frac{1}{2} \;+\; \frac{1}{(1+x)^2}\frac{1}{2}x^2\Big|_0^1 + \int_0^1 \frac{2}{(1+x)^3}\frac{1}{2}x^2 dx$$

$$= \frac{1}{2} + \frac{1}{2}\frac{1}{2^2} + \int_0^1 \underbrace{\frac{1}{(1+x)^3}}_{\mathrm{D}} \underbrace{x^2}_{\mathrm{I}} dx = \cdots\,.$$

Continuing in this way, we get

$$\ln(2) \;=\; \frac{1}{2} \;+\; \frac{1}{2}\frac{1}{2^2} \;+\; \frac{1}{3}\frac{1}{2^3} + \cdots + \frac{1}{n2^n} \;+\; \int_0^1 \frac{x^n}{(1+x)^{n+1}} dx\,.$$

Then, since

$$0 \;<\; \int_0^1 \frac{x^n}{(1+x)^{n+1}}dx \;<\; \int_0^1 x^n dx \;=\; \frac{1}{n+1}\,,$$

having $n \to \infty$, we obtain

$$\ln(2) \;=\; \frac{1}{2} \;+\; \frac{1}{2}\frac{1}{2^2} \;+\; \frac{1}{4}\frac{1}{2^4} \;+\cdots=\; \sum_{n=1}^{\infty} \frac{1}{n2^n}\,.$$

[A series with positive terms for $\pi/4$ is obtained similarly, in Exercise 7.7.]

7.3 Euler's Constant

Okay, the Harmonic Series diverges—but it does so *very* slowly. For example, the first 100 terms of the series sum to just under 5.19; the first 100,000 terms sum to just under 12.1. We can be a little more precise about this, as follows:

Have a look at Fig. 7.3. Again, since $y = 1/x$ decreases on $[1, \infty)$, we have

$$\frac{1}{n+1} \;\le\; \int_n^{n+1} \frac{1}{x}\,dx \;\le\; \frac{1}{n}\,.$$

Evaluating the integral we get the useful estimates

$$\frac{1}{n+1} \;\le\; \ln\left(\frac{n+1}{n}\right) \;\le\; \frac{1}{n}\,. \tag{7.3}$$

Now, as in [9], consider the sequence

$$\{a_n\} \;=\; \left\{1\,,\; \ln\left(\tfrac{2}{1}\right)\,,\; \tfrac{1}{2}\,,\; \ln\left(\tfrac{3}{2}\right)\,,\; \tfrac{1}{3}\,,\; \ln\left(\tfrac{4}{3}\right)\,,\; \tfrac{1}{4}\,,\; \ln\left(\tfrac{5}{4}\right)\,,\ldots.\right\}.$$

With an eye on (7.3), the Alternating Series Test (Sect. 2.3) tells us that the series

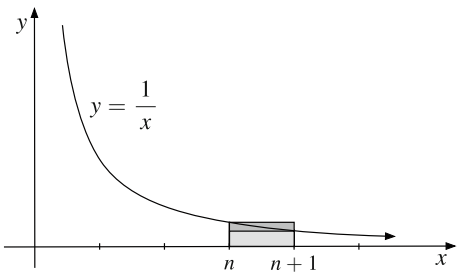


Fig. 7.3 Area of the smaller rectangle $< \int_n^{n+1} \frac{1}{x}dx <$ area of the larger one

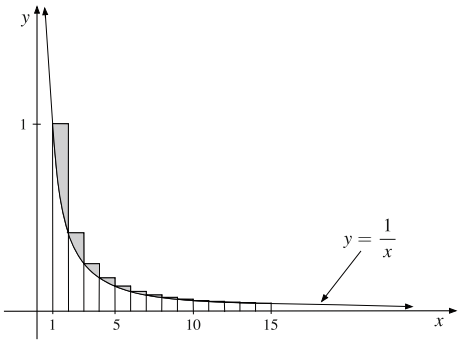


Fig. 7.4 γ is the total area of *all* of these shaded regions

$$\sum_{n=1}^{\infty}(-1)^{n+1}a_n$$

converges to some number γ. This number γ is called *Euler's Constant*, or the Euler-Mascheroni[3] constant. The coarsest of estimates give $1 - \ln(2) < \gamma < 1$.

We obtain a manageable expression for γ, as follows: Denote by γ_{2n-1} a partial sum having an odd number of terms and by γ_{2n} the next partial sum, having an even number of terms:

$$\gamma_{2n-1} = 1 - \ln\left(\frac{2}{1}\right) + \frac{1}{2} - \ln\left(\frac{3}{2}\right) + \cdots + \frac{1}{n-1} - \ln\left(\frac{n}{n-1}\right) + \frac{1}{n},$$

$$\gamma_{2n} = 1 - \ln\left(\frac{2}{1}\right) + \frac{1}{2} - \ln\left(\frac{3}{2}\right) + \cdots + \frac{1}{n-1} - \ln\left(\frac{n}{n-1}\right) + \frac{1}{n} - \ln\left(\frac{n+1}{n}\right).$$

Here then,

$$\lim_{n\to\infty}\gamma_{2n-1} \;=\; \lim_{n\to\infty}\gamma_{2n} \;=\; \gamma\,,$$

with γ_{2n-1} *decreasing* and γ_{2n} *increasing*. There are lots of cancellation, and we get

$$1+\frac{1}{2}+\frac{1}{3}+\cdots+\frac{1}{n}-\ln(n+1) \;=\; \gamma_{2n} \;\le\; \gamma \;\le\; \gamma_{2n-1} \;=\; 1+\frac{1}{2}+\frac{1}{3}+\cdots+\frac{1}{n}-\ln(n).$$

Since $\displaystyle\int_n^{n+1}\frac{1}{x}\,dx \;=\; \ln\!\left(\frac{n+1}{n}\right)$, γ is the area of the shaded regions in Fig. 7.4.

Any informal survey of mathematicians would probably reveal γ as the third most intriguing constant in classical analysis, after π and e. That $\gamma > 0$ exists is an

[3] Lorenzo Mascheroni (Italy, 1750–1800) was best known for his work in geometry. He was (probably) the first to denote this constant by γ. He computed γ to 19 decimal places.

indication of how slowly the Harmonic Series diverges: $\ln(x)$ goes to infinity, yes, but leisurely! Computing γ_N for very large N gives

$$\gamma \;=\; 0.57721566\cdots\cdots .$$

Though very doubtful, it is not currently known whether γ is rational.[4]

Example
One can find the sum of the Alternating Harmonic Series (7.1) using Euler's Constant γ, as follows: For n even, and with γ_{2n} as above,

$$\begin{aligned}
&1-\tfrac{1}{2}+\tfrac{1}{3}-\tfrac{1}{4}+\tfrac{1}{5}-\tfrac{1}{6}+\cdots-\tfrac{1}{n}\\
&= 1+\left(\tfrac{1}{2}-\tfrac{2}{2}\right)+\tfrac{1}{3}+\left(\tfrac{1}{4}-\tfrac{2}{4}\right)+\tfrac{1}{5}+\left(\tfrac{1}{6}-\tfrac{2}{6}\right)+\cdots+\left(\tfrac{1}{n}-\tfrac{2}{n}\right)\\
&= 1+\tfrac{1}{2}+\tfrac{1}{3}+\tfrac{1}{4}+\tfrac{1}{5}+\tfrac{1}{6}+\cdots+\tfrac{1}{n}-\left(1+\tfrac{1}{2}+\tfrac{1}{3}+\tfrac{1}{4}+\tfrac{1}{5}+\tfrac{1}{6}+\cdots+\tfrac{1}{n/2}\right)\\
&= \left(\gamma_{2n}+\ln(n+1)\right)-\left(\gamma_n+\ln\left(\tfrac{n}{2}+1\right)\right).
\end{aligned}$$

Now as $n\to\infty$, we have $\gamma_{2n}\to\gamma$, $\gamma_n\to\gamma$, and $\ln(n+1)-\ln\left(\frac{n}{2}+1\right)\to\ln(2)$. So as claimed, we see again that $\sum_{n=1}^{\infty}\frac{(-1)^{n-1}}{n}=\ln(2)$. ✓

Exercises

7.1.

(a) Let P be a prime number. Show that $\sum_{n=1}^{P}\frac{1}{n}$ is not an integer.
[This is true also if $P\geq 2$ is not prime, but is trickier to prove...]

(b) Let $p_1, p_2, \ldots, p_n$ be distinct (not necessarily consecutive) prime numbers. Show that $\sum_{k=1}^{n}\frac{1}{p_k}$ is not an integer.

7.2 ([6, 8]). Consider the Harmonic Series but modified by omitting all terms which have the digit 9 appearing in the denominator. Show that this series *converges*. How about omitting instead the ones with an 8? 7? 6? .... 0? [Such modifications, having some particular *string* of digits removed, are called *Kempner Series*.]

[4] "...ah that they dont know neither do I so there you are."—Molly Bloom.

7.3 ([1]). Since the Harmonic Series diverges, for each integer $n \geq 1$, there is a least integer $a_n \geq 2$ such that $\frac{1}{n} + \frac{1}{n+1} + \cdots + \frac{1}{a_n} > 1$. Show that $\frac{a_n}{n} \to e$ as $n \to \infty$.

7.4 ([11]). Here's another proof that $\sum_{n=1}^{\infty} \frac{1}{n^p}$ converges for $p > 1$.

(a) Agree that Bernoulli's Inequality from Sect. 6.1 gives the following:
For $\alpha \geq 1$ and $0 \leq x \leq 1$, we have $(1-x)^\alpha \geq 1 - \alpha x$.

(b) Let $p > 1$ and choose $r \geq 1$ so that $1 + 1/r < p$. Show that Bernoulli's Inequality as above, with $x = \frac{1}{r}\frac{1}{n+1}$ and $\alpha = r$, after some rearranging, gives

$$r\left(\frac{1}{n^{1/r}} - \frac{1}{(n+1)^{1/r}}\right) \geq \frac{1}{(n+1)\,n^{1/r}}.$$

(c) Deduce that $r\sum_{n=1}^{N}\left(\frac{1}{n^{1/r}} - \frac{1}{(n+1)^{1/r}}\right) > \sum_{n=1}^{N}\frac{1}{(n+1)^p}$.

(d) Observing that the sum on the left-hand side telescopes completes the proof.

7.5 ([3]). Here's another proof that $\sum_{n=1}^{\infty} \frac{1}{n^p}$ converges for $p > 1$. Let $S_N = \sum_{n=1}^{N} \frac{1}{n^p}$.

(a) Fill in the steps which show that

$$S_{2N+1} = 1 + \left(\frac{1}{2^p} + \frac{1}{4^p} + \cdots + \frac{1}{(2N)^p}\right) + \left(\frac{1}{3^p} + \frac{1}{5^p} + \cdots + \frac{1}{(2N+1)^p}\right)$$

$$< 1 + \frac{1}{2^{p-1}} S_{2N+1}.$$

(b) Conclude then that $\{S_N\}$ converges—that is, $\sum_{n=1}^{\infty} \frac{1}{n^p} < \infty$.

7.6. Find the sum of the series

$$1 - \frac{1}{2} - \frac{1}{4} + \frac{1}{3} - \frac{1}{5} - \frac{1}{7} + \frac{1}{6} - \frac{1}{8} - \frac{1}{10} + \frac{1}{9} - \cdots.$$

[The answer is not $\ln(2)$; the series in (7.1) is *alternating*. See [4] for more on this.]

7.7. Begin with $\frac{\pi}{4} = \tan^{-1}(1) = \int_0^1 \frac{dx}{1+x^2}$ and then integrate by parts repeatedly, to obtain (also by Euler!)

$$\frac{\pi}{4} = \frac{1}{2}\left(1 + \frac{1}{3} + \frac{1\cdot 2}{3\cdot 5} + \frac{1\cdot 2\cdot 3}{3\cdot 5\cdot 7} + \frac{1\cdot 2\cdot 3\cdot 4}{3\cdot 5\cdot 7\cdot 9} + \cdots\right).$$

7.8. Show how inequalities (7.3), i.e., $\frac{1}{n+1} \le \ln\left(\frac{n+1}{n}\right) \le \frac{1}{n}$, follow very quickly from Lemma 6.2.

7.9. Show that for any $\alpha > 0$,

$$\gamma = \lim_{n\to\infty}\left(1 + \tfrac{1}{2} + \tfrac{1}{3} + \cdots + \tfrac{1}{n} - \ln(n+\alpha)\right).$$

Note: Taking $\alpha = 1/2$ is a particularly good idea—see [5, 12].

7.10. Show that $\lim_{n\to\infty}\left[\sum_{k=0}^{n} \frac{2}{2k+1} - \ln\left(\frac{2n+1}{2}\right)\right] = \gamma + \ln(4)$.

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Geometric Probability

8

An excellent introduction to the subject of Geometric Probability is the following problem, which is by now a bit of a classic.

Glass Rod Problem A A glass rod of length 1 is broken randomly into three pieces. What is the probability that the three pieces can be arranged to form a triangle?

Solution. We think of the two breaks of the rod as points x and y with, say, $0 < x < y < 1$. Then the three pieces have lengths x, $y - x$, and $1 - y$. For these three numbers to form the sides of a triangle, it is necessary and sufficient that each is less than the sum of the other two (think Triangle Inequality):

$$x < (y-x)+(1-y), \qquad y-x < x+(1-y), \qquad \text{and} \quad 1-y < x+(y-x).$$

That is, tidying up, we require that

$$x < \frac{1}{2}, \qquad y < x+\frac{1}{2}, \quad \text{and} \quad \frac{1}{2} < y.$$

The intersection of these three regions is the shaded triangle as shown in Fig. 8.1.

Breaks of the rod after which a triangle can be formed correspond exactly to points (x, y) which lie in the shaded triangle. All possible breaks correspond to points (x, y) in the region $0 < x < y < 1$. Since the shaded triangle has area equal to one-fourth that of the region $0 < x < y < 1$, the probability that the three pieces can be arranged to form a triangle is $1/4$. ✓

The regions appearing in the solution above are triangles, but if they were more complicated (or if they were infinite in number!), we might have needed calculus to compute their areas. In this chapter we consider such problems.

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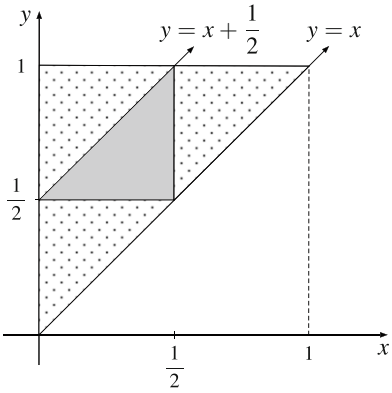


Fig. 8.1 The intersection of $x < \frac{1}{2}$, $y < x + \frac{1}{2}$, and $\frac{1}{2} < y$

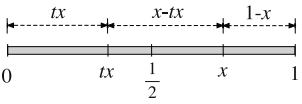


Fig. 8.2 A break at x, then the longer piece $[0, x]$, is broken at tx

8.1 Enter: Calculus

Glass Rod Problem B (e.g., [9]) A glass rod of length 1 is broken randomly into two pieces. Then, the longer piece is broken randomly into two pieces. What is the probability that the resulting three pieces can be arranged to form a triangle?

Solution. We think of the first break as $x \in (0, 1)$ with the left-hand piece (say) being the longer,[1] so that $x \geq \frac{1}{2}$. Then the second break is $y \in (0, x)$, with $y = tx$ for some $t \in (0, 1)$. See Fig. 8.2.

Then the three pieces have lengths

$$tx\,, \quad x - tx = x(1-t), \quad \text{and} \quad 1 - x.$$

For these three lengths to form the sides of a triangle, it is necessary and sufficient that each is less than the sum of the other two. After some simplifying this gives

$$x > \frac{1}{2} \quad \text{and} \quad 1 - \frac{1}{2x} < t < \frac{1}{2x}.$$

The intersection of these three regions is shown Fig. 8.3.

The area of the shaded region is

[1] If the right-hand piece is the longer one, then look at the glass rod from the opposite side!

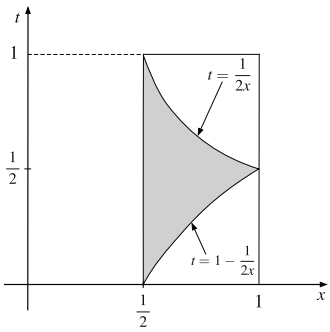


Fig. 8.3 The region $1 - \frac{1}{2x} < t < \frac{1}{2x}$, in the rectangle $[\frac{1}{2}, 1] \times [0, 1]$

$$\int_{1/2}^{1} \left(\frac{1}{2x} - \left(1 - \frac{1}{2x}\right)\right) dx \;=\; \int_{1/2}^{1} \left(\frac{1}{x} - 1\right) dx \;=\; \big(\ln(x) - x\big)\Big|_{1/2}^{1} \;=\; \ln(2) - \frac{1}{2}.$$

The area of the rectangle $[1/2, 1] \times [0, 1]$ is $1/2$. Therefore the probability that the three pieces can be arranged to form a triangle is $(\ln(2) - 1/2) \div 1/2 \approx 0.3863$.[2] ✓

For the next problem, we'll use the *floor function*, defined for each $x \in \mathbb{R}$ as the largest integer $\leq x$. That is,

$$\lfloor\, x \,\rfloor \;=\; \max\big\{\, m \in \mathbb{Z} \,:\, m \leq x \,\big\}.$$

For example, $\lfloor\, 0.0001 \,\rfloor = 0$, $\lfloor\, 1.2 \,\rfloor = 1$, $\lfloor\, \pi \,\rfloor = 3$, $\lfloor\, -1/2 \,\rfloor = -1$, $\lfloor\, 5 \,\rfloor = 5$.

Floor Function Problem Suppose that two real numbers p and q are chosen randomly in the interval $(0, 1)$. What is the probability P that $\lfloor\, p/q \,\rfloor$ is an even number?

Solution. Clearly $0 < \frac{p}{q}$. If also $\frac{p}{q} < 1$, then $\lfloor \frac{p}{q} \rfloor = 0$ (even). And then

If $1 \leq \frac{p}{q} < 2$, then $\lfloor \frac{p}{q} \rfloor = 1$ (odd); if $2 \leq \frac{p}{q} < 3$, then $\lfloor \frac{p}{q} \rfloor = 2$ (even)

If $3 \leq \frac{p}{q} < 4$, then $\lfloor \frac{p}{q} \rfloor = 3$ (odd); if $4 \leq \frac{p}{q} < 5$, then $\lfloor \frac{p}{q} \rfloor = 4$ (even)

Etcetera: For $n \geq 0$, if $2n \leq \frac{p}{q} < 2n + 1$, then $\lfloor \frac{p}{q} \rfloor = 2n$ (even).

[2] Alternatively, if after the first break the *right-hand* piece is the longer one (i.e., $x < 1/2$), and we obstinately refuse to look at the glass rod from the opposite side, then we would obtain the same shaded region above but reflected in the line $x = 1/2$. Then the probability that the three pieces can be arranged to form a triangle is $2(\ln(2) - 1/2) \div 1 \approx 0.3863$. Whew!

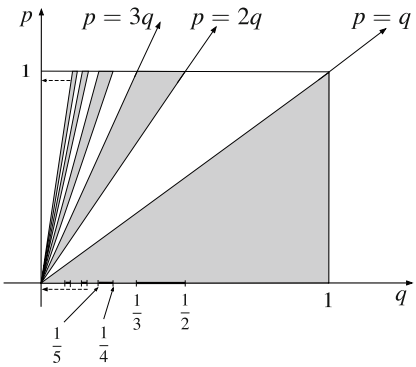


Fig. 8.4 Triangular regions $p < q$, $2q \le p < 3q$, $4q \le p < 5q$, . . . in $[0, 1] \times [0, 1]$

The cases in which $\lfloor p/q \rfloor$ is even form an infinite sequence of triangular regions in the square $[0, 1] \times [0, 1]$. The first five such regions are shown shaded in Fig. 8.4.

The probability P that we seek is the sum of the areas of all these shaded regions:

$$\begin{aligned} P &= \frac{1}{2} + \frac{1}{2}\left(\frac{1}{2} - \frac{1}{3}\right) + \frac{1}{2}\left(\frac{1}{4} - \frac{1}{5}\right) + \frac{1}{2}\left(\frac{1}{6} - \frac{1}{7}\right) + \cdots \\ &= \frac{1}{2}\left(1 + \frac{1}{2} - \frac{1}{3} + \frac{1}{4} - \frac{1}{5} + \frac{1}{6} - \frac{1}{7} + \cdots\right). \end{aligned}$$

Now we saw in Sect. 7.2 that the Alternating Harmonic Series

$$1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \cdots = \ln(2),$$

and therefore $P = \frac{1}{2}(2 - \ln(2)) \approx 0.6534$. ✓

8.2 Buffon's Needle

Geometric Probability goes back at least as far as the following [1, 6]:

Buffon's Needle Problem[3] A needle of length 1 is dropped randomly on a (huge) "plank-work" of parallel lines which are distance 1 apart. What is the probability that the landed needle touches one of the lines?

[3] Georges-Louis Leclerc, Comte de Buffon (France, 1707–1788). Apparently, in the original problem, the plank-work was a wooden floor and the needle was a loaf of French bread [6].

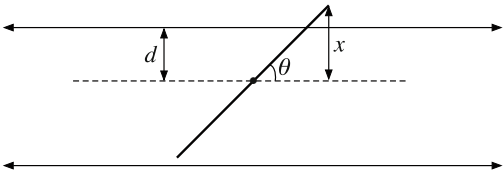


Fig. 8.5 Buffon's needle touches one of the lines precisely when $d \le x$

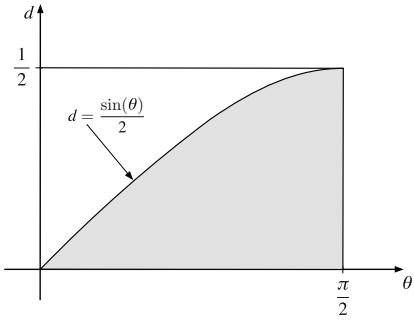


Fig. 8.6 The shaded region is $d \le \frac{1}{2}\sin(\theta)$, for $0 \le \theta \le \frac{\pi}{2}$

Solution. Let d be the distance from the center of the needle to the nearest line, so that $0 \le d \le 1/2$. Let θ be the acute angle that the needle makes with the plank-work, so that $0 \le \theta \le \pi/2$. See Fig. 8.5—the needle touches a line $\Leftrightarrow$ $d \le x$.

The needle has half-length $1/2$, so that $x = \frac{1}{2}\sin\theta$. Therefore the needle touches the plank-work $\Leftrightarrow$ $d \le \frac{1}{2}\sin\theta$. That is, the needle touches the plank-work precisely when the point (θ, d) is in the shaded region shown in Fig. 8.6.

The area of this shaded region is $\int_0^{\pi/2} \frac{1}{2}\sin\theta d\theta = \frac{1}{2}$, and the area of rectangle $[0, \pi/2]\times[0, 1/2]$ is $\pi/4$. So the probability that the landed needle touches one of the lines is

$$\frac{1/2}{\pi/4} = \frac{2}{\pi} \approx 0.6366\,. \qquad \checkmark$$

By performing *Buffon's needle experiment*, that is, by actually tossing a needle many (many!) times one could, in principle, obtain an approximation for π. In 1901, Italian mathematician Mario Lazzarini claimed to have done exactly that: tossing a needle 3408 times to obtain the approximation $355/113$, which is accurate to 6 decimal places! Alas, it is unlikely that he truly carried out the experiment [1].

8.3 Epilogue

We simply cannot let you go without mentioning a very cool solution to Glass Rod Problem A above [7]. We'll use the following fact from geometry, which is pretty neat all by itself.

Viviani's Theorem[4] *Let $\triangle$ be an equilateral triangle with height h, and let P be a point inside $\triangle$. Then the sum of the (perpendicular) distances from P to each of the three sides of $\triangle$ equals h.*

Proof. See Fig. 8.7. Let b be the length of each side of $\triangle$. Then the area of $\triangle$ is $A = \frac{1}{2}bh$. But if $\triangle$ is partitioned into three smaller triangles, with heights h_1, h_2, and h_3, then the area of $\triangle$ is also $A = \frac{1}{2}bh_1 + \frac{1}{2}bh_2 + \frac{1}{2}bh_3$. Therefore $\frac{1}{2}bh = \frac{1}{2}bh_1 + \frac{1}{2}bh_2 + \frac{1}{2}bh_3 \Rightarrow h = h_1 + h_2 + h_3$. □

Another Solution to Glass Rod Problem A Since the sum of the lengths of the three pieces is 1, using Viviani's Theorem we may interpret the two breaks of the glass rod into three pieces as a unique point P in an equilateral triangle with height $h = 1 = h_1 + h_2 + h_3$. For the three pieces to form a triangle, it is necessary and sufficient that $h_1 \leq 1/2$, $h_2 \leq 1/2$, and $h_3 \leq 1/2$. This means that P must be a point in the smaller shaded triangle in Fig. 8.8.

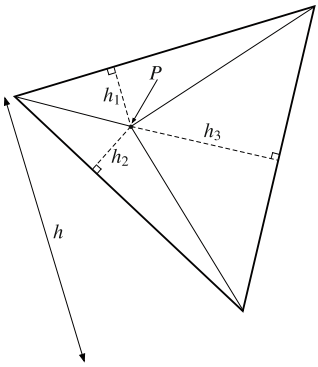


Fig. 8.7 $h = h_1 + h_2 + h_3$. Cool

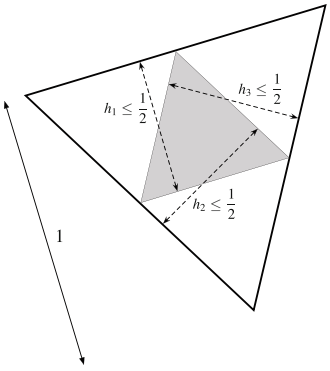


Fig. 8.8 Region $h_1 \leq \frac{1}{2}$, $h_2 \leq \frac{1}{2}$, $h_3 \leq \frac{1}{2}$, in an equilateral triangle, height 1

[4] Vincenzo Viviani (Italy, 1622–1703) was a student of Galileo.

The area of the shaded triangle is one quarter of the area of the entire triangle, so the probability that the three pieces can form a triangle is $1/4$. Nice. ✓

Exercises

8.1 ([2]). Three angles x, y, z are chosen randomly in $[0, \pi]$ to form a triangle, that is, $x + y + z = \pi$. What is the probability that the triangle is acute?

8.2 ([2]). Three points are chosen on the circumference of a circle, and a triangle is thus formed. What is the probability that the triangle contains the circle's center?

8.3 ([8]).

(a) Two glass rods of length 1 are *each* broken randomly to obtain pieces having lengths x, y, z, w. What is the probability that we can form a triangle using just the first three pieces—the ones with lengths x, y, z? [Answer: $1/2$.]

(b) What is the probability that we can form a triangle with any three pieces? [Answer: $1/8$.]

8.4. A glass rod of length 1 is broken randomly into two pieces—left and right, say. The left piece is then randomly broken into two pieces. What is the probability that we can form a triangle with these three pieces?

8.5. This is a problem from the 1993 Putnam Competition—North America's preeminent mathematics competition for undergraduates. Suppose that two real numbers p and q are chosen randomly in the interval $(0, 1)$. What is the probability that the integer m closest to p/q is even? [Answer: $5/4 - \pi/4 \approx 0.6440$. Hint: You'll need the Madhavan Series (7.2).]

8.6. Solve Buffon's needle problem but with a needle length of L and with a being the distance between the parallel lines. [Answer: $2L/\pi a$.]

8.7 ([4]). An equilateral triangle with side length 1 is dropped randomly on a plankwork of parallel lines which are distance 1 apart. What is the probability that the landed triangle touches one of the lines?

8.8 ([5]).

(a) A needle of length 1 is dropped randomly on a (huge) grid, that is, a set of parallel lines of length 1 apart, all perpendicular to another set of parallel lines length 1 apart. What is the probability that the landed needle touches one of the grid lines? [Answer: $3/\pi$.]

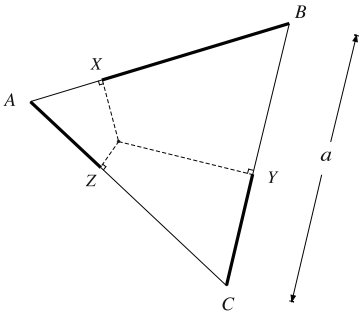


Fig. 8.9 (For Exercise 8.10) Equilateral triangle; $XB + YC + ZA = 3a/2$

(b) How about a needle of length L on a grid with parallel lines of length a apart, perpendicular to parallel lines length b apart (with $L < a$ and $L < b$)? [Answer: $2L(a + b - (1/2)r)/\pi ab$.]

8.9 ([6]). (Bertrand's[5] Paradox: There are several pretty convincing solutions to this problem; only one is correct.) A straight line is drawn randomly through the unit circle. What is the probability that the line's intercepted length with the circle has length greater than $\sqrt{3}$?

8.10 (E.g., [10]). As regards Viviani's Theorem: For the equilateral triangle in Fig. 8.9, with side length a, show that $XB + YC + ZA = 3a/2$. (That this sum is constant was conjectured in 2003 by an 11th grade student in Cape Town, named Duncan Clough. That it is constant, and is $3a/2$, was proved soon after by de Villers [3].)

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[5] Named for Joseph Bertrand (France, 1822–1900), not to be confused with British mathematician and logician Bertrand Russell (1872–1970), who had a paradox of his own...

Check for updates

Convexity—Chords

9

Recall that if $f''(x) \geq 0$ for all $x \in I$, then f is *convex* on I, and if $f'' > 0$, then f is *strictly convex* there. And f is (strictly) concave $\Leftrightarrow -f$ is (strictly) convex. Many of the everyday functions that we meet in calculus are either convex or concave, at least when restricted to some particular interval(s).

In Sect. 6.1 we obtained the following useful result, which says that the graph of a convex function lies above (or on) its tangent lines.

Convex Tangents Lemma *Let f be such that $f'' \geq 0$ on an open interval I and let $x_0 \in I$. Then for each $x \in I$, $f(x) \geq f(x_0) + f'(x_0)(x - x_0)$.*

This is illustrated (again) in Fig. 9.1. It is also useful to observe that the graph of a convex function lies below (or on) any chord drawn from one point on the graph to any other point on the graph. See Fig. 9.2.

In this chapter we shall verify this observation and then see that it too has far-reaching consequences. Again, Mr. Jensen was not mistaken.

9.1 Convexity and Chords

Let $x, y \in \mathbb{R}$ with $x < y$. For $t \in [0, 1]$, the point $(1 - t)x + ty$ divides the interval $[x, y]$ in the ratio $t : (1 - t)$. For example, $t = 0$ gives x; $t = 1/4$ gives $\frac{3}{4}x + \frac{1}{4}y$; $t = 1/2$ gives the midpoint $\frac{x+y}{2}$; $t = 4/5$ gives $\frac{1}{5}x + \frac{4}{5}y$; $t = 1$ gives the right endpoint y. See Fig. 9.3.

With this *parametrization* in place, we prove the following, thus verifying the observation illustrated in Fig. 9.2. We add (yet) another picture (Fig. 9.4), to go along with the proof.

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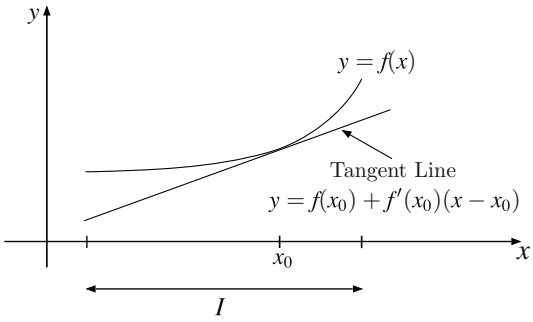


Fig. 9.1 The graph of a convex function lies above (or on) its tangent lines

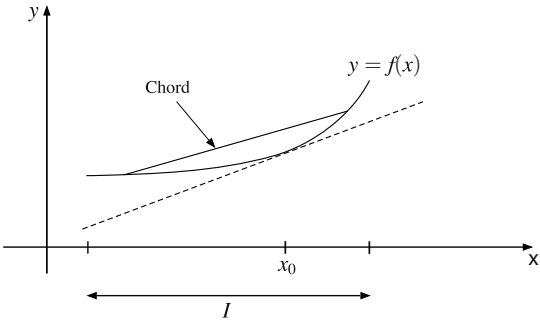


Fig. 9.2 The graph of a convex function lies below (or on) its chords

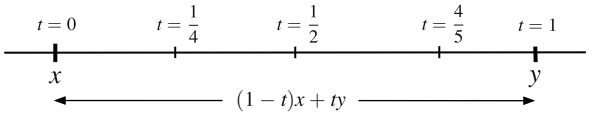


Fig. 9.3 $0 \le t \le 1$ gets us anywhere between x and y

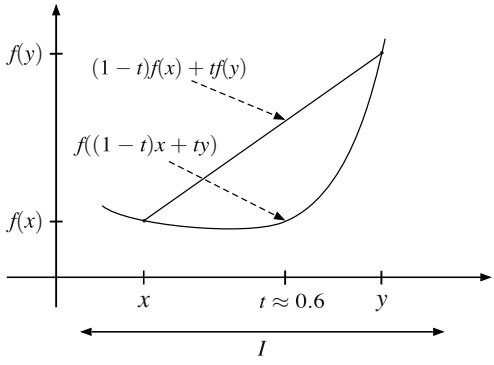


Fig. 9.4 For f convex and $t \in [0, 1]$, $f((1-t)x + ty) \le (1-t)f(x) + tf(y)$

Convex Chords Lemma *Let f be convex on some interval I. Then for any pair of points $x < y$ in I, we have*

$$f((1-t)x + ty) \le (1-t)f(x) + tf(y) \quad \textit{for every } t \in [0, 1].$$

Proof. Let $x < y$ in I, let $t \in [0, 1]$, and set $z = (1-t)x + ty$ (somewhere between x and y). Then by the Convex Tangents Lemma, we have

$$f(x) \geq f(z) + f'(z)(x - z) \quad \text{and} \quad f(y) \geq f(z) + f'(z)(y - z).$$

Multiplying the first inequality by $(1 - t)$ and the second by t, and then adding, we get

$$\begin{aligned}(1-t)f(x) + tf(y) &\geq (1-t)f(z) + (1-t)f'(z)(x-z) + tf(z) + tf'(z)(y-z) \\ &= f(z) + f'(z)\big[(1-t)x + ty - \big((1-t)z + tz\big)\big] \\ &= f(z) + f'(z)\big[z - z\big] = f(z) = f((1-t)x + ty).\end{aligned}$$

□

The following useful pair of inequalities is named for C. Jordan.[1]

$$\frac{2}{\pi}x \leq \sin(x) \leq x \qquad \text{for } x \in [0, \pi/2].$$

The right-hand inequality is true for all $x \geq 0$.

Jordan's Inequalities are illustrated in Fig. 9.5. Jordan's Inequality (singular) is just the one on the left.

Proof. The function $f(x) = \sin(x)$ is concave on $[0, \pi/2]$: $f''(x) = -\sin(x) \leq 0$ there. By the Convex Tangents Lemma, the graph of f lies below its tangent at $x = 0$; this is the right-hand inequality. By the Convex Chords Lemma, the graph of

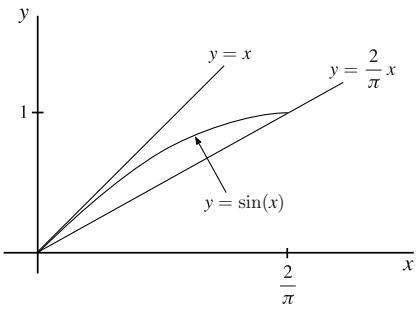


Fig. 9.5 Jordan's Inequalities: $\frac{2}{\pi}x \leq \sin(x) \leq x$, for $x \in [0, \pi/2]$

[1] Camille Jordan (France, 1838–1922) was responsible for many groundbreaking results in analysis and algebra, not just this wee inequality! (But it is nice to have an inequality named for you.)

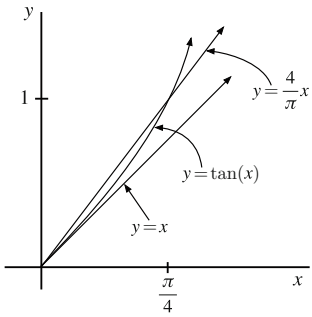


Fig. 9.6 Just like Jordan: $x \le \tan(x) \le \frac{4}{\pi}x$, for $x \in [0, \pi/4]$

f lies above the chord between the points $(0, 0)$ and $(\pi/2, \sin(\pi/2)) = (\pi/2, 1)$. This chord has equation $y = \frac{2}{\pi}x$, which gives the left-hand inequality. □

Example
Another pair of inequalities, very similar in spirit to Jordan's, is often useful:

$$x \le \tan(x) \le \frac{4}{\pi}x \qquad \text{for } x \in [0, \pi/4]. \tag{9.1}$$

The left-hand inequality is true for $x \in [0, \pi/2)$. These are illustrated in Fig. 9.6.

The proof is much the same as for Jordan's Inequalities—it's contained in Exercise 9.1. [You really should do this one—we shall use it in Chap. 14.]

9.2 Jensen's Inequality

If we take $t = 1/2$ in the Convex Chords Lemma, we get

$$f\left(\tfrac{x+y}{2}\right) \le \frac{f(x) + f(y)}{2}.$$

The famous Jensen's Inequality extends this much further. The idea is that the two points x, y and their average and $f(x)$, $f(y)$ and *their* average can be replaced by any number of points $x_1, x_2, \ldots, x_n$ and their average and $f(x_1), f(x_2), \ldots, f(x_n)$ points and *their* average. Whew!

Our proof of Jensen's Inequality is a very natural extension of our proof of the Convex Chords Lemma.

Jensen's Inequality *Let f be convex on I. For $n \ge 2$, let $x_1, x_2, \ldots, x_n \in I$. Then*

$$f\left(\frac{1}{n}\sum_{j=1}^{n} x_j\right) \le \frac{1}{n}\sum_{j=1}^{n} f(x_j).$$

Proof. Let $A = \frac{1}{n}\sum_{j=1}^{n} x_j$. This is the average of all the x_js, and so is in I. By the Convex Tangents Lemma, for each x_j we have

$$f(x_j) \;\geq\; f(A) + f'(A)(x_j - A).$$

Now multiplying by $1/n$ and summing from 1 to n, we get

$$\begin{aligned}\frac{1}{n}\sum_{j=1}^{n} f(x_j) &\geq \frac{1}{n}\sum_{j=1}^{n} f(A) \;+\; \frac{1}{n}\sum_{j=1}^{n} f'(A)(x_j - A)\\ &= \frac{1}{n}\sum_{j=1}^{n} f(A) \;+\; f'(A)\,\frac{1}{n}\sum_{j=1}^{n}(x_j - A)\\ &= \frac{1}{n}\sum_{j=1}^{n} f(A) \;+\; f'(A)(0).\end{aligned}$$

That is, $\frac{1}{n}\sum_{j=1}^{n} f(x_j) \;\geq\; \frac{1}{n}\sum_{j=1}^{n} f(A) \;=\; \frac{1}{n}nf(A) = f\big(\frac{1}{n}\sum_{j=1}^{n} x_j\big)$, as desired. □

Many important—and obscure!—inequalities follow from Jensen's. Here follows probably its best-known consequence. But first we need just a little background.

Let $x_1,\, x_2,\, \ldots,\, x_n \;>\; 0$. Their *Arithmetic Mean* is their usual average

$$A \;=\; A(x_1,\, x_2,\, \cdots,\, x_n) \;=\; \frac{1}{n}\sum_{j=1}^{n} x_j\,,$$

and their *Geometric Mean* is

$$G \;=\; G(x_1,\, x_2,\, \cdots,\, x_n) \;=\; \prod_{j=1}^{n} x_j^{\,1/n}\,.$$

[Notice that $\ln(G(x_1,\, x_2,\, \cdots,\, x_n)) \;=\; A(\ln(x_1),\, \ln(x_2),\, \cdots,\, \ln(x_n))$.]

The AM-GM Inequality[2] *For any* $x_1,\, x_2,\, \ldots,\, x_n \;>\; 0$*, we have* $G \leq A$*. And equality occurs here* $\Leftrightarrow\; x_1 \;=\; x_2 \;=\; \cdots \;=\; x_n$.

[2] The first proof of this was by Cauchy (1821). It's tricky but a beauty. Maclaurin had one earlier (1729) that wasn't quite correct. Cauchy's proof inspired Jensen's proof of Jensen's Inequality.

Proof. Since e^x is convex, Jensen's Inequality gives

$$e^{\left(\frac{1}{n}\sum_{j=1}^{n} t_j\right)} \leq \frac{1}{n}\sum_{j=1}^{n} e^{t_j},$$

or

$$\prod_{j=1}^{n} e^{\frac{1}{n}t_j} \leq \frac{1}{n}\sum_{j=1}^{n} e^{t_j}.$$

Setting $x_j = e^{t_j}$, that is, $t_j = \ln(x_j)$, this reads

$$\prod_{j=1}^{n} x_j^{\frac{1}{n}} \leq \frac{1}{n}\sum_{j=1}^{n} x_j,$$

as desired. The equality condition holds since e^x is in fact *strictly* convex. □

9.3 Means Means Means

The AM-GM Inequality is part of a much bigger picture, as follows: Again, let $x_1, x_2, \ldots, x_n > 0$ and $p \in (-\infty, 0) \cup (0, \infty)$. The Hölder Mean[3] or *Power Mean* M_p is given by

$$M_p = M_p(x_1, x_2, \ldots, x_n) = \left(\frac{1}{n}\sum_{j=1}^{n} {x_j}^p\right)^{\frac{1}{p}}.$$

Clearly $p = 1$ gives the Arithmetic Mean $A = M_1$. Less obvious (this is Exercise 9.9) is that, by L'Hospital's Rule,

$$\lim_{p\to 0} M_p(x_1, x_2, \ldots, x_n) = G(x_1, x_2, \ldots, x_n) \quad \text{(the Geometric Mean)}.$$

So for the sake of continuity (suppressing the x_js), we take $M_0 = G$.

At the other extremes (also in Exercise 9.9), again taking limits, we get

$$M_\infty(x_1, x_2, \ldots, x_n) = \lim_{p\to\infty} M_p(x_1, x_2, \ldots, x_n) = \max\{x_1, x_2, \ldots, x_n\}$$

and

[3] Otto Hölder (Germany, 1859–1937) was the one who really put these means on the map.

$$M_{-\infty}(x_1,\, x_2,\, \ldots,\, x_n) \;=\; \lim_{p\to-\infty} M_p(x_1,\, x_2,\, \ldots,\, x_n) \;=\; \min\{x_1,\, x_2,\, \ldots,\, x_n\}.$$

So using this notation, we have that (again suppressing the x_js)

$$M_{-\infty} \;\le\; M_0 \;\le\; M_1 \;\le\; M_\infty\,.$$

But using Jensen's Inequality, we can say much more.

Increasing Means Theorem *For fixed $x_1,\, x_2,\, \ldots,\, x_n \;>\; 0$, the means M_p are increasing, for $p \in (-\infty, \infty)$.*

Proof. Let $0 \;<\; r \;<\; s$. Then $f(x) \;=\; x^{\,s/r}$ is a convex function. So by Jensen's Inequality,

$$\left(\frac{1}{n}\sum_{j=1}^{n} x_j^r\right)^{s/r} \;\le\; \frac{1}{n}\sum_{j=1}^{n}\left(x_j^r\right)^{s/r}.$$

That is,

$$\left(\frac{1}{n}\sum_{j=1}^{n} x_j^r\right)^{1/r} \;\le\; \left(\frac{1}{n}\sum_{j=1}^{n} x_j^s\right)^{1/s},$$

or $M_r \;\le\; M_s$. Therefore M_p is increasing, for $p \in (0, \infty)$. For $p \in (-\infty, 0)$, we observe

$$M_{-p}(1/x_1,\, 1/x_2,\, \cdots,\, 1/x_n) \;=\; \frac{1}{M_p(x_1, x_2, \ldots, x_n)}\,,$$

and appeal to the case $p \in (0, \infty)$. □

According to [3], the Increasing Means Theorem was discovered by Oskar Schlömilch (Germany, 1823–1901), whose proof used Bernoulli's Inequality (Sect. 6.1) rather than Jensen's.

A few other means which get a name are the Harmonic Mean[4] M_{-1}, the Lorentz Mean $M_{1/3}$, and the Root Mean Square M_2. The Increasing Means Theorem really puts these means in their place!

[4] In the Harmonic Series, each term (after the first) is the Harmonic Mean of the term just before it and the term just after it—same deal for a geometric series and the Geometric Mean.

Exercises

9.1.

(a) Use Jordan's Inequality to show that for $x \in [0, \pi/2]$,

$$1 - \frac{2}{\pi}x \le \cos(x) \le \frac{\pi}{2} - x.$$

(b) Show that $x \le \tan(x)$ for $x \in [0, \pi/2)$.

(c) Show that $\cos(x) \le \dfrac{1}{\sqrt{1+x^2}}$ for $x \in [0, \pi/2]$.

(d) Let $\beta \in (0, \pi/2)$. Show that $\tan(x) \le \dfrac{\tan(\beta)}{\beta}x$, for $x \in [0, \beta]$.

9.2. Let f be convex on I and let $a < x < b$ belong to I. Show that

$$\frac{f(x) - f(a)}{x - a} \le \frac{f(b) - f(a)}{b - a} \le \frac{f(b) - f(x)}{b - x}.$$

Draw a picture which shows what these inequalities say.
[Hint: Write $x = (1 - \frac{x-a}{b-a})a + \frac{x-a}{b-a}\,b$, and notice that $\frac{x-a}{b-a} \in (0, 1)$.]

9.3.

(a) Verify that the outside inequality, in $M_{-1} \le M_0 \le M_1$, can be written (very agreeably) as $\displaystyle\sum_{j=1}^{n} x_j \sum_{j=1}^{n} \frac{1}{x_j} \ge n^2$.

(b) Apply this to the *three* numbers $a + b$, $a + c$, and $b + c$ to prove **Nesbitt's Inequality**:

$$\frac{a}{b+c} + \frac{b}{a+c} + \frac{c}{a+b} \ge \frac{3}{2}.$$

9.4. Prove the AM-GM Inequality using the concavity of $\ln(x)$, rather than—as we did above—the convexity of e^x.

9.5 ([2]). Fill in the details, as follows, of G. Pólya's wonderful proof of the AM-GM Inequality.[5] This is one that everybody should know.

(a) Set $x = x_j/A - 1$, in $1 + x \le e^x$, to get $x_j/A \le e^{x_k/A - 1}$. Conclude

[5] George Pólya (Hungarian-American, 1887–1985) said that this proof came to him in a dream.

$$\left(\frac{x_j}{A}\right)^{1/n} \;\le\; \mathrm{e}^{x_k/(nA)-1/n}.$$

(b) Take products on both sides, from $j = 1$ to n, to get

$$\prod_{j=1}^{n}\left(\frac{x_j}{A}\right)^{1/n} \;\le\; \prod_{j=1}^{n}\mathrm{e}^{x_k/(nA)-1/n}.$$

(c) Show that the left-hand side is G/A and the right-hand side is 1. Done!

9.6 ([2]). Prove the AM-GM Inequality using $\ln(x) \le x - 1$ for $x > 0$, by first setting $x = x_j/A$. This is the logarithmic version, due to Frigyes Riesz (Hungary, 1880–1956), of Pólya's proof in the exercise above.

9.7 ([5]). Fill in the details of another slick proof of the AM-GM Inequality:

(a) We have seen (Exercise 6.4) that $x^{\mathrm{e}} \le \mathrm{e}^x$. Substitute $x = x_j\mathrm{e}/G$ into this, for $j = 1, 2, \ldots n$, and multiply the n resulting inequalities together to get

$$\mathrm{e}^{\frac{\mathrm{e}}{G}\sum_{j=1}^{n} x_j} \;\ge\; \left(\frac{x_1\mathrm{e}}{G}\cdot\frac{x_2\mathrm{e}}{G}\cdot\cdots\cdot\frac{x_n\mathrm{e}}{G}\right)^{\mathrm{e}}.$$

(b) Show that the left-hand side is $\mathrm{e}^{n\mathrm{e}A/G}$ and the right-hand side is $\mathrm{e}^{n\mathrm{e}}$.

So $A/G \ge 1$ and that's it!

9.8 ([4]). Bernoulli's Inequality in its earliest form (Exercise 6.5) is as follows: *For* $x \ge 0$,

$$x^n \ge 1 + n(x-1) \quad \textit{for } n \in \mathbb{N}.$$

(a) Set $x = G_n/G_{n-1}$ here; then sum from $n = 1$ to N to prove (again!) the AM-GM Inequality. [Hint: Verify first that $(G_n/G_{n-1})^n = x_n/G_{n-1}$.]

(b) Instead set $x = \left(1 + \frac{1}{n}\right)G_n/G_{n-1}$ in Bernoulli's Inequality as above; then have $N \to \infty$ to obtain *Carleman's Inequality* [1]: *For* $x_1, x_2, x_3, \ldots > 0$, *we have*

$$\sum_{n=1}^{\infty} G_n \;<\; \mathrm{e}\sum_{n=1}^{\infty} x_n\,.$$

(c) Show that the e on the right-hand side above is the best possible constant.

9.9. Let $x_1, x_2, \ldots, x_n > 0$.

(a) Verify that the Geometric Mean

$$G = G(x_1, x_2, \ldots, x_n) = \lim_{p \to 0} M_p(x_1, x_2, \ldots, x_n).$$

(b) Verify that

$$\lim_{p \to \infty} M_p(x_1, x_2, \ldots, x_n) = \max\{x_1, x_2, \ldots, x_n\}$$

and that

$$\lim_{p \to -\infty} M_p(x_1, x_2, \ldots, x_n) = \min\{x_1, x_2, \ldots, x_n\}.$$

9.10.

(a) Prove the following more general version of Jensen's Inequality: Let f be convex on (a, b). For $n \geq 2$, let $x_1, x_2, \ldots, x_n \in (a, b)$. Let $w_1, w_2, \ldots, w_n$ be a set of *weights*: each $w_j > 0$, with $\sum_{j=1}^{n} w_j = 1$. Then

$$f\Big(\sum_{j=1}^{n} w_j x_j\Big) \leq \sum_{j=1}^{n} w_j f(x_j).$$

(b) State and prove the corresponding version of the Increasing Means Theorem.

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Interlude: Minkowski Distance

10

The xy-plane has what is called a *vector space structure*, by way of addition and scalar multiplication. These two operations enable us to sort of get around the plane: Let $\mathbf{u} = (x_1, y_1)$, $\mathbf{v} = (x_2, y_2)$, and $\alpha \in \mathbb{R}$. Then

$$\mathbf{u} + \mathbf{v} = (x_1 + x_2,\ y_1 + y_2) \quad \text{and} \quad \alpha\mathbf{v} = (\alpha x_1,\ \alpha y_1).$$

As is customary, we denote by $\mathbf{0} = (0, 0)$ the origin. The *Euclidean distance* in the plane, between $\mathbf{u}$ and $\mathbf{v}$, is

$$d(\mathbf{u}, \mathbf{v}) = d(\mathbf{0}, \mathbf{v} - \mathbf{u}) = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}\,.$$

This definition arises directly from the Pythagorean Theorem. The (closed) *unit disk in the Euclidean plane* is the set

$$\triangle = \big\{\mathbf{w} : d(\mathbf{0}, \mathbf{w}) \le 1\big\} = \big\{(x, y) : x^2 + y^2 \le 1\big\}.$$

This is just the unit circle, together with its interior. But the Euclidean distance above is not the only distance in town...

For $p > 0$, consider

$$d_p(\mathbf{u}, \mathbf{v}) = d_p(\mathbf{0}, \mathbf{v} - \mathbf{u}) = \Big[\,|x_2 - x_1|^p + |y_2 - y_1|^p\,\Big]^{1/p}.$$

Evidently, the Euclidean distance d above is d_p with $p = 2$. So the unit disk using d_p, that is, the (closed) *unit disk in the Minkowski plane*,[1] is the set

[1] Hermann Minkowski (Germany, 1864–1909) was Einstein's professor. Minkowski made major contributions to the Theory of Relativity, along with this Einstein fellow.

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$$\triangle_p \;=\; \big\{\mathbf{w} : d_p(\mathbf{0}, \mathbf{w}) \le 1\big\} \;=\; \big\{(x, y) : |x|^p + |y|^p \le 1\big\}.$$

In this interlude we study the d_ps and their associated unit disks $\triangle_p$. We'll see how they relate to some of the things that we already know.

We shall rely on the fact (verify!) that for $\alpha \in \mathbb{R}$, we have

$$d_p(\alpha\mathbf{u}, \alpha\mathbf{v}) \;=\; |\alpha|\, d_p(\mathbf{0}, \mathbf{v} - \mathbf{u}). \tag{10.1}$$

We shall also rely on the following: For $p \ge 1$, the function $f(x) = x^p$ is convex on $[0, \infty)$. So the Convex Chords Lemma (Sect. 9.1) says that for $a, b \ge 0$

$$\big((1-t)a + tb\big)^p \le (1-t)a^p + tb^p \quad \forall\, t \in [0, 1]. \tag{10.2}$$

10.1 Convex Sets

A *set* S in the plane is *convex* if whenever $\mathbf{u}$ and $\mathbf{v}$ are in S, then the entire line segment joining $\mathbf{u}$ and $\mathbf{v}$ is also in S. That is,

$$\mathbf{u},\, \mathbf{v} \in S \;\Rightarrow\; (1-t)\mathbf{u} + t\mathbf{v} \;\in\; S \quad \forall\, t \in [0, 1].$$

In Fig. 10.1, the first two sets are convex. The third is not: the indicated line segment exits (and reenters) the set depicted.

The relationship between convex sets and convex functions is as follows: The *epigraph* of any function f defined on $[a, b]$ is the set

$$E_f \;=\; \big\{(x, y) : a \le x \le b \text{ and } f(x) \le y\big\}.$$

If a function is convex, then its epigraph is a convex set. See Fig. 10.2. (The boundary of a convex set is, at least in bits, the graph of a convex function.)

Again for $p > 0$, consider d_p and $\triangle_p$ as defined above. See Fig. 10.3 for unit disks $\triangle_p$ for various values of p. For example, $\triangle_2$ is the boundary and interior of the unit circle; $\triangle_1$ is the boundary and interior of a diamond; $\triangle_{2/3}$ is boundary and interior of an *astroid*, a curve you may have met in calculus.

Fig. 10.1 The first two sets are convex. The third is not

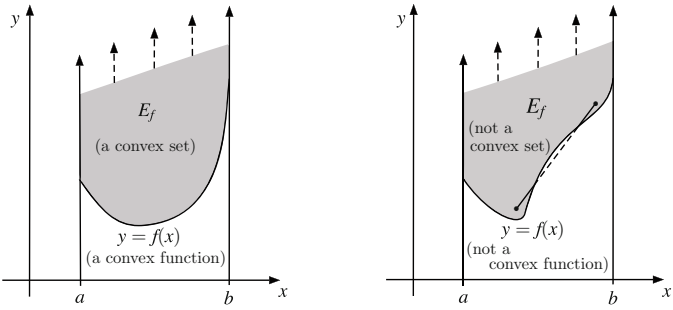


Fig. 10.2 Epigraph of a convex function; epigraph of a non-convex function

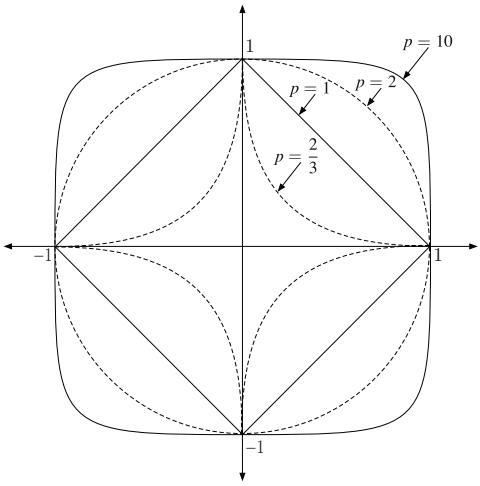


Fig. 10.3 Unit disks $\triangle_p$, for four values of p

Figure 10.3 seems to suggest that the unit disks are convex sets precisely when $p \geq 1$. This is indeed the case, as the following two lemmas show.

Lemma 10.1
If $p \geq 1$, then the unit disk $\triangle_p$ is a convex set.

Proof. Let each of $\mathbf{u} = (x_1, y_1) \in \triangle_p$ and $\mathbf{v} = (x_2, y_2) \in \triangle_p$, so that

$$d_p(\mathbf{0}, \mathbf{u}) \leq 1 \quad \text{and} \quad d_p(\mathbf{0}, \mathbf{v}) \leq 1.$$

We want to show that for any $t \in [0, 1]$, the point $(1 - t)\mathbf{u} + t\mathbf{v}$ remains in $\triangle_p$. So we compute how far such a point can possibly get from the origin $\mathbf{0}$. To avoid the p-th root as long as possible, it is convenient to take p-th powers:

$$\left[\, d_p(\mathbf{0}, (1-t)\mathbf{u} + t\mathbf{v}) \,\right]^p \;=\; \left|(1-t)x_1 + tx_2\right|^p \;+\; \left|(1-t)y_1 + ty_2\right|^p.$$

Then by (10.2) we have

$$\begin{aligned}
&\left|(1-t)x_1 + tx_2\right|^p + \left|(1-t)y_1 + ty_2\right|^p \\
&\qquad\qquad \leq \;\; (1-t)|x_1|^p + t|x_2|^p + (1-t)|y_1|^p + t|y_2|^p \\
&\qquad\qquad = \;\; (1-t)(|x_1|^p + |y_1|^p) + t(|x_2|^p \;+\; |y_2|^p) \\
&\qquad\qquad = \;\; (1-t)\left[\, d_p(\mathbf{0}, \mathbf{u}) \,\right]^p \;+\; t\left[\, d_p(\mathbf{0}, \mathbf{v}) \,\right]^p \\
&\qquad\qquad \leq \;\; (1-t)(1) \;+\; t(1) \;=\; 1.
\end{aligned}$$

Therefore $d_p(\mathbf{0}, (1-t)\mathbf{u} + t\mathbf{v}) \leq 1$, and so $(1-t)\mathbf{u} + t\mathbf{v} \in \triangle_p$, as desired. □

Lemma 10.2
If $p < 1$, then the unit disk $\triangle_p$ is not a convex set.

Proof. Consider the points $\mathbf{u} = (1, 0)$ and $\mathbf{v} = (0, 1)$, each in $\triangle_p$. Then the midpoint $\frac{\mathbf{u}+\mathbf{v}}{2}$ of the line segment between $\mathbf{u}$ and $\mathbf{v}$ satisfies

$$\left[\, d_p(\mathbf{0}, \tfrac{\mathbf{u}+\mathbf{v}}{2}) \,\right]^p \;=\; \left[\, d_p(\mathbf{0}, (\tfrac{1}{2}, \tfrac{1}{2})) \,\right]^p \;=\; \left(\tfrac{1}{2}\right)^p + \left(\tfrac{1}{2}\right)^p \;=\; 2\left(\tfrac{1}{2}\right)^p \;=\; \left(\tfrac{1}{2}\right)^{p-1}.$$

Now when $p < 1$, $\left(\frac{1}{2}\right)^{p-1} > 1$. This means that $\frac{\mathbf{u}+\mathbf{v}}{2}$ lies *outside* $\triangle_p$. Therefore $\triangle_p$ is not a convex set. □

10.2 The Triangle Inequality

The Euclidean distance is officially called a *distance* because it satisfies the following requirements, for any $\mathbf{u}$ and $\mathbf{v}$ in the plane:

(i) $d(\mathbf{u}, \mathbf{v}) \geq 0$, and $d(\mathbf{u}, \mathbf{v}) = 0 \;\Leftrightarrow\; \mathbf{u} = \mathbf{v}$,
(ii) $d(\mathbf{u}, \mathbf{v}) = d(\mathbf{v}, \mathbf{u})$,
(iii) $d(\mathbf{0}, \mathbf{u} + \mathbf{v}) \;\leq\; d(\mathbf{0}, \mathbf{u}) + d(\mathbf{0}, \mathbf{v})$.

That the d_ps satisfy items (i) and (ii) is clear enough. Less clear is whether they satisfy item (iii), which is called the *Triangle Inequality*. In settling this, we shall see that we have done most of the work already.

Lemma 10.3
If $p \geq 1$, then d_p satisfies the Triangle Inequality. (So d_p is a distance.)
[Here, the Triangle Inequality is often called *Minkowski's Inequality*.]

Proof. For $\mathbf{u} = (x_1, y_1)$ and $\mathbf{v} = (x_2, y_2)$ (neither of which is $\mathbf{0}$; else there's nothing to prove), set

$$\widehat{\mathbf{u}} = \frac{\mathbf{u}}{d_p(\mathbf{0}, \mathbf{u})}, \qquad \widehat{\mathbf{v}} = \frac{\mathbf{v}}{d_p(\mathbf{0}, \mathbf{v})}.$$

Then $\widehat{\mathbf{u}} \in \triangle_p$ and $\widehat{\mathbf{v}} \in \triangle_p$. Therefore, since $\triangle_p$ is a convex set for $p \geq 1$,

$$d_p(\mathbf{0}, (1-t)\widehat{\mathbf{u}} + t\widehat{\mathbf{v}}) \leq 1 \quad \text{for any } t \in [0, 1].$$

Setting $t = \frac{d_p(\mathbf{0},\mathbf{v})}{d_p(\mathbf{0},\mathbf{u})+d_p(\mathbf{0},\mathbf{v})}$, so that $1 - t = \frac{d_p(\mathbf{0},\mathbf{u})}{d_p(\mathbf{0},\mathbf{u})+d_p(\mathbf{0},\mathbf{v})}$, we get

$$d_p\left(\mathbf{0}, \frac{d_p(\mathbf{0},\mathbf{u})}{d_p(\mathbf{0},\mathbf{u})+d_p(\mathbf{0},\mathbf{v})}\widehat{\mathbf{u}} + \frac{d_p(\mathbf{0},\mathbf{v})}{d_p(\mathbf{0},\mathbf{u})+d_p(\mathbf{0},\mathbf{v})}\widehat{\mathbf{v}}\right) \leq 1. \tag{10.3}$$

And we're finished because $d_p(\mathbf{0}, \mathbf{u})\,\widehat{\mathbf{u}} = \mathbf{u}$ and $d_p(\mathbf{0}, \mathbf{v})\,\widehat{\mathbf{v}} = \mathbf{v}$. Indeed, using (10.1), (10.3) reads

$$\frac{1}{d_p(\mathbf{0}, \mathbf{u}) + d_p(\mathbf{0}, \mathbf{v})} d_p(\mathbf{0}, \mathbf{u} + \mathbf{v}) \leq 1,$$

so that $d_p(\mathbf{0}, \mathbf{u} + \mathbf{v}) \leq d_p(\mathbf{0}, \mathbf{u}) + d_p(\mathbf{0}, \mathbf{v})$, as desired. □

Lemma 10.4
If $p < 1$, then d_p does not satisfy the Triangle Inequality. (So d_p is not a distance.)

Proof. Consider the points $\mathbf{u} = (1, 0)$, $\mathbf{v} = (0, 1)$, so $\mathbf{u} + \mathbf{v} = (1, 1)$. Here,

$$d_p(\mathbf{0}, \mathbf{u}) + d_p(\mathbf{0}, \mathbf{v}) = 1 + 1 = 2.$$

However,

$$d_p(\mathbf{0}, \mathbf{u} + \mathbf{v}) = \left(1^p + 1^p\right)^{1/p} = 2^{1/p}.$$

And again, $2^{1/p} > 2$ when $p < 1$. □

The Triangle Inequality from plane geometry says that for any triangle in the plane, the length of each side is less than or equal to the sum of the lengths of the other two sides. This actually holds in many geometries. In *Euclidean* geometry, it is the Pythagorean Theorem (one of the cornerstones!) which engenders the distance d_2 above, which, as we have seen, satisfies the Triangle Inequality. For d_1, the Triangle Inequality holds—with equality. The distance d_1 is often called the **taxicab distance**. See [4, 5], for example.

10.3 From One Extreme to the Other

First off, looking at the unit disks $\triangle_p$ in Fig. 10.3, it *appears* that $\triangle_p \subset \triangle_q$ when $p < q$. This is indeed the case:

Lemma 10.5
Let $\mathbf{w} = (x, y)$. *Then* $d_p(\mathbf{0}, \mathbf{w})$ *is a decreasing function of* p. *That is,* $p < q \Rightarrow d_q(\mathbf{0}, \mathbf{w}) \leq d_p(\mathbf{0}, \mathbf{w})$.

Proof. If either of x or y is zero, then $d_p(\mathbf{0}, \mathbf{w})$ is a constant function of p and we're good. By rotating the plane about $(0, 0)$ if necessary, we may suppose that $0 < x \leq y$. So with a clear conscience, we can write

$$d_p(\mathbf{0}, \mathbf{w}) = \left(x^p + y^p\right)^{1/p} = x\left(1 + \left(\frac{y}{x}\right)^p\right)^{1/p}.$$

Therefore it suffices to show that for *fixed* $t \geq 1$, the function f below decreases as p increases:

$$f(p) = \left(1 + t^p\right)^{1/p} \quad (p > 0).$$

Recall that $\frac{d}{dp} t^p = t^p \ln(t)$. Taking the logarithm of f and then differentiating with respect to p, we get

$$\begin{aligned} \frac{f'(p)}{f(p)} &= -\frac{1}{p^2} \ln(1 + t^p) + \frac{1}{p} \frac{1}{1 + t^p} t^p \ln(t) \\ &= -\frac{1}{p^2} \left[\ln(1 + t^p) - \frac{t^p}{1 + t^p} \ln(t^p) \right]. \end{aligned}$$

So now it suffices to show that what is in the square brackets $[\cdots]$ above satisfies $[\cdots] > 0$, for then we'll have $f'(p) < 0$, so that f is decreasing. It does: $u \ln(u)$ is increasing for $u \geq 1$, and $t^p \geq 1$, so that $(1 + t^p)\ln(1 + t^p) > t^p \ln(t^p)$. □

▶ **Remark** Lemma 10.5 is kind of an opposite of the Increasing Means Theorem from Sect. 9.3: *For fixed $x_1, x_2, \ldots, x_n > 0$, the means M_p increase with p.*

Looking again at Fig. 10.3, it also appears that the disks $\triangle_p$ are sort of filling out toward becoming a square, as $p \to \infty$. This is exactly what they are doing:

Lemma 10.6
$\triangle_\infty$ is the unit square centered at the origin.

Proof. Let $\mathbf{w} = (x, y)$. We show that

$$d_\infty(\mathbf{0}, \mathbf{w}) = \lim_{p\to\infty} d_p(\mathbf{0}, \mathbf{w}) = \max(|x|, |y|),$$

so that this is indeed the case. One of $|x|$ or $|y|$ is at least as large as the other, so we may as well suppose that $\max(|x|, |y|) = |x|$. Then

$$|x| = \Big[|x|^p\Big]^{1/p} \leq \Big[|x|^p + |y|^p\Big]^{1/p} \leq \Big[|x|^p + |x|^p\Big]^{1/p} = 2^{1/p}|x|.$$

Therefore, since $2^{1/p} \to 1$ as $p \to \infty$, we get $\lim_{p\to\infty} \Big[|x|^p + |y|^p\Big]^{1/p} = |x|$. That is,

$$\lim_{p\to\infty} d_p(\mathbf{0}, \mathbf{w}) = \max\big(|x|, |y|\big).$$

□

At the other extreme, Fig. 10.3 suggests that the disks $\triangle_p$ shrink down to a sort of plus sign, as $p \to 0$. They do:

Lemma 10.7
$\triangle_0$ is the union of $[-1, 1]$ on the x-axis and $[-1, 1]$ on the y-axis.

Proof. Again let $\mathbf{w} = (x, y)$. If $y = 0$, then $d_p(\mathbf{0}, \mathbf{w}) = \big[|x|^p + |0|^p\big]^{1/p} = |x|$, so that $d_p(\mathbf{0}, \mathbf{w}) \leq 1 \Leftrightarrow x \in [-1, 1]$. In the same way, $x = 0 \Leftrightarrow y \in [-1, 1]$.

Otherwise, for any other $\mathbf{w} = (x, y)$, we may suppose that $\min(|x|, |y|) = |x| \neq 0$. Then

$$d_0(\mathbf{0}, \mathbf{w}) = \lim_{p \to 0^+} \left[|x|^p + |y|^p \right]^{1/p} \geq \lim_{p \to 0^+} \left[2|x|^p \right]^{1/p} = \lim_{p \to 0^+} 2^{1/p} |x| = \infty,$$

so that $d_0(\mathbf{0}, \mathbf{w}) > 1$, whence $\mathbf{w} \notin \triangle_0$. □

▶ **Remarks** $\pi = 3.14159\ldots$ is, of course, the ratio of the circumference of $\triangle_2$ to the diameter of $\triangle_2$. (That π is even constant—that is, the same for all circles—is believable but not obvious. See [6, 8].) The circumference and diameter here are measured via d_2. If we measure the circumference and diameter of $\triangle_p$ ($p \geq 1$) via d_p, then it's a different story. Going from $p = 1$ to $p = \infty$, the value of π for $\triangle_p$ begins at 4 ($p = 1$), decreases to $\pi = 3.14159\ldots$ ($p = 2$), and then increases back up to 4 ($p = \infty$). Pretty neat. See [1–3, 7].

10.4 Epilogue

Suppose that $\mathbf{u} = (u_1, u_2, \ldots, u_n)$, $\mathbf{v} = (v_1, v_2, \ldots, v_n) \in \mathbb{R}^n$. Then for $p > 0$,

$$d_p(\mathbf{u}, \mathbf{v}) = d_p(\mathbf{0}, \mathbf{v} - \mathbf{u}) = \left[|v_1 - u_1|^p + |v_2 - u_2|^p + \cdots + |v_n - u_n|^p \right]^{1/p}.$$

Here, the (closed) unit ball in n-dimensional Minkowski space is the set

$$B_p = \left\{ \mathbf{u} : d_p(\mathbf{0}, \mathbf{u}) \leq 1 \right\} = \left\{ (u_1, u_2, \ldots, u_n) : |u_1|^p + |u_2|^p + \cdots + |u_n|^p \leq 1 \right\}.$$

Many of the properties that we have established extend to this setting, with appropriate modifications to their proofs. In particular, the d_ps are decreasing, and d_p is a distance $\Leftrightarrow$ B_p is a convex set $\Leftrightarrow$ $p \geq 1$.

But things get really interesting when we allow $n = \infty$. (And even more so if we also replace $\mathbb{R}$ with $\mathbb{C}$!) Then one must be very careful; many of the properties of the B_ps in $\mathbb{R}^n$ which we might take for granted are no longer true. And there are other important distances to consider, besides the d_ps. This is about where *functional analysis* begins...

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The Basel Problem

11

We have seen by now at least three proofs that the Harmonic Series diverges:

$$1 + \frac{1}{2} + \frac{1}{3} + \frac{1}{4} + \frac{1}{5} + \frac{1}{6} + \cdots = \sum_{n=1}^{\infty} \frac{1}{n} = \infty .$$

But in case you don't feel like flipping back at all, here's another [4].

If we suppose that $S = \sum_{n=1}^{\infty} \frac{1}{n}$ is some finite number, then

$$\begin{aligned} \frac{1}{2}S &= \frac{1}{2}\left(1 + \frac{1}{2} + \frac{1}{3} + \frac{1}{4} + \frac{1}{5} + \frac{1}{6} + \cdots\right) \\ &= \frac{1}{2} + \frac{1}{4} + \frac{1}{6} + \frac{1}{8} + \frac{1}{10} + \cdots . \end{aligned}$$

Therefore, looking at the rest of the series S, we must have

$$\frac{1}{2}S = 1 + \frac{1}{3} + \frac{1}{5} + \frac{1}{7} + \frac{1}{9} + \frac{1}{11} + \cdots .$$

Comparing each of these term by term,

$$\frac{1}{2} < 1 , \qquad \frac{1}{4} < \frac{1}{3} , \qquad \frac{1}{6} < \frac{1}{5} , \qquad \frac{1}{8} < \frac{1}{7} , \qquad \ldots . \text{etc.}$$

we get $\frac{1}{2}S < \frac{1}{2}S$, which is a problem. So we must have $S = \infty$. ✓

On the other hand, we have seen (Sect. 7.1) that $\sum_{n=1}^{\infty} \frac{1}{n^p}$ converges for $p > 1$. Here is a simple proof for $p = 2$ (and hence for $p \geq 2$):

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$$\sum_{n=1}^{N} \frac{1}{n^2} = 1 + \sum_{n=2}^{N} \frac{1}{n^2} < 1 + \sum_{n=2}^{N} \frac{1}{n(n-1)}$$

$$= 1 + \sum_{n=2}^{N} \left(\frac{1}{n-1} - \frac{1}{n} \right), \quad \text{and this sum telescopes:}$$

$$= 1 + 1 - \frac{1}{2} + \frac{1}{2} - \frac{1}{3} + \frac{1}{3} - \frac{1}{4} + \cdots - \frac{1}{N-1} + \frac{1}{N-1} - \frac{1}{N}$$

$$= 2 - \frac{1}{N}, \quad \text{which} \to 2 \ \text{ as } N \to \infty .$$

Therefore, indeed, $\sum_{n=1}^{\infty} \frac{1}{n^2}$ converges (and its sum is < 2). ✓

Originally posed by Mengoli[1] around 1650 was the thus reasonable question:

$$\textit{Qual é il valore di} \quad \sum_{n=1}^{\infty} \frac{1}{n^2} \ ?$$

Mengoli's question came to be known as the *Basel Problem*, named for the Swiss hometown of the mathematically formidable Bernoulli family, many of whom attempted (unsuccessfully) to answer it.[2] Other great minds tackled the problem as well—notably Stirling, de Moivre, and Leibniz—but the question was correctly answered by Euler, also of Basel, in 1734:[3]

$$\sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6} . \tag{11.1}$$

A good number of proofs of (11.1) have since been discovered, but many of them are quite difficult or are a consequence of very sophisticated ventures. The proof we present mainly follows [2], but see also [3, 5, 9, 10]. It leans, as many do, on the following computation.

[1] Pietro Mengoli: the summer of the Alternating Harmonic Series (Sect. 7.2).

[2] The Bernoullis (brothers Jacob and Johann, whom we have met, their sons/nephews Daniel, two Nicolauses, two more Johanns, and another Jacob) were a clan of great Swiss mathematicians.

[3] Euler was married that year too!

Lemma 11.1

$$\sum_{k\ odd} \frac{1}{k^2} \;=\; \frac{3}{4} \sum_{k=1}^{\infty} \frac{1}{k^2}\,.$$

Proof. Indeed,

$$\begin{aligned}
\sum_{k\ odd} \frac{1}{k^2} &= 1 + \frac{1}{3^2} + \frac{1}{5^2} + \frac{1}{7^2} + \cdots \\
&= \sum_{k=1}^{\infty} \frac{1}{k^2} - \left(\frac{1}{2^2} + \frac{1}{4^2} + \frac{1}{6^2} + \frac{1}{8^2} \cdots \right) \\
&= \sum_{k=1}^{\infty} \frac{1}{k^2} - \frac{1}{4} \left(1 + \frac{1}{4} + \frac{1}{3^2} + \frac{1}{4^2} + \cdots \right) \\
&= \sum_{k=1}^{\infty} \frac{1}{k^2} - \frac{1}{4} \sum_{k=1}^{\infty} \frac{1}{k^2} \;=\; \frac{3}{4} \sum_{k=1}^{\infty} \frac{1}{k^2}\,.
\end{aligned}$$

□

We shall also use, as we have done before, the important calculus limit:

$$\lim_{\theta \to 0} \frac{\sin(\theta)}{\theta} \;=\; 1\,. \tag{11.2}$$

11.1 The Basel Problem—Solution

The following extended trig identity shall be very useful.

Lemma 11.2

$$\frac{1}{2} + \sum_{k=1}^{n} \cos(k\pi x) \;=\; \frac{\sin\left(\frac{2n+1}{2}\pi x\right)}{2\sin\left(\frac{1}{2}\pi x\right)}\,. \tag{11.3}$$

Proof. Subtracting the familiar trig identities for $\sin(A+B)$ and $\sin(A-B)$, we get

$$\cos(A)\sin(B) \;=\; \frac{1}{2}\big[\sin(A+B) \;-\; \sin(A-B)\big].$$

In this, we substitute $A = k\pi x$ and $B = \frac{1}{2}\pi x$, to obtain

$$\begin{aligned}\cos(k\pi x)\sin\big(\tfrac{1}{2}\pi x\big) \;&=\; \frac{1}{2}\left[\sin\Big(k\pi x + \tfrac{1}{2}\pi x\Big) \;-\; \sin\big(k\pi x - \tfrac{1}{2}\pi x\big)\right]\\ &=\; \frac{1}{2}\left[\sin\Big(\tfrac{2k+1}{2}\pi x\Big) \;-\; \sin\Big(\tfrac{2k-1}{2}\pi x\Big)\right].\end{aligned}$$

Then summing over k, there is telescoping, which nicely gives

$$\sum_{k=1}^{n}\cos(k\pi x)\sin\big(\tfrac{1}{2}\pi x\big) \;=\; \frac{1}{2}\left[\sin\Big(\tfrac{2n+1}{2}\pi x\Big) \;-\; \sin\big(\tfrac{1}{2}\pi x\big)\right].$$

That is,

$$\sin\big(\tfrac{1}{2}\pi x\big)\sum_{k=1}^{n}\cos(kx\pi) \;+\; \frac{1}{2}\sin\big(\tfrac{1}{2}\pi x\big) \;=\; \frac{1}{2}\sin\Big(\tfrac{2n+1}{2}\pi x\Big).$$

And after a bit more tidying, the proof is complete. □

Now We Can Prove (11.1) Multiplying (11.3) by x and integrating, we get

$$\frac{1}{2}\int_0^1 x\,dx \;+\; \sum_{k=1}^{n}\int_0^1 x\cos(k\pi x)\,dx \;=\; \int_0^1 \frac{x}{2\sin\left(\frac{1}{2}\pi x\right)}\sin\Big(\tfrac{2n+1}{2}\pi x\Big)\,dx\,. \tag{11.4}$$

Something pretty good happens here:

$$\frac{1}{2}\int_0^1 x\,dx \;=\; \frac{1}{4}\,,$$

and for $k = 1, 2, 3, \cdots$, we have (see Sect. 2.2) that

$$\int_0^1 x\cos(k\pi x)\,dx \;=\; \begin{cases} 0 & \text{if } k \text{ is even}\\[2ex] -\dfrac{1}{\pi^2}\dfrac{2}{k^2} & \text{if } k \text{ is odd}\,. \end{cases}$$

Therefore, letting $n \to \infty$ in the left-hand side of (11.4), we get

$$\frac{1}{4} \; - \; \frac{2}{\pi^2} \sum_{k\,odd} \frac{1}{k^2} \, .$$

So if we can show that the right-hand side of (11.4) $\to \; 0$ as $n \; \to \infty,$ then we'll be all set because by Lemma 11.1 we have

$$\frac{1}{4} - \frac{2}{\pi^2} \sum_{k\,odd} \frac{1}{k^2} \; = \; \frac{1}{4} - \frac{2}{\pi^2} \frac{3}{4} \sum_{k=1}^{\infty} \frac{1}{k^2} \; = \; 0 \quad \Rightarrow \quad \sum_{k=1}^{\infty} \frac{1}{k^2} = \frac{\pi^2}{6} \, .$$

It does: Let's denote the right-hand side of (11.4) by R_n :

$$R_n \; = \; \int_0^1 \frac{x}{2 \sin\left(\frac{1}{2}\pi x\right)} \sin\Big(\frac{2n+1}{2}\pi x\Big) \, dx \; = \; \int_0^1 f(x) \sin\Big(\frac{2n+1}{2} x\Big) \, dx,$$

where

$$f(x) \; = \frac{x}{2 \sin\left(\frac{1}{2}\pi x\right)} \; = \; \frac{1}{\pi} \frac{\frac{1}{2}\pi x}{\sin\left(\frac{1}{2}\pi x\right)} \, .$$

For $x \in (0, 1)$ the only trouble that $f(x)$ might cause is near $x = 0$. But there is no trouble—thanks to the limit (11.2): $\lim\limits_{x \to 0^+} f(x) = \frac{1}{\pi}$. So integrating by parts gives

$$\begin{aligned} R_n \; &= \; \int_0^1 \underbrace{f(x)}_{\mathrm{D}} \underbrace{\sin\Big(\frac{2n+1}{2}\pi x\Big)}_{\mathrm{I}} \, dx \\ &= \; f(x) \frac{-2}{(2n+1)\pi} \cos\Big(\tfrac{2n+1}{2}\pi x\Big) \Big|_0^1 \; + \; \frac{2}{(2n+1)\pi} \int_0^1 f'(x) \cos\Big(\tfrac{2n+1}{2}\pi x\Big) dx \\ &= \; \frac{2}{(2n+1)\pi^2} \; + \; \frac{2}{(2n+1)\pi} \int_0^1 f'(x) \cos\Big(\tfrac{2n+1}{2}\pi x\Big) dx. \end{aligned}$$

By the Quotient Rule (after much simplification!), we get

$$f'(x) \; = \; \frac{1}{4} \frac{1 - \frac{\frac{1}{2}\pi x}{\sin(\frac{1}{2}\pi x)} \cos(\frac{1}{2}\pi x)}{\sin(\frac{1}{2}\pi x)} \, .$$

For $x \in (0, 1)$ the only trouble that $f'(x)$ may cause is again near $x = 0$. But thanks to the limit (11.2), and to L'Hospital's Rule, there is no trouble. (The limit at $x = 0$ is 0.) So f' is bounded on $(0, 1)$. Therefore the $\frac{2}{(2n+1)\pi^2}$ and $\frac{2}{(2n+1)\pi}$ above make $R_n \to 0$ as $n \to \infty$, and we're all set.

▶ **Remark** The alternating version of the series in Lemma 11.1, that is,

$$\sum_{n=0}^{\infty} \frac{(-1)^n}{(2n+1)^2} = 1 - \frac{1}{3^2} + \frac{1}{5^2} - \frac{1}{7^2} + \cdots,$$

converges, by the Alternating Series Test, to *Catalan's constant* $\kappa \approx 0.915\cdots$, named for Eugène Catalan (France/Belgium, 1814–1894). No simple expression for κ is known; it is not even known whether κ is irrational.[4]

Euler was able to obtain formulas for computing *all* of the sums $\sum_{n=1}^{\infty} \frac{1}{n^p}$, for p even. For example, $\sum_{n=1}^{\infty} \frac{1}{n^4} = \frac{\pi^4}{90}$ and $\sum_{n=1}^{\infty} \frac{1}{n^6} = \frac{\pi^6}{945}$ (the focus of Chap. 12).

11.2 A Cool Example

Here follows a neat application of (11.1).

Problem (e.g., [1]). Let m and n be two given natural numbers. What is the probability that they have no common factors, i.e., that their greatest common divisor is 1? That is,

$$P\big(\gcd(m, n) = 1\big) = \textit{what?}$$

Solution. Let p be a prime number. The probability that p divides a given natural number m is $1/p$. Therefore the probability that p divides given numbers m and n is $1/p^2$. So the probability that p *does not* divide given natural numbers m and n is $1 - 1/p^2$. The probability that m and n have *no* two prime factors in common is therefore

$$\prod_{p \text{ prime}} \Big(1 - \frac{1}{p^2}\Big).$$

[4] "It is better to know some of the questions than all of the answers."—James Thurber.

Since $1-\frac{1}{p^2} = \left(\frac{1}{1-\frac{1}{p^2}}\right)^{-1}$, we write $1-\frac{1}{p^2}$ as the *reciprocal* of a geometric series:

$$1-\frac{1}{p^2} = \left(\sum_{k=0}^{\infty}\left(\frac{1}{p^2}\right)^k\right)^{-1} = \left(1+\frac{1}{p^2}+\frac{1}{p^4}+\frac{1}{p^6}+\cdots\right)^{-1}.$$

Therefore

$$\prod_{p \text{ prime}}\left(1-\tfrac{1}{p^2}\right) = \prod_{p \text{ prime}}\left(1+\tfrac{1}{p^2}+\tfrac{1}{p^4}+\cdots\right)^{-1}$$

$$= \left[\prod_{p \text{ prime}}\left(1+\tfrac{1}{p^2}+\tfrac{1}{p^4}+\cdots\right)\right]^{-1}$$

$$= \left[\left(1+\tfrac{1}{2^2}+\tfrac{1}{2^4}+\cdots\right)\left(1+\tfrac{1}{3^2}+\tfrac{1}{3^4}+\cdots\right)\cdots\right]^{-1}.$$

Now we think of the expression inside the [ ], i.e., the

$$\left(1+\tfrac{1}{2^2}+\tfrac{1}{2^4}+\tfrac{1}{2^6}+\cdots\right)\left(1+\tfrac{1}{3^2}+\tfrac{1}{3^4}+\tfrac{1}{3^6}+\cdots\right) \times\left(1+\tfrac{1}{5^2}+\tfrac{1}{5^4}+\tfrac{1}{5^6}+\cdots\right)\cdots,$$

as being *entirely* multiplied out, which would give us something that looks like

$$1+\sum\frac{1}{d^2}. \tag{11.5}$$

The *Fundamental Theorem of Arithmetic* says that each natural number can be factored into a product of prime numbers—and uniquely so, up to order that the factors are written down. For example,

$$42 = 2\times3\times7,\quad 244 = 2^2\times61,\quad 41743 = 13^3\times19,\quad 339405 = 3\times5\times11^3\times17, \text{ etc.}$$

So for each natural number $k > 1$, the number k^2 must appear *exactly once* as a denominator d^2 in the sum (11.5). Therefore we have

$$\left(1+\frac{1}{2^2}+\frac{1}{2^4}+\cdots\right)\left(1+\frac{1}{3^2}+\frac{1}{3^4}+\cdots\right)\left(1+\frac{1}{5^2}+\frac{1}{5^4}+\cdots\right)\cdots = \sum_{k=1}^{\infty}\frac{1}{k^2}.$$

By (11.1), this is $\pi^2/6$. So the probability that m and n do not have a common factor is

$$\left(\frac{\pi^2}{6}\right)^{-1} = \frac{6}{\pi^2} \approx 0.6079. \qquad \checkmark$$

11.3 The Sum of the Reciprocals of the Primes

While the Fundamental Theorem of Arithmetic is still warm, we might as well reuse it in a similar way. We prove the following classic fact [8, also [13]].

Theorem 11.1

$$\sum_{p \text{ prime}} \frac{1}{p} = \frac{1}{2} + \frac{1}{3} + \frac{1}{5} + \frac{1}{7} + \frac{1}{11} + \frac{1}{13} + \frac{1}{17} + \cdots \text{ diverges}.$$

Proof. Since $\frac{1}{p} \geq \frac{1}{2}\frac{1}{p-1}$ for $p \geq 2$, it's enough to show that $\sum_{p \text{ prime}} \frac{1}{p-1}$ diverges. For p prime, set $x = \frac{1}{1-\frac{1}{p}}$ in the inequality (Lemma 6.4) $x - 1 \geq \log(x)$ to get

$$\frac{1}{p-1} > \log\left(\frac{1}{1-\frac{1}{p}}\right).$$

Then we have

$$\sum_{p \text{ prime}} \frac{1}{p-1} \geq \sum_{p \text{ prime}} \log\left(\frac{1}{1-\frac{1}{p}}\right) = \log \prod_{p \text{ prime}} \left(\frac{1}{1-\frac{1}{p}}\right).$$

Since $p \geq 2$, the fraction $\frac{1}{1-\frac{1}{p}}$ can be written as a geometric series (just like before):

$$\begin{aligned}
\prod_{p \text{ prime}} \frac{1}{1-\frac{1}{p}} &= \prod_{p \text{ prime}} \sum_{k=0}^{\infty} \left(\frac{1}{p}\right)^k \\
&= \prod_{p \text{ prime}} \left(1 + \frac{1}{p} + \frac{1}{p^2} + \frac{1}{p^3} + \cdots\right) \\
&= \left(1 + \frac{1}{2} + \frac{1}{2^2} + \frac{1}{2^3} + \cdots\right)\left(1 + \frac{1}{3} + \frac{1}{3^2} + \frac{1}{3^3} + \cdots\right) \cdots .
\end{aligned}$$

Now again, as we did in the previous section, we think of the expression

$$\left(1+\frac{1}{2}+\frac{1}{2^2}+\frac{1}{2^3}+\cdots\right)\left(1+\frac{1}{3}+\frac{1}{3^2}+\frac{1}{3^3}+\cdots\right)\cdots\cdots$$

as being entirely multiplied out, which would give us something that looks like

$$1+\sum\frac{1}{d}\,. \tag{11.6}$$

And by the Fundamental Theorem of Arithmetic, for each natural number $k>1$ in the sum (11.6) above, the number k must appear *exactly once* as a denominator d. Therefore we have

$$\left(1+\frac{1}{2}+\frac{1}{2^2}+\frac{1}{2^3}+\cdots\right)\left(1+\frac{1}{3}+\frac{1}{3^2}+\frac{1}{3^3}+\cdots\right)\cdots\cdots=\sum_{k=1}^{\infty}\frac{1}{k}\,.$$

This is the Harmonic Series, which diverges (!), and so $\sum\limits_{p\text{ prime}}\frac{1}{p-1}$ diverges. □

An immediate corollary of Theorem 11.1 is Euclid's[5] theorem: *There are infinitely many prime numbers.* Indeed, if there were only finitely many prime numbers, then the sum in the theorem would have to be finite.

But this is an unnecessarily circuitous way to obtain Euclid's great theorem. His own original proof (which everyone should know!) would be difficult to beat—for brevity, for beauty, or for just about anything else.

▶ **Remark** *Twin primes* are pairs of prime numbers $\langle p,\,q\rangle$ such that $p+2=q$. Some examples are $\langle 3,5\rangle$; $\langle 5,7\rangle$; $\langle 11,13\rangle$; $\langle 59,61\rangle$; $\langle 107,109\rangle$; and $\langle 70121,70123\rangle$. V. Brun (Norway, 1885–1978) proved in 1919 that $\sum 1/p$ converges, where the sum is taken over all twin primes. It is not known[6] whether there are infinitely many twin primes—if there are not, then Brun's result is trivial. Either way, it is known that this sum (called *Brun's Constant*) is somewhere between 1.9021 and 1.9022.

Exercises

11.1.

(a) Use $\displaystyle\int_q^{\infty}\frac{1}{x^2}\,dx=\frac{1}{q}$ (for $q>0$) to show that

[5] Euclid of Alexandria ($\sim$ 300 BC), author of the greatest-ever mathematics textbook.

[6] "We know there are some things we do not know."—Donald Rumsfeld.

$$\frac{1}{n+1} < \frac{\pi^2}{6} - \sum_{k=1}^{n} \frac{1}{k^2} < \frac{1}{n}.$$

(b) Verify, and then use, the partial fraction decomposition [12]

$$\frac{1}{(n(n+1))^3} = \left(\frac{1}{n^3} - \frac{1}{(n+1)^3}\right) - 3\left(\frac{1}{n^2} + \frac{1}{(n+1)^2}\right) + 6\left(\frac{1}{n} - \frac{1}{n+1}\right)$$

to get $\pi^2 = 10 - \sum_{n=1}^{\infty} \frac{1}{(n(n+1))^3} < 10 - \frac{1}{8}$. [This is better than Archimedes' $\pi < \frac{22}{7}$.]

11.2. Show that for $k = 2, 3, 4, \ldots,$

$$\sum_{n=1}^{\infty} \frac{1}{n^k} = \frac{2^k}{2^k - 1} \sum_{n=1}^{\infty} \frac{1}{(2n-1)^k}.$$

11.3.

(a) Use Lemma 11.1 to show that $\sum_{n=1}^{\infty} \frac{(-1)^{n-1}}{n^2} = \frac{\pi^2}{12}$.

(b) Consider concentric circles with radii $1, \frac{1}{2}, \frac{1}{3}, \frac{1}{4}, \cdots$ (on forever), with alternate rings shaded, as in Fig. 11.1 (left side). The area inside the largest circle is π. What is the total area of all the shaded regions?

(c) Write your own problem, and solve it, using instead Fig. 11.1 (right side).

11.4. Here is another proof of (11.1), closer to the one in [10].

(a) Begin by multiplying (11.3) by $x(x - 2\pi)$; then integrate over $[0, 1]$.

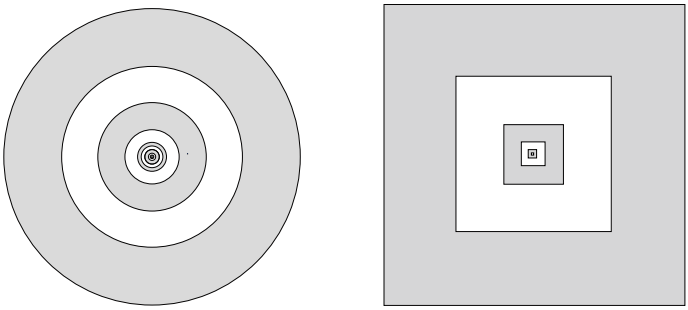


Fig. 11.1 (For Exercise 11.3) Infinitely many concentric rings

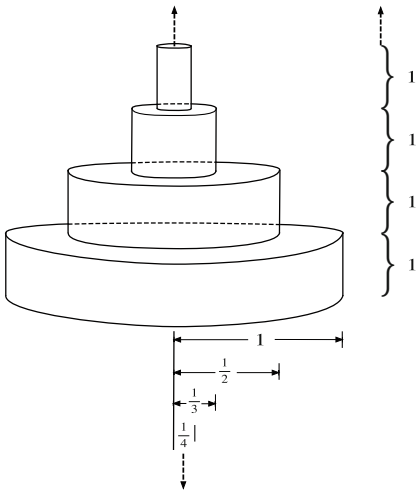


Fig. 11.2 (For Exercise 11.5) Enough to share—an infinitely tall wedding cake

(b) Show that the resulting left-hand side equals $\sum_{k=1}^{n} \frac{1}{k^2} - \frac{\pi^2}{6}$.

(c) Show that the resulting right-hand side $\to 0$ as $n \to \infty$.

11.5 ([6]). Consider the infinitely tall wedding cake, as shown in Fig. 11.2. Find its volume (the amount of filling) and its surface area (the amount of frosting).

11.6.

(a) What is the probability that a given natural number has *no* prime factors? How does this fit into the same sort of analysis as in Sect. 11.2?

(b) What is the probability that four given numbers have gcd $= 1$? Explain.

11.7. A natural number q is *square-free* if any prime factor of q appears exactly once. Find the probability that a given natural number is square-free.

11.8. A *lattice point* in the first quadrant is any point $(n, m) \in \mathbb{R}^2$, where $n, m \in \mathbb{N}$. Let $a, b \in \mathbb{N}$. Show that the probability that the line segment from $(0, 0)$ to (a, b) passes though no (other) lattice points is $6/\pi^2$.

11.9 ([11]). Call a positive integer *coarse* if it has a prime factor larger than 3. Show that the sum of the reciprocals of the squares of the coarse numbers

$$\frac{1}{25} + \frac{1}{49} + \frac{1}{100} + \frac{1}{121} + \frac{1}{169} + \frac{1}{196} + \cdots = \frac{\pi^2}{6} - \frac{3}{2} \approx 0.1449.$$

11.10. Let $N \in \mathbb{N}$. We saw that $\sum_{p \text{ prime}} \frac{1}{p}$ diverges, so there is a partial sum of $\sum_{p \text{ prime}} \frac{1}{p}$ which exceeds N. Show that there is no partial sum of $\sum_{p \text{ prime}} \frac{1}{p}$ which $= N$.

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Interlude: Beyond Basel

12

The famous *Zeta Function* is given by

$$\zeta(x) = \sum_{n=1}^{\infty} \frac{1}{n^x} \quad (\text{for } x > 1).$$

If $x = 1$, this is the Harmonic Series, which diverges.

[What the heck—here's another proof [5]: If we assume that $S = \sum_{n=1}^{\infty} \frac{1}{n}$ is some finite number, then

$$\begin{aligned} S &= \left(1 + \frac{1}{2}\right) + \left(\frac{1}{3} + \frac{1}{4}\right) + \left(\frac{1}{5} + \frac{1}{6}\right) + \cdots \\ &> \left(\frac{1}{2} + \frac{1}{2}\right) + \left(\frac{1}{4} + \frac{1}{4}\right) + \left(\frac{1}{6} + \frac{1}{6}\right) + \cdots \\ &= 1 + \frac{1}{2} + \frac{1}{3} + \cdots \\ &= S, \end{aligned}$$

which is clearly a problem. So we must have $S = \infty$.]

So the solution to the Basel Problem is, in short,

$$\zeta(2) = \frac{\pi^2}{6}.$$

Beyond this, Euler actually obtained formulas for computing *all* of the sums

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P. R. Mercer, *A Compact Capstone Course in Classical Calculus*, Compact Textbooks in Mathematics, https://doi.org/10.1007/978-3-031-43914-8_12

$$\zeta(p) = \sum_{n=1}^{\infty} \frac{1}{n^p}, \quad \text{for } p \text{ even}.$$

For example,

$$\zeta(4) = \sum_{n=1}^{\infty} \frac{1}{n^4} = \frac{\pi^4}{90} \qquad \text{and} \qquad \zeta(6) = \sum_{n=1}^{\infty} \frac{1}{n^6} = \frac{\pi^6}{945}.$$

In this interlude we derive Euler's formulas for $\zeta(2m)$, for $m = 1, 2, 3, \ldots$.

12.1 The Bernoulli Polynomials

Euler's formulas involve a family of polynomials named for Jacob Bernoulli.[1] There are many equivalent ways of defining these, the *Bernoulli Polynomials*. The following (e.g., [1]) will be the most useful for our purposes.

Set $B_0(x) = 1$ and $b_0 = 1$. Then for $m = 0, 1, 2, 3, \ldots$,

$$B_{m+1}(x) = (m+1) \int_0^x B_m(u)du + b_{m+1}, \tag{12.1}$$

where each constant b_{m+1} is chosen so that

$$\int_0^1 B_{m+1}(x)dx = 0. \tag{12.2}$$

For example,

$$B_1(x) = \int_0^x B_0(u)du = x + b_1, \quad \text{then} \int_0^1 B_1(x)dx = 0 \quad \text{gives } b_1 = -\frac{1}{2},$$

and

$$B_2(x) = 2\int_0^x B_1(u)du = x^2 - x + b_2, \quad \text{then} \int_0^1 B_2(x)dx = 0 \quad \text{gives } b_2 = \frac{1}{6}.$$

[1] We have met Jacob before, in Chap. 6. He was also the discoverer of that special number e while considering continuously compounded interest. His father wanted him to become a minister.

Etcetera: The first few Bernoulli Polynomials are as follows:

$$B_0(x) = 1, \quad B_1(x) = x - \tfrac{1}{2}, \quad B_2(x) = x^2 - x + \tfrac{1}{6}, \quad B_3(x) = x^3 - \tfrac{3}{2}x^2 + \tfrac{1}{2}x,$$

$$B_4(x) = x^4 - 2x^3 + x^2 - \tfrac{1}{30}, \quad B_5(x) = x^5 - \tfrac{5}{2}x^4 + \tfrac{5}{3}x^3 - \tfrac{1}{6}x.$$

$$B_6(x) = x^6 - 3x^5 + \tfrac{5}{2}x^4 - \tfrac{1}{2}x^2 + \tfrac{1}{42}, \quad B_7(x) = x^7 - \tfrac{7}{2}x^6 + \tfrac{7}{2}x^5 - \tfrac{7}{6}x^3 + \tfrac{1}{6}x.$$

The relationships (12.1) and (12.2) give immediately

$$b_m = B_m(0) = B_m(1) \quad \text{for } m = 0 \ \& \ m \geq 2.$$

But much more than this is true, as follows:

Lemma 12.1
For $m \geq 0$, we have $B_m(1-x) = (-1)^m B_m(x)$ for all x.

Proof. The cases $m = 0$ and $m = 1$ are clear. Proceeding by induction, we assume first that the Lemma holds with $m = k$. Then since $\int_0^1 B_k(u)du = 0$,

$$\begin{aligned}
B_{k+1}(1-x) - b_{k+1} &= (k+1)\int_0^{1-x} B_k(u)du = -(k+1)\int_{1-x}^{1} B_k(u)du \\
&= (k+1)\int_x^0 B_k(1-t)dt \quad \text{(after setting } u = 1-t\text{)} \\
&= -(k+1)\int_0^x (-1)^k B_k(t)dt \quad \text{(induction hypothesis)} \\
&= (-1)^{k+1}(k+1)\int_0^x B_k(t)dt \\
&= (-1)^{k+1}\big[B_{k+1}(x) - b_{k+1}\big].
\end{aligned}$$

If $k = 2N - 1$, this gives

$$B_{2N}(1-x) - b_{2N} = B_{2N}(x) - b_{2N} \quad \Rightarrow \quad B_{2N}(1-x) = B_{2N}(x). \quad \text{Good.}$$

If $k = 2N$, this gives

$$B_{2N+1}(1-x) - b_{2N+1} = -B_{2N+1}(x) + b_{2N+1} \Rightarrow B_{2N+1}(1-x) + B_{2N+1}(x) = 2b_{2N+1}\,.$$

Integrating, we get

$$\int_0^1 B_{2N+1}(1-x)dx \; + \; \int_0^1 B_{2N+1}(x)dx \;\; = \;\; 2b_{2N+1},$$

or

$$2\int_0^1 B_{2N+1}(x)dx \;\; = \;\; 2b_{2N+1},$$

so that

$$b_{2N+1} = 0\,. \tag{12.3}$$

In either case ($k = 2N - 1$ or $k = 2N$), $B_{k+1}(1-x) = (-1)^{k+1} B_{k+1}(x)$. □

Lemma 12.1 says that the B_ms with even subscripts are *even functions with respect to* $1/2$, while the B_ms with odd subscripts are *odd functions with respect to* $1/2$. The *nonzero* b_{2m}s, that is, $b_0 = 1$, $b_2 = 1/6$, $b_4 = -1/30$, $b_6 = 1/42$, etc., are called the *Bernoulli Numbers*. These are simply the constant terms (i.e., the y-intercepts) of each of the $B_{2m}(x)$s. The only nonzero constant term, for Bernoulli Polynomials having odd subscripts, is $b_1 = -1/2$. Figure 12.1 has graphs of six of the Bernoulli Polynomials.

12.2 Toward $\zeta(2m)$

Okay, let us reflect on the proof from Chap. 11 that $\zeta(2) = \pi^2/6$. We began with line (11.3) from Lemma 11.2, which reads

$$\frac{1}{2} \; + \; \sum_{k=1}^{n} \cos(k\pi x) \;\; = \;\; \frac{\sin\left(\frac{2n+1}{2}\pi x\right)}{2\sin\left(\frac{1}{2}\pi x\right)}\,. \tag{12.4}$$

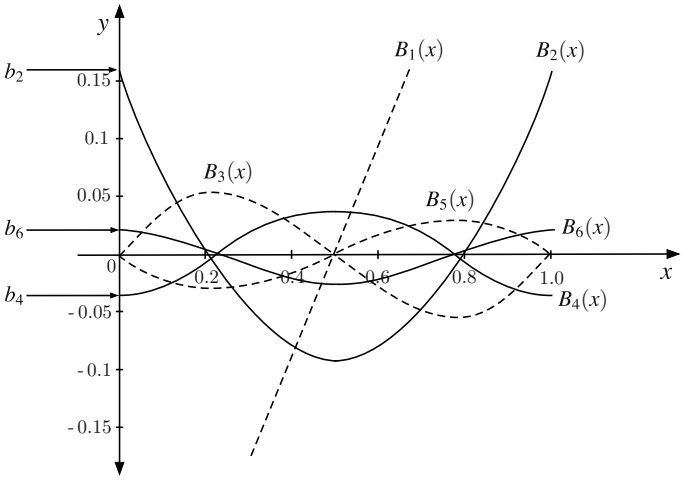


Fig. 12.1 After $B_0(x) = 1$, the next six Bernoulli Polynomials

Then we multiplied through by x and readied ourselves to integrate in (11.4):

$$\frac{1}{2}\int_0^1 x\,dx \;+\; \sum_{k=1}^{n}\int_0^1 x\cos(k\pi x)\,dx \;=\; \int_0^1 \frac{x/2}{\sin\left(\frac{1}{2}\pi x\right)}\sin\left(\frac{2n+1}{2}\pi x\right)\,dx\,.$$

Integration by parts, and some fiddling by way of Lemma 11.1, showed that the left-hand side $\to \frac{1}{4} - \frac{3}{2\pi^2}\sum_{k=1}^{\infty}\frac{1}{k^2}$ as $n \to \infty$. And thanks to the $\frac{x/2}{\sin(\pi x/2)}$ in the integrand on the right-hand side, the entire right-hand side $\to 0$ as $n \to \infty$. Very good.

You can verify for yourself, perhaps during the next rainy day, that multiplying (12.4) by x^3 and integrating over [0, 1] ultimately gives, after more integrations by parts and more fiddling, $\zeta(4) = \pi^4/90$. Multiplying by x^5 gives (even more integrations and even more fiddling!) $\zeta(6) = \pi^6/945$. Etcetera.[2]

But as you can imagine, manipulations such as these become increasingly nasty. Influenced by [2], however, a systematic attack that leads to Euler's formulas for $\zeta(2m)$ was found in [3]; see also [8].

The crucial observation is that instead of x, x^3, x^5, etc., we should be multiplying through by certain Bernoulli Polynomials! By their very definition, these polynomials behave well under integration by parts—after which, just a bit more fiddling is required.

[2] Conspicuous by their absence are the even powers of x. Alas, multiplying (12.4) by x^2 just leads again (oh snap!) to $\zeta(2) = \pi^2/6$. (Try it.) Multiplying by x^4?: $\zeta(4) = \pi^4/90$. Oh well.

12.3 The Formula for $\zeta(2m)$

We multiply (12.4) through by $B_{2m}(x)$, with $m > 0$ and then integrate

$$\frac{1}{2}\int_0^1 B_{2m}(x)\,dx + \sum_{k=1}^{n}\int_0^1 B_{2m}(x)\cos(k\pi x)\,dx = \int_0^1 \frac{B_{2m}(x)}{2\sin(\frac{1}{2}\pi x)}\sin\Big(\frac{2n+1}{2}\pi x\Big)\,dx. \tag{12.5}$$

Since $\int_0^1 B_{2m}(x)\,dx = 0$, the first term is zero. For the second term, we integrate by parts. By (12.1), $\frac{d}{dx}B_n(x) = nB_{n-1}(x)$ and so

$$\int_0^1 B_{2m}(x)\cos(k\pi x)dx = \int_0^1 \underbrace{B_{2m}(x)}_{\mathrm{D}}\underbrace{\cos(k\pi x)}_{\mathrm{I}}\,dx$$

$$= B_{2m}(x)\frac{1}{k\pi}\sin(k\pi x)\Big|_0^1 - \frac{2m}{k\pi}\int_0^1 \underbrace{B_{2m-1}(x)}_{\mathrm{D}}\underbrace{\sin(k\pi x)}_{\mathrm{I}}\,dx$$

$$= 0 + \frac{2m}{k^2\pi^2}B_{2m-1}(x)\cos(k\pi x)\Big|_0^1 - \frac{2m(2m-1)}{k^2\pi^2}\int_0^1 B_{2m-2}(x)\cos(k\pi x)dx\,.$$

By (12.3), $B_j(0) = B_j(1) = 0$ for j odd ≥ 3. So continuing on in this way, we come all the way down to $m = 1$:

$$\int_0^1 B_{2m}(x)\cos(k\pi x)dx = \frac{(-1)^m 2m(2m-1)(2m-2)\cdots(2)}{k^{2m-1}\pi^{2m-1}}\int_0^1 B_1(x)\sin(k\pi x)dx\,.$$

That is,

$$\int_0^1 B_{2m}(x)\cos(k\pi x)dx = \frac{(-1)^m(2m)!}{k^{2m-1}\pi^{2m-1}}\int_0^1 \Big(x-\frac{1}{2}\Big)\sin(k\pi x)dx.$$

We met the integral on the right-hand side in Sect. 2.2, so that

$$\int_0^1 B_{2m}(x)\cos(k\pi x)dx = \frac{(-1)^m(2m)!}{k^{2m-1}\pi^{2m-1}} \times \begin{cases} 0 & \text{if } k \text{ is odd} \\ -\frac{1}{k\pi} & \text{if } k \text{ is even} \end{cases}$$

$$= \frac{(-1)^{m+1}(2m)!}{k^{2m}\pi^{2m}} \times \begin{cases} 0 & \text{if } k \text{ is odd} \\ 1 & \text{if } k \text{ is even} \end{cases}$$

$$= \frac{(-1)^{m+1}(2m)!}{(2k)^{2m}\pi^{2m}} \quad \text{for} \quad k = 1, 2, 3, \ldots .$$

Okay let's take a breather. We have so far that the left-hand side of (12.5) is

$$\sum_{k=1}^{n} \int_0^1 B_{2m}(x)\cos(k\pi x)\,dx = \frac{(-1)^{m+1}(2m)!}{2^{2m}\pi^{2m}} \sum_{k=1}^{n} \frac{1}{k^{2m}}.$$

So if we can show that the right-hand side of (12.5) $\to$ *something,* as $n \to \infty$, then we'll have a formula for $\zeta(2m)$. Indeed it does—now we'll go find that *something*.

The right-hand side of (12.5) reads

$$\int_0^1 \frac{B_{2m}(x)}{2\sin(\frac{1}{2}\pi x)} \sin\Big(\frac{2n+1}{2}\pi x\Big)\,dx.$$

For x near zero, the integrand here presents a problem because $B_{2m}(0) \neq 0$. So we use the old calculus/analysis trick of subtracting something we'd *like to have* and then adding it in again:[3] we write the right-hand side of (12.5) as

$$\int_0^1 \frac{B_{2m}(x) - B_{2m}(0)}{2\sin(\frac{1}{2}\pi x)} \sin\Big(\frac{2n+1}{2}\pi x\Big)\,dx + \int_0^1 \frac{B_{2m}(0)}{2\sin(\frac{1}{2}\pi x)} \sin\Big(\frac{2n+1}{2}\pi x\Big)\,dx.$$

Just as in our proof that $\zeta(2) = \pi^2/6$, the term $\frac{B_{2m}(x)-B_{2m}(0)}{2\sin(\frac{1}{2}\pi x)}$ in the first integrand has a bounded derivative, and so after integrating by parts, we see that the first integral $\to 0$ as $n \to \infty$. So now we must deal with the second integral—this takes us back to (12.4):

$$\frac{\sin\left(\frac{2n+1}{2}\pi x\right)}{2\sin\left(\frac{1}{2}\pi x\right)} = \frac{1}{2} + \sum_{k=1}^{n} \cos(k\pi x),$$

[3] Remember how you proved the Product Rule, back in the day?

which gives

$$\int_0^1 \frac{B_{2m}(0)}{2\sin(\frac{1}{2}\pi x)} \sin\Big(\frac{2n+1}{2}\pi x\Big)\,dx = \frac{1}{2}\int_0^1 B_{2m}(0)dx + B_{2m}(0)\sum_{k=1}^{n}\int_0^1 \cos(k\pi x)\,dx$$

$$= \frac{B_{2m}(0)}{2} + B_{2m}(0)\sum_{k=1}^{n}\frac{1}{k\pi}\sin(k\pi x)\Big|_0^1$$

$$= \frac{1}{2}b_{2m} + 0\,.$$

Therefore, finally, we have

$$\frac{(-1)^{m+1}(2m)!}{2^{2m}\pi^{2m}}\sum_{k=1}^{\infty}\frac{1}{k^{2m}} = \frac{1}{2}b_{2m}\,,$$

or

$$\zeta(2m) = \sum_{k=1}^{\infty}\frac{1}{k^{2m}} = \frac{(-1)^{m+1}2^{2m-1}\pi^{2m}}{(2m)!}b_{2m} \quad (m = 1, 2, 3, \ldots).$$

Here again, the b_{2m} are the *Bernoulli Numbers*:

$$b_2 = \frac{1}{6}\,,\ b_4 = \frac{-1}{30}\,,\ b_6 = \frac{1}{42}\,,\ b_8 = \frac{-1}{30}\,,\ b_{10} = \frac{5}{66}\,,\ b_{12} = \frac{-691}{2730}\,,\ \ldots\,. \quad \checkmark$$

12.4 Sums of Powers

We must mention the connection between the Bernoulli Numbers b_m and something that you have probably seen before, in calculus and/or discrete mathematics. Using induction, it's easy to see (try it!) that the sums

$$\sum_{k=1}^{n}k = \frac{n(n+1)}{2}\,,\quad \sum_{k=1}^{n}k^2 = \frac{n(n+1)(2n+1)}{6}\,,\ \text{and}\ \sum_{k=1}^{n}k^3 = \frac{n^2(n+1)^2}{4}\,.$$

The b_ms first came up while Jacob Bernoulli was looking for a general formula for *all* such sums. The formula he sought is [1, 7, cf. [6]]

$$\sum_{k=1}^{n}k^m = \frac{B_{m+1}(n+1) - b_{m+1}}{m+1} \quad (m, n = 1, 2, 3, \ldots). \tag{12.6}$$

(Keep in mind that the only nonzero b_j having an odd subscript is $b_1 = -1/2$.)

We'll prove (12.6) in two steps. First we show that for all x, we have

$$B_m(x+1) - B_m(x) = m\,x^{m-1} \quad (m = 1, 2, 3, \ldots). \tag{12.7}$$

Observe that $B_1(x) = x - 1/2$ and so (12.7) holds for $m = 1$. Proceeding by induction, we assume that (12.7) holds as stated, for m.[4] Then for $m+1$,

$$\begin{aligned}
B_{m+1}(x+1) - B_{m+1}(x) &= (m+1)\left(\int_0^{x+1} B_m(t)dt - \int_0^x B_m(t)dt\right) \\
&= (m+1)\int_0^1 B_m(t)dt \\
&\quad + (m+1)\left(\int_1^{x+1} B_m(t)dt - \int_0^x B_m(t)dt\right) \\
&= 0 + (m+1)\left(\int_1^{x+1} B_m(t)dt - \int_0^x B_m(t)dt\right) \\
&= (m+1)\left(\int_0^x \big(B_m(t+1) - B_m(t)\big)dt\right) \\
&= (m+1)\left(\int_0^x m\,u^{m-1}du\right) \\
&= (m+1)\,x^m,
\end{aligned}$$

which is precisely what we want, for proving (12.7). ✓

Now we can verify (12.6). Replacing m with $m+1$ and x with k in (12.7), we get

$$B_{m+1}(k+1) - B_{m+1}(k) = (m+1)\,k^m.$$

[4] Admittedly this is bad form, but many mathematicians do it when their students aren't around.

Summing over k, from 0 to n, we get

$$\sum_{k=0}^{n} [B_{m+1}(k+1) - B_{m+1}(k)] \;=\; (m+1)\sum_{k=0}^{n} k^m \;=\; (m+1)\sum_{k=1}^{n} k^m.$$

The left-hand side telescopes in a very satisfying fashion, leaving us with

$$B_{m+1}(n+1) - B_{m+1}(0) \;=\; (m+1)\sum_{k=1}^{n} k^m\,,$$

which is (12.6). ✓

12.5 Epilogue

Bernoulli Numbers and Polynomials also arise in the *Euler-Maclaurin Summation Formula*. This is a formula for estimating integrals using sums, or vice versa. It leads to relationships among the Bernoulli Numbers, Euler's Constant γ, and Stirling's Formula (for example). Just about anyone who does classical analysis carries the Euler-Maclaurin Summation Formula around with them. See [7], for example, for more on this. But we have touched on a few other important topics from classical analysis as well.

The function $f(u) = \dfrac{\sin\left(\frac{2n+1}{2}u\right)}{2\sin(\frac{1}{2}u)}$ on the right-hand side of (12.4) is called the *Dirichlet Kernel*.[5] It is an important function in mathematical analysis and Fourier analysis in particular.

The Zeta Function extended to (much of) the complex plane $\mathbb{C}$ is the extraordinary *Riemann Zeta Function* $\zeta(z)$.[6] It holds—often very tightly!—some of the mysteries of physics, probability, number theory, and the prime numbers in particular. Many monumental open problems in mathematics involve the Riemann Zeta Function.

Still, surprisingly little is known about the sums $\zeta(p)$ for p odd, despite the efforts of countless great mathematicians.[7] It was shown in 1979, by Roger Apéry (France, 1916–1994), that $\zeta(3) \approx 1.202$ is irrational. This number has come to be called *Apéry's Constant*. See [4] for an interesting account.

[5] Johann Peter Gustav Lejeune Dirichlet (Germany, 1805–1859), a giant in number theory and analysis, was married to the sister of the famous composer Felix Mendelssohn. A music book might say that Mendelssohn's sister was married to the famous mathematician Dirichlet.

[6] Named for Bernhard Riemann (Germany, 1826–1866), perhaps best known for his rigorous definition of the definite integral. He was a master of analysis, number theory, and geometry.

[7] "We're not doing so well Patrick; we need a new approach, a new tactic."—SpongeBob SquarePants.

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Stirling's Formula

13

Stirling's Formula[1] is

$$n! \;\sim\; \sqrt{2\pi n}\,\frac{n^n}{\mathrm{e}^n}\,. \tag{13.1}$$

The $\sim$ above means that the *quotient* of the left- and right-hand sides $\to 1$ as $n \to \infty$. This means that the relative error

$$\frac{n! \;-\; \sqrt{2\pi n}\,\frac{n^n}{\mathrm{e}^n}}{n!} \quad \to \quad 0 \qquad \text{as} \;\; n \to \infty\,.$$

Formula (13.1) is, in fact, due to de Moivre.[2] But instead of the exact $C = \sqrt{2\pi} \approx 2.5066$ which Stirling obtained, de Moivre had $C \approx 2.5074$. There is a slightly more complicated (and more accurate) formula by Stirling [8,16] which we consider in Exercise 13.8(b) and again, more fully, in Chap. 15.

Messy limits can often be evaluated, or at least estimated, using Stirling's Formula. But numerically, the point of Stirling's Formula is that $n!$ grows ridiculously fast—fast even for a computer to process. (Have a look at [7].) However, the approximation given by the right-hand side of (13.1) involves only elementary functions, which are relatively easy to handle.

Example
Recall from Sect. 8.1 that for $x \in \mathbb{R}$ the *floor* of x, denoted by $\lfloor x \rfloor$, is the largest integer $\le x$. For example, $\lfloor 0.1 \rfloor = 0$, $\lfloor \mathrm{e} \rfloor = 2$, $\lfloor -2.2 \rfloor = -3$, etc. The reader should agree (we leave it as Exercise 13.1) that the number of digits of a number $x > 0$ to the left of the decimal point is

[1] James Stirling (Scotland, 1692–1770), who rubbed elbows with the intelligentsia of Venice in the 1710s, was nicknamed *the Venetian* by family and friends [17]. "Hey Venetian, pass the butter."

[2] Abraham de Moivre (France, 1667–1754) lived most of his life in England. He and Stirling were good friends who, presumably, seldom diverged [8,14].

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P. R. Mercer, *A Compact Capstone Course in Classical Calculus*, Compact Textbooks in Mathematics, https://doi.org/10.1007/978-3-031-43914-8_13

$$d(x) = \lfloor \log_{10}(x) \rfloor + 1\,.$$

Therefore, taking $\log_{10}(\cdot)$ of both sides, (13.1) predicts that the (extremely large) number $10{,}000{,}000! = 10^7!$ has

$$d(x) \;=\; \left\lfloor \frac{1}{2}\log_{10}(2\pi) + \frac{1}{2}\log_{10}(10^7) + 10^7 \log_{10}(10^7) - 10^7 \log_{10}(\mathrm{e}) \right\rfloor \;+\; 1 \quad \text{digits.}$$

A little simplifying (log rules + calculator) gives

$$\begin{aligned} d(x) \;&=\; \left\lfloor \frac{1}{2}\log_{10}(2\pi) \;+\; \frac{7}{2} \;+\; 7\times 10^7 \;-\; 10^7 \log_{10}(\mathrm{e}) \right\rfloor + 1 \\ &=\; 65{,}657{,}060\,. \end{aligned}$$

And $10^7!$, in fact, has *precisely* this number of digits. Not bad. ✓

In this chapter we verify (13.1). In doing so, Euler's Constant γ, which we met in Sect. 7.3, makes an appearance. Recall that this is

$$\gamma \;=\; \lim_{n\to\infty}\left[\sum_{k=1}^{n}\frac{1}{k} - \ln(n+1)\right] \;\approx\; 0.577\,.$$

Equivalently (after exponentiating), as $n \to \infty$,

$$\frac{1}{n+1}\prod_{k=1}^{n}\mathrm{e}^{1/k} \;\to\; \mathrm{e}^{\gamma} \;\approx\; 1.781\,. \tag{13.2}$$

13.1 Prep Work

We shall use the *Trapezoid Rule* $\displaystyle\int_a^b f(x)dx \;\approx\; \frac{f(a)+f(b)}{2}[b-a]$, which approximates the area under the curve $y = f(x)$ from $x = a$ to $x = b$ with the area of the trapezoid determined by $(a, 0)$, $(a, f(a))$, $(b, f(b))$, $(b, 0)$. It's easier to look at a picture, like Fig. 13.1.

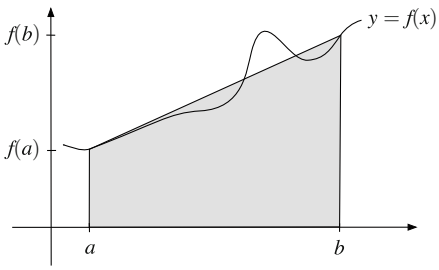


Fig. 13.1 The Trapezoid Rule approximates the integral $\int_a^b f(x)dx$ with the area of the shaded trapezoid

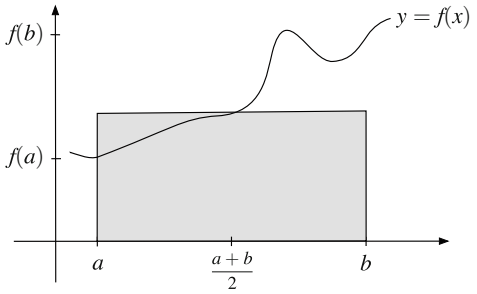


Fig. 13.2 The Midpoint Rule approximates the integral $\int_a^b f(x)dx$ with the area of the shaded rectangle

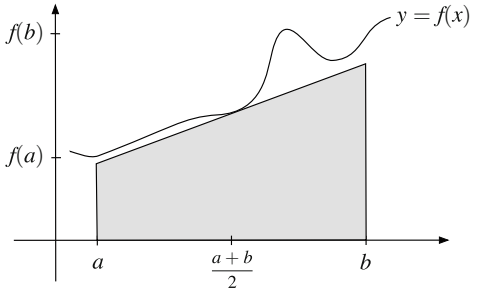


Fig. 13.3 The Midpoint Rule (also) approximates the integral $\int_a^b f(x)dx$ with the area of the shaded trapezoid; some call it the Tangent Rule

We shall also use the *Midpoint Rule* $\int_a^b f(x)dx \approx f(\frac{a+b}{2})[b-a]$, which approximates the area under the curve $y = f(x)$ from $x = a$ to $x = b$ with the area of the rectangle having length $b - a$ and height $f(\frac{a+b}{2})$, as in Fig. 13.2.

It is a simple but useful observation that the area of the rectangle in the Midpoint Rule is equal to the area of any trapezoid with base $[a, b]$ and *mid-height* $f(\frac{a+b}{2})$. If we take the trapezoid which is tangent to the graph at $x = \frac{a+b}{2}$, we might (and some do!) call the Midpoint Rule the *Tangent Rule*; see Fig. 13.3.

Okay, now let f be a convex function on $[a, b]$. By the Convex Tangents Lemma from Sect. 6.1 (i.e., that f lies above its tangent lines), and our simple observation just above, the Midpoint Rule is an *underestimate* of the integral. By the Convex Chords Lemma from Sect. 9.1 (i.e., that f lies below its chords), the Trapezoid Rule is an *overestimate*. Together

$$f(\tfrac{a+b}{2})[b-a] \le \int_a^b f(x)dx \le \frac{f(a)+f(b)}{2}[b-a]\,. \tag{13.3}$$

Inequalities (13.3) are sometimes called the Hermite-Hadamard Inequalities.[3] See Fig. 13.4 (and see [1] for a *proof without words*).

[3] Mathematical immortals Charles Hermite (France, 1822–1901) and Jacques Hadamard (France, 1865–1963) might have been bemused at having such pedestrian inequalities named for them.

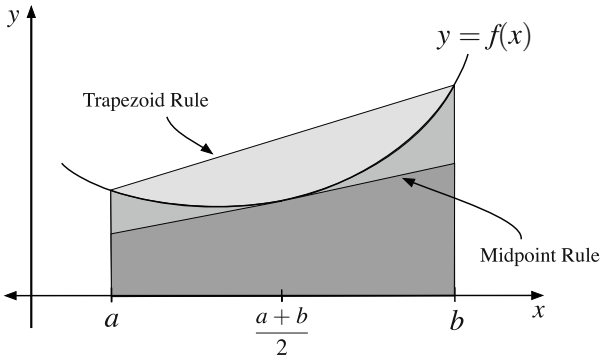


Fig. 13.4 The Hermite-Hadamard Inequalities (13.3), for a convex function

Example
The function $f(t) = 1/t$ is convex on $(0, \infty)$. Therefore by (13.3), with $a = x > 0$ and $b = x+1$, we have

$$\frac{1}{\frac{x+x+1}{2}} \;\le\; \int_x^{x+1} \frac{1}{t}\,dt \;\le\; \frac{1}{2}\left(\frac{1}{x} + \frac{1}{x+1}\right).$$

That is,

$$\frac{2}{2x+1} \;\le\; \ln\left(1+\frac{1}{x}\right) \;\le\; \frac{2x+1}{2x(x+1)} \qquad (x > 0). \tag{13.4}$$

Exponentiating throughout, this is equivalent with

$$\left(1+\frac{1}{x}\right)^{x\left(1+\frac{1}{2x+1}\right)} \;\le\; \mathrm{e} \;\le\; \left(1+\frac{1}{x}\right)^{x+1/2} \qquad (x > 0). \tag{13.5}$$

These estimates improve the

$$\left(1+\frac{1}{x}\right)^{x} \;\le\; \mathrm{e} \;\le\; \left(1+\frac{1}{x}\right)^{x+1}$$

which we met in Lemma 6.2 and then again in Sect. 7.3.

Next we show how some careful manipulations of the improvements (13.5) yield a proof of (13.1).

13.2 de Moivre's Version of Stirling's Formula

Most proofs of Stirling's Formula (13.1) come in two parts, and ours (similar to [9]) is no exception. We show

(a) $\lim_{n\to\infty} \dfrac{\mathrm{e}^n\, n!}{n^n\sqrt{n}} \;=\; C \;>\; 0\,,$ and then **(b)** $C = \sqrt{2\pi}\,.$

Part (a) that $\lim_{n\to\infty} \dfrac{\mathrm{e}^n\, n!}{n^n\sqrt{n}} \;>\; 0$ exists: For $n = 1, 2, 3, \ldots,$ set

$$a_n \;=\; \frac{\mathrm{e}^n\, n!}{n^n\sqrt{n}}\,.$$

Here we verify (i) that $\{a_n\}$ is *decreasing* and (ii) that $\{a_n\}$ is *bounded below* by some positive constant. Therefore, by the fundamental property of $\mathbb{R}$ (Sect. 2.3), there shall exist a number $C > 0$ such that $a_n \to C$ as $n \to \infty$.

(i) It is easy to see, after a bit of tidying, that

$$\frac{a_{n+1}}{a_n} \;=\; \frac{\mathrm{e}}{\left(1+\frac{1}{n}\right)^{n+1/2}}\,.$$

This is ≤ 1, by the right-hand side of (13.5), and so $\{a_n\}$ is indeed decreasing.

(ii) Here we shall use the left-hand side of (13.5), twice in fact. Replacing x with k there, it reads

$$\left(\frac{k+1}{k}\right)^k \left(\frac{k+1}{k}\right)^{\frac{k}{2k+1}} \;\le\; \mathrm{e}\,. \tag{13.6}$$

Taking products in (13.6), from $k = 1$ to n, and using the fact that

$$\prod_{k=1}^{n} \left(\frac{k+1}{k}\right)^k \;=\; \frac{(n+1)^n}{n!}\,,$$

we get

$$\frac{(n+1)^n}{n!} \prod_{k=1}^{n} \left(\frac{k+1}{k}\right)^{\frac{k}{2k+1}} \;\le\; \mathrm{e}^n\,.$$

Clearly now

$$\frac{\sqrt{n}}{\sqrt{n+1}}\,\frac{n^n}{n!} \;<\; \frac{(n+1)^n}{n!}\,,$$

and so we have

$$\frac{1}{\sqrt{n+1}} \prod_{k=1}^{n} \left(\frac{k+1}{k} \right)^{\frac{k}{2k+1}} \leq \frac{e^n n!}{n^n \sqrt{n}} = a_n .$$

However, $\sqrt{n+1} = \prod_{k=1}^{n} \left(\frac{k+1}{k} \right)^{1/2}$, so this reads

$$\prod_{k=1}^{n} \left(\frac{k+1}{k} \right)^{\frac{k}{2k+1} - \frac{1}{2}} \leq a_n ,$$

which is the same as

$$\prod_{k=1}^{n} \left(\frac{k+1}{k} \right)^{\frac{-1}{2k+1}} \leq a_n^2 . \tag{13.7}$$

Now we (lower) bound the left-hand side of (13.7). Beginning again with (13.6), and taking kth roots, we have

$$\left(\frac{k+1}{k} \right) \left(\frac{k+1}{k} \right)^{\frac{1}{2k+1}} \leq e^{\frac{1}{k}} ,$$

or

$$\left(\frac{k+1}{k} \right) e^{-\frac{1}{k}} \leq \left(\frac{k+1}{k} \right)^{\frac{-1}{2k+1}} .$$

Taking products, from $k = 1$ to n, this is

$$(n+1) \left(\prod_{k=1}^{n} e^{-\frac{1}{k}} \right) \leq \prod_{k=1}^{n} \left(\frac{k+1}{k} \right)^{\frac{-1}{2k+1}} .$$

So looking at (13.7), we get

$$(n+1) \left(\prod_{k=1}^{n} e^{-\frac{1}{k}} \right) \leq a_n^2 .$$

By (13.2), the left-hand side goes to $e^{-\gamma}$ as $n \to \infty$, and so

$$e^{-\gamma/2} \leq a_n .$$

Therefore, $\{a_n\}$ is indeed bounded below, by $e^{-\gamma/2} \approx 0.7493$. Good.

Part (b), that $C = \sqrt{2\pi}$: Using Wallis's Product, we saw in line (4.2) that

$$\frac{2^{2n}}{\sqrt{2n+1}} \cdot \frac{(n!)^2}{(2n)!} \quad \sim \quad \sqrt{\frac{\pi}{2}} \qquad (\text{as } n \to \infty).$$

By Part (a), we may substitute

$$n! \ \sim \ \frac{C\, n^n \sqrt{n}}{\mathrm{e}^n} \qquad \text{and} \qquad \frac{1}{(2n)!} \ \sim \ \frac{\mathrm{e}^{2n}}{C\,(2n)^{2n}\sqrt{2n}}.$$

This gives

$$\sqrt{\frac{\pi}{2}} \quad \sim \quad \frac{2^{2n}}{\sqrt{2n+1}} \cdot \frac{C^2 n^{2n}\, n}{\mathrm{e}^{2n}} \cdot \frac{\mathrm{e}^{2n}}{C(2n)^{2n}\sqrt{2n}}.$$

There is a lot of cancellation here—after the dust settles, this is

$$\sqrt{\frac{\pi}{2}} \quad \sim \quad \frac{C\, n}{\sqrt{2n+1}\sqrt{2n}}, \quad \text{which is} \ \sim \ \frac{C}{2},$$

and so indeed, $C = \sqrt{2\pi}$. ✓

▶ **Remark** We point out that since the sequence $\{a_n\}$ in the proof above is decreasing, (13.1) is an *underestimate*: $\sqrt{2\pi n}\,\dfrac{n^n}{\mathrm{e}^n} \ < \ n!$.

Exercises

13.1. Show that the number of digits of a number $x > 0$ to the left of the decimal point is

$$d(x) = \lfloor \log_{10}(x) \rfloor + 1.$$

13.2.

(a) Use (13.4) to show that $\left\{\left(1 + \frac{1}{n}\right)^n\right\}$ *increases* to e and that $\left\{\left(1 + \frac{1}{n}\right)^{n+1}\right\}$ *decreases* to e [3, 10].

(b) Use (13.4) to show that $\gamma = \lim_{n\to\infty} \left(\sum_{k=1}^{n} \frac{1}{k} - \ln(n) \right)$ exists.

(c) Recall that γ is the sum of the shaded regions in Fig. 13.5. Use (13.3) to show that $1/2 < \gamma < 2(1 - \ln(2))$ [9, 15].

13.3 ([4, 11, 13]). We saw that the Midpoint Rule is sometimes called the *Tangent Rule*, because it approximates the integral using the tangent line to f at the midpoint

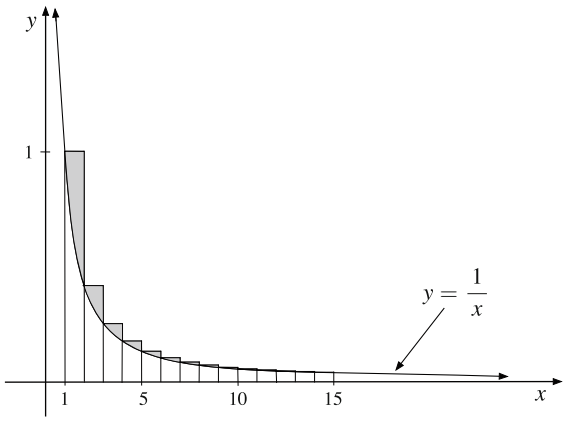


Fig. 13.5 (For Exercise 13.2) γ is the total area of *all* these shaded regions

of $[a, b]$. Let f be convex on $[a, b]$. Show that the Midpoint Rule yields a better approximation than one obtained by taking a tangent to f any other point of $[a, b]$.

13.4 (E.g., [12]). Show analytically, as follows, that for convex functions, the Midpoint Rule is always at least as good as the Trapezoid Rule:

$$0 \;\le\; \int_a^b f(x)\,dx - f\big(\tfrac{a+b}{2}\big)\big[b-a\big] \;\le\; \tfrac{f(a)+f(b)}{2}\big[b-a\big] - \int_a^b f(x)\,dx\,.$$

(So the left-hand side of (13.4) is at least as sharp as the right side of (13.4).)

(a) Let $c = (a + b)/2$. Apply the Trapezoid Rule on each of $[a, c]$ and $[c, b]$ to get

$$\int_a^c f(x)\,dx \;+\; \int_c^b f(x)\,dx \;\le\; \tfrac{f(a)+f(c)}{2}\big[c-a\big] \;+\; \tfrac{f(c)+f(b)}{2}\big[b-c\big].$$

(b) Do some manipulating, and use $c - a = b - c = (b - a)/2$, to get

$$2\int_a^b f(x)\,dx \;\le\; f(c)\big[b-a\big] \;+\; \tfrac{f(a)+f(b)}{2}\big[b-a\big]\,.$$

(c) Now show that this yields the desired inequalities.

13.5 ([5, 18]). Use the Trapezoid and Midpoint Rules to show $\frac{2n+1}{2n^2} < \sum_{k=n}^{\infty} \frac{1}{k^2} < \frac{1}{n-1/2}$.

13.6. Here's another way to obtain the right-hand side of (13.5): $\mathrm{e} \le \left(1+\frac{1}{x}\right)^{x+1/2}$.

(a) Verify that $\tanh(t) = \frac{\mathrm{e}^t - \mathrm{e}^{-t}}{\mathrm{e}^t + \mathrm{e}^{-t}}$ is concave for $t \ge 0$.
(b) Conclude that $\tanh(t) \le t$ for $t \ge 0$.
(c) Verify that $\tanh^{-1}(t) = \frac{1}{2}\ln\left(\frac{1+t}{1-t}\right)$.
(d) Conclude that $\mathrm{e}^{2t} \le \frac{1+t}{1-t}$ for $0 \le t < 1$.
(e) Set $t = \frac{1}{2x+1}$.

13.7. Show how (13.1), i.e.,

$$n! \sim \sqrt{2\pi n}\,\frac{n^n}{\mathrm{e}^n}$$

can be used to obtain Wallis's Product (Chap. 4)

$$\frac{2\cdot 2}{1\cdot 3}\frac{4\cdot 4}{3\cdot 5}\frac{6\cdot 6}{5\cdot 7}\cdots\cdot\frac{2n\cdot 2n}{(2n-1)(2n+1)} \sim \frac{\pi}{2}.$$

13.8.

(a) Discuss the relationship (if any!) between

$$f(n) \sim g(n) \quad \text{and} \quad f(n) - g(n) \to 0, \quad \text{as } n \to \infty.$$

(b) (E.g., [6, 8, 16]) Show that Stirling's Formula (due to de Moivre)

$$n! \sim \sqrt{2\pi n}\,\frac{n^n}{\mathrm{e}^n}$$

is equivalent with the formula (due to Stirling)

$$n! \sim \sqrt{2\pi}\left(\frac{n+1/2}{\mathrm{e}}\right)^{n+1/2}.$$

13.9. Denote by A_n the Arithmetic Mean of the first n natural numbers and by G_n their Geometric Mean (Sect. 9.2). Show that $\lim_{n\to\infty}\frac{G_n}{A_n} = \frac{2}{\mathrm{e}}$. (This is a rich little fact. There are many proofs, and it has been generalized almost beyond recognition.)

13.10.

(a) Let $0 < a < b$. Apply the Midpoint Rule, and then the Trapezoid Rule, to $f(x) = e^x$, on $[\ln(a), \ln(b)]$ to obtain the following relationship between the Geometric Mean, the *Logarithmic Mean*, and the Arithmetic Mean [2]:

$$G(a,b) = \sqrt{ab} \le \frac{b-a}{\ln(b)-\ln(a)} \le \frac{a+b}{2} = A(a,b)\,.$$

(b) Take $a = x$, $b = x+1$ to obtain the rather pleasing fact that

$$\left(\frac{x+1}{x}\right)^{G(x+1,\,x)} \le e \le \left(\frac{x+1}{x}\right)^{A(x+1,\,x)}.$$

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Euler's Sine Product

14

The function $\sin(\pi x)$ has infinitely many zeros: one at each $n \in \mathbb{Z}$. Just as we do with polynomials, we might think of factoring $\sin(\pi x)$ according to its zeros, like this (where C is some number $\neq 0$):

$$\sin(\pi x) \;=\; Cx\,(1-x^2)(4-x^2)(9-x^2)(16-x^2)\cdots \tag{14.1}$$

The right-hand side here is an odd function, as is sine ($\sin(\pi(-x)) = -\sin(\pi x)$), with zeros occurring at all the right places. However, the by-now-very-familiar limit

$$\lim_{x\to 0} \frac{\sin(\pi x)}{\pi x} \;=\; 1$$

dictates that in (14.1), we must have

$$1 \;=\; \lim_{x\to 0} \frac{\sin(\pi x)}{\pi x} \;=\; \frac{C}{\pi}\,(1)(4)(9)(16)\cdots\,.$$

Therefore we require—in some sense—that

$$C \;=\; \pi \cdot \frac{1}{1}\cdot\frac{1}{4}\cdot\frac{1}{9}\cdot\frac{1}{16}\cdots\,.$$

So we substitute this C into (14.1) but by distributing its factors *in every other spot* throughout:

$$\begin{aligned}\sin(\pi x) \;&=\; C\,x(1-x^2)(4-x^2)(9-x^2)(16-x^2)\cdots \\ &=\; \pi\,x\,\frac{1}{1}(1-x^2)\,\frac{1}{4}(4-x^2)\,\frac{1}{9}(9-x^2)\,\frac{1}{16}(16-x^2)\cdots\end{aligned}$$

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$$= \pi x \left(1 - \frac{x^2}{1}\right)\left(1 - \frac{x^2}{4}\right)\left(1 - \frac{x^2}{9}\right)\left(1 - \frac{x^2}{16}\right)\cdots$$

$$= \pi x \prod_{n=1}^{\infty}\left(1 - \frac{x^2}{n^2}\right). \tag{14.2}$$

Now (14.2) yields the correct answer for the limit above. Things are looking up.

Let us denote by $\phi(x)$ the right-hand side of (14.2):

$$\phi(x) = \pi x \prod_{n=1}^{\infty}\left(1 - \frac{x^2}{n^2}\right).$$

Further evidence that (14.2) *might* be correct is provided by the fact that $\phi(x)$ has the same periodicity as $\sin(\pi x)$, viz. $\phi(x+2) = \phi(x)$. We verify this presently. It's a little messy, so you can be forgiven for skipping it on your first go-through and going directly to $\hookrightarrow$ below.

$$\begin{aligned}
\phi(x+1) &= \pi(x+1)\prod_{n=1}^{\infty}\left(1 - \frac{(x+1)^2}{n^2}\right)\\
&= \pi(x+1)\prod_{n=1}^{\infty}\left(1 - \frac{(x+1)}{n}\right)\left(1 + \frac{(x+1)}{n}\right)\\
&= \pi(x+1)\prod_{n=1}^{\infty}\left[\left(1 - \frac{x}{n}\right) - \frac{1}{n}\right]\left[\left(1 + \frac{x}{n}\right) + \frac{1}{n}\right]\\
&= \pi(x+1)\prod_{n=1}^{\infty}\left[\left(1 - \frac{x^2}{n^2}\right) + \frac{1}{n}\left(1 - \frac{x}{n}\right) - \frac{1}{n}\left(1 + \frac{x}{n}\right) - \frac{1}{n^2}\right]\\
&= \pi(x+1)\prod_{n=1}^{\infty}\left(1 - \frac{x^2}{n^2}\right)\left[1 - \frac{1}{n\left(1 - \frac{x}{n}\right)}\right]\left[1 + \frac{1}{n\left(1 + \frac{x}{n}\right)}\right]\\
&= \pi(x+1)\prod_{n=1}^{\infty}\left(1 - \frac{x^2}{n^2}\right)\left[\frac{n-x-1}{n-x}\right]\left[\frac{n+x+1}{n+x}\right].
\end{aligned}$$

The latter two pieces telescope:

$$\prod_{n=1}^{N}\left[\frac{n-x-1}{n-x}\right]\left[\frac{n+x+1}{n+x}\right] = \frac{-x}{N-x}\cdot\frac{N+x+1}{1+x} \underbrace{\longrightarrow}_{\text{as } N\to\infty} \frac{-x}{1+x}.$$

Therefore

$$\phi(x+1) \;=\; \pi\,(x+1)\prod_{n=1}^{\infty}\left(1-\frac{x^2}{n^2}\right)\left(\frac{-x}{1+x}\right) \;=\; -\phi(x)\,.$$

Finally, $\phi(x+2)=\phi(x+1+1)=-\phi(x+1)=\phi(x)$, and so ϕ is 2-periodic. ✓

↪ Euler[1] was the first to conjecture, and then prove, that $\phi(x)$ indeed equals $\sin(\pi x)$ for all x. In this chapter we shall prove (14.2) and then bask in some of its amazing consequences.

14.1 The Sine Product

Theorem 14.1
For all $x \in \mathbb{R}$, we have

$$\sin(\pi x) \;=\; \pi x\prod_{n=1}^{\infty}\left(1-\frac{x^2}{n^2}\right). \tag{14.3}$$

Proof. Our proof follows [4], which anticipates similar approaches rediscovered in [2,3]. For a fixed $x \in \mathbb{R}$ and $N \in \mathbb{N}$, set

$$J_N(x) \;=\; \int_0^{\pi/2} \cos^N(t)\cos(2xt)\,dt\,.$$

The 2 in the integrand here is not entirely necessary; it just makes subsequent formulas come out a little nicer.[2] More importantly, observe that

$$J_N(0) \;=\; \int_0^{\pi/2} \cos^N(t)\,dt \;=\; I_N\,.$$

These are the same integrals that we considered on the way to obtaining Wallis's Product in Chap. 4. There we saw that $I_0=\frac{\pi}{2}$ and $I_1=1$, and Lemma 4.1 gave

[1] Euler obtained the product in 1734. Famous Americans Daniel Boone, Paul Revere, and Thomas Sumter were all born that year.

[2] You will have noticed by now that proofs often seem mysterious because we only see the shiny finished product. There's never a mention of the countless failed efforts and bits of recycled paper.

$$I_{2n} \;=\; \frac{\pi}{2}\prod_{k=1}^{n}\frac{2k-1}{2k} \qquad \text{and} \qquad I_{2n+1} \;=\; \prod_{k=1}^{n}\frac{2k}{2k+1}\,. \tag{14.4}$$

Multiplying these together (as we did in Chap. 5), we get

$$I_N \cdot I_{N-1} \;=\; \frac{\pi/2}{N}. \tag{14.5}$$

Now the proof gets pretty messy. We break it into three bits, so we might take a few deep breaths in between.

(i) We obtain a reduction formula for the $J_N(x)$s, as we did with the I_Ns in Chap. 4: For $N \geq 2$ and $x \neq 0$, integrating $J_N(x)$ by parts twice, we get

$$\begin{aligned}
J_N(x) \;&=\; \int_0^{\pi/2} \cos^N(t)\cos(2xt)dt \;=\; \int_0^{\pi/2} \underbrace{\cos^N(t)}_{\mathrm{D}}\,\underbrace{\cos(2xt)}_{\mathrm{I}}\,dt \\
&=\; \cos^N(t)\frac{\sin(2xt)}{2x}\,\Big|_{t=0}^{\pi/2} \;-\; \frac{N}{2x}\int_0^{\pi/2}\cos^{N-1}(t)(-\sin(t))\sin(2xt)dt \\
&=\; 0 \;+\; \frac{N}{2x}\int_0^{\pi/2} \underbrace{\cos^{N-1}(t)\sin(t)}_{\mathrm{D}}\,\underbrace{\sin(2xt)}_{\mathrm{I}}\,dt\,. \\
&=\; \frac{N}{2x}\left(\cos^{N-1}(t)\sin(t)\frac{-\cos(2xt)}{2x}\,\Big|_{t=0}^{\pi/2}\right. \\
&\quad -\; \left.\frac{N}{2x}\int_0^{\pi/2}\Big[(N-1)\cos^{N-2}(t)\big(-\sin^2(t)\big) \;+\; \cos^N(t)\Big]\frac{-\cos(2xt)}{2x}dt\right) \\
&=\; 0 \;+\; \frac{N}{4x^2}\int_0^{\pi/2}\Big[(N-1)\cos^{N-2}(t)\big(-1+\cos^2(t)\big) \;+\; \cos^N(t)\Big] \\
&\quad \times \cos(2xt)dt\,.
\end{aligned}$$

From here, manipulations and tidying give

$$J_N(x) = \frac{N}{4x^2}\int_0^{\pi/2}\Big[-(N-1)\cos^{N-2}(t)+(N-1)\cos^N(t)+\cos^N(t)\Big]$$
$$\times\cos(2xt)dt$$
$$= -\frac{N(N-1)}{(2x)^2}\int_0^{\pi/2}\cos^{N-2}(t)\cos(2xt)dt$$
$$+\frac{N^2}{(2x)^2}\int_0^{\pi/2}\cos^N(t)\cos(2xt)dt,$$

which is

$$J_N(x) = -\frac{N(N-1)}{(2x)^2}J_{N-2}(x) + \frac{N^2}{(2x)^2}J_N(x)\,.$$

Solving for $J_N(x)$ we obtain the reduction formula

$$J_N(x) = \frac{N-1}{N}\frac{1}{\left(1-(\frac{2x}{N})^2\right)}J_{N-2}(x) \quad (\text{for } N\geq 2)\,. \tag{14.6}$$

[Having $x = 0$ gives the reduction formula $I_N = \frac{N-1}{N}I_{N-2}$ from Chap. 4. Nice!]

(ii) Using (14.6) repeatedly, with $N = 2n$ (even), we get

$$J_{2n}(x) = \frac{2n-1}{2n}\frac{1}{1-\left(\frac{2x}{2n}\right)^2}J_{2n-2}(x) = \frac{2n-1}{2n}\frac{1}{1-\frac{x^2}{n^2}}J_{2n-2}(x)$$
$$= \frac{2n-1}{2n}\frac{2n-3}{2n-2}\frac{1}{1-\frac{x^2}{n^2}}\frac{1}{1-\frac{x^2}{(n-1)^2}}J_{2n-4}(x)$$
$$= \frac{2n-1}{2n}\frac{2n-3}{2n-2}\frac{2n-5}{2n-4}\frac{1}{1-\frac{x^2}{n^2}}\frac{1}{1-\frac{x^2}{(n-1)^2}}\frac{1}{1-\frac{x^2}{(n-2)^2}}J_{2n-6}(x)$$
$$\vdots$$
$$\downarrow$$
$$= \prod_{k=1}^{n}\frac{2k-1}{2k}\frac{1}{\prod_{k=1}^{n}\left(1-\frac{x^2}{k^2}\right)}\cdot J_0(x)\,.$$

Now, looking at (14.4), we recognize the first product here as $\frac{2}{\pi} I_{2n}$, and

$$J_0(x) \;=\; \int_0^{\pi/2} \cos(2xt)\,dt \;=\; \frac{\sin(2xt)}{2x}\,\Big|_{t=0}^{\pi/2} \;=\; \frac{\sin(\pi x)}{2x}\,,$$

so we have

$$J_{2n}(x) \;=\; \frac{2}{\pi}\, I_{2n}\, \frac{1}{\prod_{k=1}^{n}\left(1-\frac{x^2}{k^2}\right)}\, \frac{\sin(\pi x)}{2x}\,.$$

After some rearranging, this is

$$\sin(\pi x) \;=\; \pi x \prod_{k=1}^{n}\left(1-\frac{x^2}{k^2}\right)\, \frac{J_{2n}(x)}{I_{2n}}\,. \tag{14.7}$$

(iii) Finally, we claim that

$$\frac{J_{2n}(x)}{I_{2n}} \;\to\; 1 \quad \text{as} \quad n \to \infty\,.$$

So looking at (14.7), this will complete the proof. To verify the claim (e.g., [1], p127–128), for $t \in [0, \pi/2]$ and $x \geq 0$, we have

$$\big|\cos(2xt)-1\big| \;=\; \Big|\int_0^{2xt} -\sin(u)du\Big| \;\leq\; \int_0^{2xt} \big|\sin(u)\big|du \;\leq\; \int_0^{2xt} 1\,du \;=\; 2xt.$$

And for $x \leq 0$, we have

$$\big|\cos(2xt)-1\big| \;=\; \Big|\int_0^{2xt} -\sin(u)du\Big| \;\leq\; \int_{2xt}^{0} \big|\sin(u)\big|du \;\leq\; \int_{2xt}^{0} 1du \;=\; -2xt.$$

Either way,

$$\big|\cos(2xt)-1\big| \;\leq\; 2\big|x\big|t \quad \text{for} \quad t \in [0, \pi/2].$$

Therefore

$$\left| J_{2n}(x) - I_{2n} \right| \;\le\; \int_0^{\pi/2} \left| \cos(2xt) - 1 \right| \cos^{2n}(t)\, dt \;\le\; 2|x| \int_0^{\pi/2} t \cos^{2n}(t)\, dt.$$

Now (from Sect. 9.1), $t \le \tan(t)$ on $[0, \pi/2)$, and so $t\cos(t) \le \sin(t)$ on $[0, \pi/2]$.

Therefore

$$\begin{aligned}
\left| J_{2n}(x) - I_{2n} \right| &\le 2|x| \int_0^{\pi/2} t \cos^{2n}(t)\, dt \\
&= 2|x| \int_0^{\pi/2} t \cos(t) \cos^{2n-1}(t)\, dt \\
&\le 2|x| \int_0^{\pi/2} \sin(t) \cos^{2n-1}(t)\, dt \\
&= \frac{|x|}{n} \qquad \text{(after setting } u = \cos(t)) \,.
\end{aligned}$$

Dividing through by $I_{2n} > 0$, we get

$$\left| \frac{J_{2n}(x)}{I_{2n}} - 1 \right| \;\le\; \frac{|x|}{n\, I_{2n}} \,.$$

Since $0 \le \cos(t) \le 1$ on $[0, \pi/2]$, the I_Ns decrease as N increases. This and (14.5) give

$$\frac{1}{I_{2n}^2} \;<\; \frac{1}{I_{2n} I_{2n+1}} \;=\; \frac{2n+1}{\pi/2} \,.$$

Therefore

$$\left| \frac{J_{2n}(x)}{I_{2n}} - 1 \right| \;\le\; \frac{|x|}{n I_{2n}} \;<\; \sqrt{\frac{2}{\pi}}\, \frac{\sqrt{2n+1}}{n} |x|$$

which $\to 0$ as $n \to \infty$, and our proof is complete. □

14.2 Amazing Consequences

- The *Cosine Product*: For all $x \in \mathbb{R}$, we have

$$\cos(\pi x) = \prod_{n=1}^{\infty} \left(1 - \frac{4x^2}{(2n-1)^2}\right).$$

Here's why: The trig identity $\sin(2\pi x) = 2\sin(\pi x)\cos(\pi x)$ gives

$$\cos(\pi x) = \frac{\sin(2\pi x)}{2\sin(\pi x)}.$$

Then by the Sine Product (14.3),

$$\begin{aligned}
\cos(\pi x) &= \frac{2\pi x \prod_{n=1}^{\infty} \left(1 - \frac{(2x)^2}{n^2}\right)}{2\pi x \prod_{n=1}^{\infty} \left(1 - \frac{x^2}{n^2}\right)} \\
&= \frac{\prod_{n \text{ even}} \left(1 - \frac{(2x)^2}{n^2}\right) \prod_{n \text{ odd}} \left(1 - \frac{(2x)^2}{n^2}\right)}{\prod_{n=1}^{\infty} \left(1 - \frac{x^2}{n^2}\right)} \\
&= \frac{\prod_{n=1}^{\infty} \left(1 - \frac{(2x)^2}{(2n)^2}\right) \prod_{n=1}^{\infty} \left(1 - \frac{(2x)^2}{(2n-1)^2}\right)}{\prod_{n=1}^{\infty} \left(1 - \frac{x^2}{n^2}\right)} \\
&= \prod_{n=1}^{\infty} \left(1 - \frac{4x^2}{(2n-1)^2}\right).
\end{aligned}$$

▶ **Remark** In our proof above of (14.3), instead of step (ii), we could have used the recursion formula (14.6) with $N = 2n+1$, and then some appropriate modifications of step (iii), to obtain the Cosine Product. We leave the details for Exercise 14.6.

- Taking $x = 1/2$ in the Sine Product (14.3), we get $\prod_{n=1}^{\infty} \left(1 - \frac{1}{4n^2}\right) = \frac{2}{\pi}$, which yields *Wallis's Product*:

$$\frac{2}{\pi} = \prod_{n=1}^{\infty} \left(\frac{4n^2-1}{4n^2}\right) = \prod_{n=1}^{\infty} \left(\frac{(2n-1)(2n+1)}{(2n)(2n)}\right) = \frac{1\cdot 3}{2\cdot 2} \cdot \frac{3\cdot 5}{4\cdot 4} \cdot \frac{5\cdot 7}{6\cdot 6} \cdot \frac{7\cdot 9}{8\cdot 8} \cdots .$$

- Taking $x = 1/4$ in the Cosine Product, we get $\frac{1}{\sqrt{2}} = \prod_{n=1}^{\infty} \left(1 - \frac{1}{4(2n-1)^2}\right)$, or

$$\begin{aligned}\frac{1}{\sqrt{2}} &= \prod_{n=1}^{\infty} \left(\frac{4(2n-1)^2 - 1}{4(2n-1)^2}\right) \\ &= \prod_{n=1}^{\infty} \left(\frac{2(2n-1) - 1}{2(2n-1)} \cdot \frac{2(2n-1) + 1}{2(2n-1)}\right) \\ &= \prod_{n=1}^{\infty} \left(\frac{4n-3}{2(2n-1)} \cdot \frac{4n-1}{2(2n-1)}\right).\end{aligned}$$

  Therefore

$$\sqrt{2} = \frac{2 \cdot 2}{1 \cdot 3} \cdot \frac{6 \cdot 6}{5 \cdot 7} \cdot \frac{10 \cdot 10}{9 \cdot 11} \cdots \frac{(4n-2)^2}{(4n-3)(4n-1)} \cdots .$$

  Other fine infinite products can be similarly obtained (e.g., Exercise 14.7).
- And here now is Euler's own *pièce de résistance*. Brace yourself. On one hand, the Maclaurin Series for sine (Sect. 2.1) yields

$$\frac{\sin(\pi x)}{\pi x} = 1 - \frac{\pi^2 x^2}{3!} + \frac{\pi^4 x^4}{5!} - \frac{\pi^6 x^6}{7!} + \cdots ,$$

  and the coefficient of x^2 here is $-\frac{\pi^2}{3!}$. On the other hand, the Sine Product reads

$$\frac{\sin(\pi x)}{\pi x} = \prod_{n=1}^{\infty} \left(1 - \frac{x^2}{n^2}\right) = \left(1 - \frac{x^2}{1^2}\right)\left(1 - \frac{x^2}{2^2}\right)\left(1 - \frac{x^2}{3^2}\right)\left(1 - \frac{x^2}{4^2}\right) \cdots ,$$

  and the coefficient of x^2 here is $-\frac{1}{1^2} - \frac{1}{2^2} - \frac{1}{3^2} - \frac{1}{4^2} - \cdots$.

  Putting our hands together (yes), we get $-\frac{\pi^2}{3!} = -\left(\frac{1}{1^2} + \frac{1}{2^2} + \frac{1}{3^2} + \cdots\right)$, which solves the *Basel Problem*:

$$\sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6}. \quad \text{Woof!}$$

Exercises

(You might consider (again?) the first few exercises from Chap. 11.)

14.1. By carefully taking logarithms and using the *Limit Comparison Test*, verify that

$$\phi(x) \;=\; \pi x \prod_{n=1}^{\infty} \left(1 - \frac{x^2}{n^2}\right)$$

even makes sense, i.e., that it converges for all x.

14.2. If you enjoyed seeing that $\phi(x+2) = \phi(x)$ above, try this one. Set

$$\psi(x) \;=\; \prod_{n=1}^{\infty} \left(1 - \frac{4x^2}{(2n-1)^2}\right),$$

and show that $\psi(x+1/2) \;=\; -\phi(1/2)\phi(x)$ for all x. Once we know what ϕ and ψ *really* are, this boils down to $\cos(x + \pi/2) = -\sin(x)$.

14.3. Use the Cosine Product and the Maclaurin Series for cosine (Sect. 2.1) to show that

$$\sum_{n\,odd} \frac{1}{n^2} \;=\; \frac{\pi^2}{8}\,.$$

14.4. Show that $\dfrac{\sin(\pi(x+\alpha))}{\sin(\pi\alpha)} = \frac{x+\alpha}{\alpha} \displaystyle\prod_{n=1}^{\infty} \left(1 - \frac{x}{n-\alpha}\right)\left(1 + \frac{x}{n+\alpha}\right).$

14.5. Write down an infinite product for $\tan(\pi x)$ and simplify. Where is it valid?

14.6. Fill in the details of another way of obtaining the Cosine Product, as follows:

(a) Using (14.6) repeatedly, with $N = 2n+1$ (odd), show that

$$J_{2n+1}(x) \;=\; I_{2n+1} \cdot \frac{1}{\displaystyle\prod_{k=1}^{n} \left(1 - \frac{x^2}{(k+1/2)^2}\right)} \cdot J_1(x)\,.$$

(b) Use the trig identity $\cos(t)\cos(2xt) = \frac{1}{2}\big(\sin(t+2xt) + \sin(t-2xt)\big)$ to show that

$$J_1(x) \;=\; \frac{\cos(\pi x)}{1-4x^2}\,.$$

(c) Conclude that

$$\cos(\pi x) \;=\; \prod_{k=1}^{n}\left(1-\frac{x^2}{(k-1/2)^2}\right)\frac{J_{2n+1}(x)}{I_{2n+1}}\,.$$

(d) Show that $\dfrac{J_{2n+1}(x)}{I_{2n+1}} \to 1$ as $n \to \infty$. Done!

14.7.

(a) Show that $\dfrac{\pi}{2\sqrt{2}} \;=\; \displaystyle\prod_{n=1}^{\infty}\frac{(4n)^2}{(4n-1)(4n+1)} \;=\; \frac{4^2}{3\cdot 5}\cdot\frac{8^2}{7\cdot 9}\cdot\frac{12^2}{11\cdot 13}\cdot\cdots\,.$

(b) Verify that the perimeter of a regular $2N$-sided polygon inscribed in the unit circle is $4N\sin(\pi/(2N))$. Express this as an infinite product.

(c) Show that $\dfrac{\sqrt{3}}{2} \;=\; \displaystyle\prod_{n=1}^{\infty}\frac{(6n-4)(6n-2)}{(6n-3)^2} \;=\; \frac{2\cdot 4}{3^2}\cdot\frac{8\cdot 10}{9^2}\cdot\frac{14\cdot 16}{15^2}\cdot\cdots\,.$

14.8. Show that $\displaystyle\sum_{n=1}^{\infty}\frac{1}{n^4} \;=\; \frac{\pi^4}{90}$. How about $\displaystyle\sum_{n=1}^{\infty}\frac{1}{n^6}$? (Or see Chap. 12 . . . .)

14.9.

(a) Show that

$$\sinh(\pi x) \;=\; \pi x\prod_{n=1}^{\infty}\left(1+\frac{x^2}{n^2}\right).$$

[Mimic the proof we used for sine, beginning with $\displaystyle\int_0^{\pi/2}\cos^N(t)\cosh(2xt)dt$, or, if you know some complex variables, $\sinh(\pi x) = -i\sin(i\pi x)$, where $i^2 = -1$.]

(b) Show that

$$\prod_{n=2}^{\infty}\frac{n^2-1}{n^2+1} \;=\; \frac{\pi}{\sinh(\pi)}.$$

14.10. Show that

$$\cosh(\pi x) \;=\; \prod_{n=1}^{\infty} \left(1 + \frac{4x^2}{(2n-1)^2}\right).$$

[Use Exercise 14.9 and a hyperbolic trig identity, or mimic the proof for cosine, or, if you know some complex variables, $\cosh(\pi x) = \cos(i\,\pi x)$, where $i^2 = -1$.]

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Postlude: Stirling's Formula Again

15

Stirling's Formula

$$n! \sim \sqrt{2\pi}\,\frac{n^{n+1/2}}{\mathrm{e}^n}\,, \tag{15.1}$$

which we verified in Chap. 13, is in fact (more or less) due to de Moivre [9, 13, 14]. The more accurate *Stirling's* Formula, which did appear in the exercises there, is

$$n! \sim \sqrt{2\pi}\left(\frac{n+1/2}{\mathrm{e}}\right)^{n+1/2}. \tag{15.2}$$

(Again, the $\sim$ means that the quotient of these two quantities $\to 1$ as $n \to \infty$.)

In this final investigation, we verify *Stirling's* Formula[1] (15.2), but without (quite) appealing to Wallis's Product, as we did when we obtained (15.1). We shall, however, use Euler's Sine Product from Chap. 14, of which Wallis's Product is a consequence:

$$\sin(\pi u) = \pi u \prod_{n=1}^{\infty}\left(1 - \frac{u^2}{n^2}\right). \tag{15.3}$$

This particular approach provides excellent connections among many of the things that we have been studying.

[1] Algebraist William Burnside (England, 1852–1927) demonstrated (15.2) in [4], so it is sometimes attributed to him. However (15.2) is indeed contained in Stirling's original work [3, 9, 13].

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15.1 Prep Work

We shall use the *Trapezoid Rule* $\int_a^b f(x)dx \approx \frac{f(a)+f(b)}{2}[b-a]$,

and the *Midpoint Rule* $\int_a^b f(x)dx \approx f(\frac{a+b}{2})[b-a]$.

Have a look at Sect. 13.1 again, to refresh if you need to. A couple of applications of the Hermite-Hadamard Inequalities (13.3) there, to the concave function $f(x) = \ln(x)$, give us the following:

- The Trapezoid Rule is an underestimate of the integral on $[k, k+1]$:

$$\frac{\ln(k) + \ln(k+1)}{2} \leq \int_k^{k+1} \ln(x)\,dx \qquad (k > 0). \tag{15.4}$$

- The Midpoint Rule is an overestimate of the integral on $[k-1/2, k+1/2]$:

$$\int_{k-1/2}^{k+1/2} \ln(x)\,dx \leq \ln\Big(\frac{k-1/2+k+1/2}{2}\Big) = \ln(k) \qquad (k > 1/2). \tag{15.5}$$

15.2 Stirling's Version of Stirling's Formula

The proof we give is a reorganization of most of the argument in [6]; see also [7]. The idea is to apply carefully the Midpoint Rule to $\ln(x)$ on each of the intervals

$$\Big[\tfrac{1}{2}, \tfrac{3}{2}\Big],\ \Big[\tfrac{3}{2}, \tfrac{5}{2}\Big],\ \Big[\tfrac{5}{2}, \tfrac{7}{2}\Big],\ \ldots,\ \Big[n-\tfrac{1}{2}, n+\tfrac{1}{2}\Big]$$

for an approximation of $\int_{1/2}^{n+1/2} \ln(x)dx$ (n large), which will inevitably contain $\ln(1) + \ln(2) + \ln(3) + \cdots + \ln(n) = \ln(n!)$. This is another messy one, so we'll break it up into bits.

(i) Observe that applying the Midpoint Rule to *any* function on $[k-1/2, k+1/2]$ is the same as applying the *Right-Rectangle Rule* on $[k-1/2, k]$, together with the *Left-Rectangle Rule* on $[k, k+1/2]$. See Fig. 15.1.

Since $f(x) = \ln(x)$ is increasing on $(0, \infty)$, the Right-Rectangle Rule is an overestimate of the integral, and the Left-Rectangle Rule is an underestimate. That is,

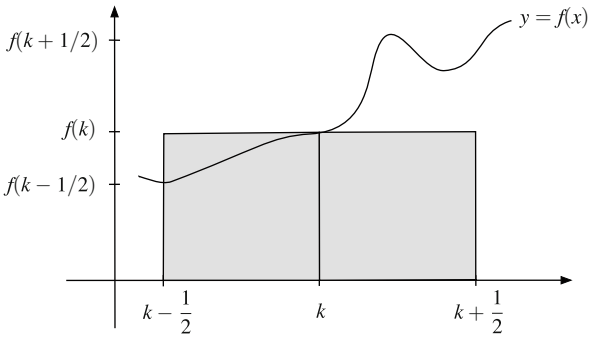


Fig. 15.1 The area of the large shaded rectangle is the sum of the areas of the two smaller shaded rectangles

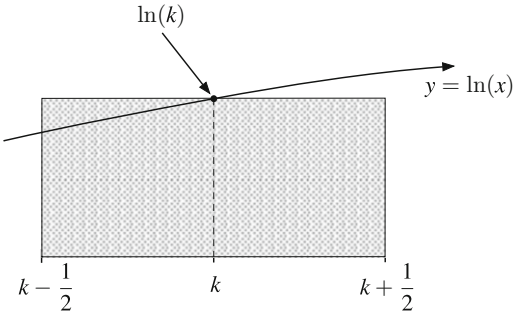


Fig. 15.2 Goldilocks Syndrome: The rectangle on the left is too big and the one on the right too small

$$\frac{1}{2}\ln(k) - \int_{k-1/2}^{k} \ln(x)dx > 0 \quad \text{and} \quad \frac{1}{2}\ln(k) - \int_{k}^{k+1/2} \ln(x)dx < 0\,. \tag{15.6}$$

See Fig. 15.2.

(ii) Over $\left[\frac{1}{2}, n+\frac{1}{2}\right]$, denote by $L(x)$ the piecewise-constant function which is equal to $\ln(1)$ on $\left[\frac{1}{2}, \frac{3}{2}\right]$, to $\ln(2)$ on $\left[\frac{3}{2}, \frac{5}{2}\right]$, to $\ln(3)$ on $\left[\frac{5}{2}, \frac{7}{2}\right]$, etc.:

$$L(x) = \ln(k) \quad \text{for} \quad k - \tfrac{1}{2} < x \le k + \tfrac{1}{2} \qquad (k = 1, 2, 3, \ldots).$$

Then (15.6) reads (see Fig. 15.3)

$$a_k := \int_{k-1/2}^{k} \big[L(x) - \ln(x)\big]dx > 0 \quad \text{and} \quad b_k := \int_{k}^{k+1/2} \big[\ln(x) - L(x)\big]dx > 0\,.$$

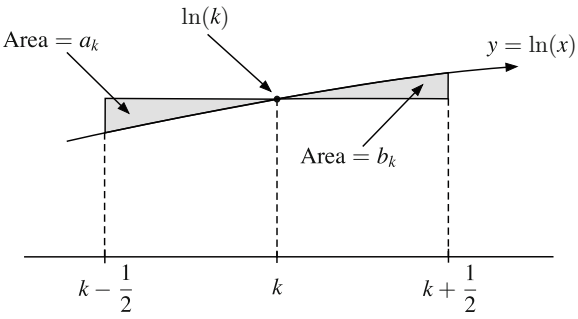


Fig. 15.3 a_k and b_k are the areas of the two shaded *curvilinear* triangles

Then we have $\int_{1/2}^{n+1/2} \big[L(x) - \ln(x)\big]dx$

$$= \left(\int_{1/2}^{1} + \int_{1}^{3/2} + \int_{3/2}^{2} + \int_{2}^{5/2} + \cdots + \int_{n-1/2}^{n} + \int_{n}^{n+1/2} \right) \big[L(x) - \ln(x)\big]dx$$

$$= \; a_1 - b_1 + a_2 - b_2 + \cdots + a_n - b_n \; := \; S_n \,, \text{ say,}$$

so that S_n is a partial sum of an alternating series.

(iii) We show that $\{S_n\}$ converges. Line (15.4) gives

$$\tfrac{1}{2}\ln(k) + \tfrac{1}{2}\ln(k+1) \;\le\; \int_{k}^{k+1} \ln(x)\,dx \;=\; \int_{k}^{k+1/2} \ln(x)dx \;+\; \int_{k+1/2}^{k+1} \ln(x)dx \,,$$

which is the same as

$$\int_{k+1/2}^{k+1} \big[L(x) - \ln(x)\big]dx \;\le\; \int_{k}^{k+1/2} \big[\ln(x) - L(x)\big]dx \,,$$

or

$$a_{k+1} \;\le\; b_k \,.$$

Line (15.5) gives

$$\int_{k-1/2}^{k} \ln(x)dx \;+\; \int_{k}^{k+1/2} \ln(x)dx \;=\; \int_{k-1/2}^{k+1/2} \ln(x)\,dx \;\le\; \ln(k) \;=\; \frac{1}{2}\ln(k) + \frac{1}{2}\ln(k) \,.$$

This is the same as

$$\int_{k}^{k+1/2} \big[\ln(x) - L(x)\big]dx \;\le\; \int_{k-1/2}^{k} \big[L(x) - \ln(x)\big]dx\,,$$

or

$$b_k \;\le\; a_k\,.$$

And since $\ln(x)$ is increasing, we also have (Right-Rectangle > Left-Rectangle)

$$a_k \;\le\; \ln(k) - \ln\Big(k - \tfrac{1}{2}\Big) \;\to\; 0 \quad \text{as } k \to \infty\,.$$

In summary then, we have

$$0 \;<\; a_{k+1} \le b_k \le a_k \;\to\; 0 \quad \text{as } k \to \infty\,.$$

Therefore, by the Alternating Series Test (Sect. 2.3), *every* sequence of partial sums of

$$a_1 - b_1 \;+\; a_2 - b_2 \;+\; \cdots \;+\; a_k - b_k + \cdots$$

converges—to S, say (with $0 < a_1 - b_1 < S < a_1$ being the coarsest of estimates). In particular then, $\{S_n\}$ converges to S.

(iv) We use the fact that $S_n \to S$ to obtain an expression for $\ln(n!)$; hence for $n!$

$$S_n = \int_{1/2}^{n+1/2} \big[L(x) - \ln(x)\big]\,dx \;=\; \int_{1/2}^{n+1/2} L(x)\;dx \;-\; \int_{1/2}^{n+1/2} \ln(x)\,dx$$

$$= \;\big(\ln(1) + \ln(2) + \cdots + \ln(n)\big) \;-\; \big(x\ln(x) - x\big)\Big|_{1/2}^{n+1/2}$$

$$= \;\ln(n!) \;-\; \Big(n + \tfrac{1}{2}\Big)\ln\Big(n + \tfrac{1}{2}\Big) \;+\; n \;+\; \tfrac{1}{2}\ln(\tfrac{1}{2})\,.$$

Therefore

$$\ln(n!) \;=\; S_n \;+\; \Big(n + \tfrac{1}{2}\Big)\ln\Big(n + \tfrac{1}{2}\Big) \;-\; n \;+\; \ln(\sqrt{2})\,.$$

Exponentiating, this is

$$n! \;=\; C_n \,\frac{\left(n+\frac{1}{2}\right)^{n+1/2}}{\mathrm{e}^{\,n}}$$

where $C_n \to C = \sqrt{2}\,\mathrm{e}^{\,S} > 0$, as $n \to \infty$.

(v) At this point we could approximate C to any desired degree of accuracy. Or we could compute C using Wallis's Product, much as we did in Sect. 13.2. But instead, we shall compute C by appealing to the Sine Product. This is pretty cool:

$$a_k \;-\; b_k \;=\; \int_{k-1/2}^{k} \ln\left(\tfrac{k}{x}\right) dx \;\;-\;\; \int_{k}^{k+1/2} \ln\left(\tfrac{x}{k}\right) dx\,.$$

Substituting $u = k - x$ and $u = x - k$ in these two integrals, respectively, we get

$$a_k - b_k \;=\; -\int_0^{1/2} \ln\left(1-u/k\right) du \;-\; \int_0^{1/2} \ln\left(1+u/k\right) du \;=\; -\int_0^{1/2} \ln\left(1-u^2/k^2\right) du\,.$$

Therefore[2]

$$S = \sum_{k=1}^{\infty}(a_k - b_k) = -\int_0^{1/2}\left(\sum_{k=1}^{\infty}\ln\left(1-u^2/k^2\right)\right) du = -\int_0^{1/2}\ln\left(\prod_{k=1}^{\infty}\left(1-u^2/k^2\right)\right) du.$$

By the Sine Product (15.3), this reads

$$S \;=\; -\int_0^{1/2}\ln\left(\frac{\sin(\pi u)}{\pi u}\right) du \;=\; -\int_0^{1/2}\ln(\sin(\pi u))du \;+\; \int_0^{1/2}\ln(\pi)du \;+\; \int_0^{1/2}\ln(u)du\,.$$

We claim now, and verify below in item (vi), that

[2] Across the second = sign, we have interchanged the order of the integral (the limit of an infinite process) and the infinite sum (the limit of an another infinite process). This is a dodgy matter, but there are theorems which tell us when we can do it with impunity. The applicable theorem here is basically *Lebesgue's Dominated Convergence Theorem* (e.g., [1], p 272).

$$\int_0^{1/2} \ln(\sin(\pi u))du \;=\; -\frac{1}{2}\ln(2)\,, \qquad (\star)$$

whence

$$S \;=\; \frac{1}{2}\ln(2) \,+\, \frac{1}{2}\ln(\pi) \,+\, \frac{1}{2}\ln(\tfrac{1}{2}) \,-\, \frac{1}{2} \;=\; \ln\!\left(\sqrt{\frac{\pi}{\mathrm{e}}}\right).$$

Finally then, $C \,=\sqrt{2}\,\mathrm{e}^S \,=\, \sqrt{2}\frac{\sqrt{\pi}}{\sqrt{\mathrm{e}}} \,=\, \frac{\sqrt{2\pi}}{\sqrt{\mathrm{e}}}$. Nice! ✓

(vi) We verify ($\star$). There are a number of ways to evaluate this integral and some of them are fairly highfalutin (e.g., [2, 10, 11]). Here's a way that is close to that in [5]. Since $\sin(\pi/2 - u) = \cos(u)$, the substitutions $u \mapsto 1/2 - u$ and $u \mapsto u - 1/2$ give

$$\int_0^{1/2} \ln(\sin(\pi u))\,du \;=\; \int_0^{1/2} \ln(\cos(\pi u)\,du \;=\; \int_{1/2}^{1} \ln(\sin(\pi u))\,du\,. \qquad (\star\star)$$

(The former has the neat consequence $\displaystyle\int_0^{\pi/2} \ln(\tan(x))dx \;=\; 0$.) Using ($\star\star$),

$$\begin{aligned} 2\int_0^{1/2} \ln(\sin(\pi u))du \;&=\; \int_0^{1/2} \ln(\sin(\pi u))du \,+\, \int_0^{1/2} \ln(\cos(\pi u))du \\ &=\; \int_0^{1/2} \ln(\sin(\pi u)\cos(\pi u))du\,. \end{aligned}$$

Then since $\sin(\pi u)\cos(\pi u) \;=\; \frac{1}{2}\sin(2\pi u)$, we get

$$\begin{aligned} 2\int_0^{1/2} \ln(\sin(\pi u))\,du \;&=\; \int_0^{1/2} \ln(\sin(2\pi u))\,du \;-\; \int_0^{1/2} \ln(2)\,du \\ &=\; \frac{1}{2}\int_0^{1} \ln(\sin(\pi u))\,du \;-\; \frac{1}{2}\ln(2) \end{aligned}$$

$$= \frac{1}{2}\int_0^{1/2} \ln(\sin(\pi u))\,du \;+\frac{1}{2}\int_{1/2}^{1} \ln(\sin(\pi u))\,du - \frac{1}{2}\ln(2)$$

$$= \int_0^{1/2} \ln(\sin(\pi u))\,du \;- \frac{1}{2}\ln(2) \quad \big(\text{again using } (\star\star)\big).$$

Therefore, as claimed,

$$\int_0^{1/2} \ln(\sin(\pi u))du \;= -\frac{1}{2}\ln(2)\,. \qquad \checkmark\checkmark$$

▶ **Remark** We saw in Chap. 13 that (15.1) is an underestimate of $n!$. Since the sequence $\{S_n\}$ in the proof above is increasing, (15.2) is an *overestimate*:

$$n! \;<\; \sqrt{2\pi}\left(\frac{n+1/2}{\mathrm{e}}\right)^{n+1/2}.$$

It is shown in [4] that for n very large, the error in approximating $n!$ with (15.2) is about one half that of approximating $n!$ with (15.1). See also [3, 8, 9].

15.3 Epilogue

The Sine Product provides yet another connection to some things that we have been studying [12]. Recall from our opening prelude (Chap. 1) that Viète's Product is

$$\frac{2}{\pi} = \prod_{n=1}^{\infty} \frac{b_n}{2}, \quad \text{where } b_1 = \sqrt{2}, \quad \text{and} \quad b_{n+1} = \sqrt{2+b_n} \quad (n = 1, 2, 3, \ldots).$$

In the course of our verification of Viète's Product (Sect. 1.2), we obtained Euler's expression

$$\frac{\sin(\theta)}{\theta} \;=\; \cos\!\left(\tfrac{\theta}{2}\right)\cos\!\left(\tfrac{\theta}{4}\right)\cos\!\left(\tfrac{\theta}{8}\right)\cdots\cos\!\left(\tfrac{\theta}{2^m}\right)\frac{\sin\!\left(\frac{\theta}{2^m}\right)}{\frac{\theta}{2^m}}. \tag{15.7}$$

Now, replacing πu with $\frac{\theta}{2^m}$ in the Sine Product (15.3), we get

$$\frac{\sin\!\left(\frac{\theta}{2^m}\right)}{\frac{\theta}{2^m}} \;=\; \prod_{n=1}^{\infty}\left(1 - \frac{(\theta/2^m)^2}{n^2\pi^2}\right)$$

$$= \prod_{n=1}^{\infty} \left(\frac{n\pi - \theta/2^m}{n\pi} \cdot \frac{n\pi + \theta/2^m}{n\pi} \right)$$

$$= \prod_{n=1}^{\infty} \left(\frac{n2^m - \theta/\pi}{n2^m} \cdot \frac{n2^m + \theta/\pi}{n2^m} \right).$$

Substituting this into (15.7) and then taking $\theta = \pi/2$, we get (after simplification):

$$\frac{2}{\pi} = \prod_{n=1}^{m} \frac{b_n}{2} \cdot \prod_{n=1}^{\infty} \left(\frac{n2^{m+1} - 1}{n2^{m+1}} \cdot \frac{n2^{m+1} + 1}{n2^{m+1}} \right). \tag{15.8}$$

Here, again, $b_1 = \sqrt{2}$, and $b_{n+1} = \sqrt{2 + b_n}$.

Taking $m = 0$ in (15.8) gives *Wallis's Product*

$$\frac{2}{\pi} = \frac{1 \cdot 3}{2 \cdot 2} \cdot \frac{3 \cdot 5}{4 \cdot 4} \cdot \frac{5 \cdot 7}{6 \cdot 6} \cdot \frac{7 \cdot 9}{8 \cdot 8} \cdot \cdots$$

and having $m \to \infty$ in (15.8) gives *Viète's Product*. Pretty good!

Each m in between gives something interesting too. For example, $m = 3$:

$$\frac{2}{\pi} = \frac{\sqrt{2}}{2} \cdot \frac{\sqrt{2+\sqrt{2}}}{2} \cdot \frac{\sqrt{2+\sqrt{2+\sqrt{2}}}}{2} \cdot \frac{15 \cdot 17}{16 \cdot 16} \cdot \frac{31 \cdot 33}{32 \cdot 32} \cdot \frac{47 \cdot 49}{48 \cdot 48} \cdot \frac{63 \cdot 65}{64 \cdot 64} \cdot \cdots .$$

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Correction to: Some Important Series

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The original version of this chapter was inadvertently published with incorrect Abstract. The chapter has been corrected and approved by the author.

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